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Introduction

Lab overview

Welcome to the watsonx Orchestrate flow and document processing lab guide. This guide is designed to help both beginners and experienced users alike, providing a step-by-step overview of the current capabilities of watsonx Orchestrate's flow and document processing capabilities.

When complete, the agent created will be able to classify documents, distinguish invoices from contracts and extract specific fields from these documents depending on their type.

What you will learn

Once you have completed this lab you will have developed skills enabling you to complete the following tasks:

- How to set up agents that use document classification
- How to identify and extract key data fields from uploaded documents
- Prompt the Model to identify key data in contracts and invoices
- Create flows inside your agent to manage complex task routing
- Test the agent you have created

How to get support

If you encounter an issue with this lab, you can request assistance through the following channels:

- IBMers can use the #ba-techlcd-support Slack channel.
- If you are an IBM Business Partner and require assistance, please open a support case at IBM Technology Zone Help.

Prerequisites

You should have completed the other labs in this learning plan:

- Completed the preparation guide activities
- Have access to a watsonx Orchestrate SaaS instance.
- You have downloaded the following files on your local computer:
  - Contract.pdf
  - Utility_bill_training.pdf
  - Utility_bill_2.png
  - Utility_bill_4.png
  - Utility_bill_5.png

Known Issues and Workarounds

As of the time of writing this lab, we have identified several issues that may affect your experience. Please read the following carefully to understand how to work around them:

- **Adding a node to a link**: In certain cases, the '+' icon used to add a node to a link may not appear—especially after adding the first extractor to the workflow. Instead, small moving arrows may be displayed, preventing you from adding any elements to the link.
  Workaround: Exit the workflow editor and reopen it.

- **Extracting fields from documents**: After creating a field or updating its definition, the UI may take time to display the newly extracted value—or may not show it at all. The same issue can occur when adding examples and clicking the *Show in document* button.
  Workaround: Wait a few seconds for the screen to refresh or click your browser's refresh button.

- **Fields showing as "Not found"**: Occasionally, fields may appear as "Not found in the document" even though they were previously extracted correctly.
  Workaround: Wait a few seconds to allow the LLM to return the correct values and for the UI to update.

Use case

This lab focuses on a practical use case involving document classification, followed by key data extraction from both unstructured documents (primarily contracts) and structured ones (such as invoices). At the outset, the guide outlines the sequence of activities required to support the user journey of extracting essential information from uploaded documents.

In this lab scenario, a payment recovery organization processes numerous contracts and invoices weekly, extracting critical information like payment terms, amounts, and due dates. Currently, this is done manually, consuming significant employee time and increasing the risk of errors. With growing document volumes, the firm faces scalability challenges and inefficiencies in their processes.

Challenges:

- Manual Process: Employees spend hours reviewing documents and entering data, which slows down operations.
- Errors: Human mistakes lead to discrepancies, delays, and potential client issues.
- Scalability: As the business expands, the volume of documents grows, requiring more manual labor, which is costly and unsustainable.

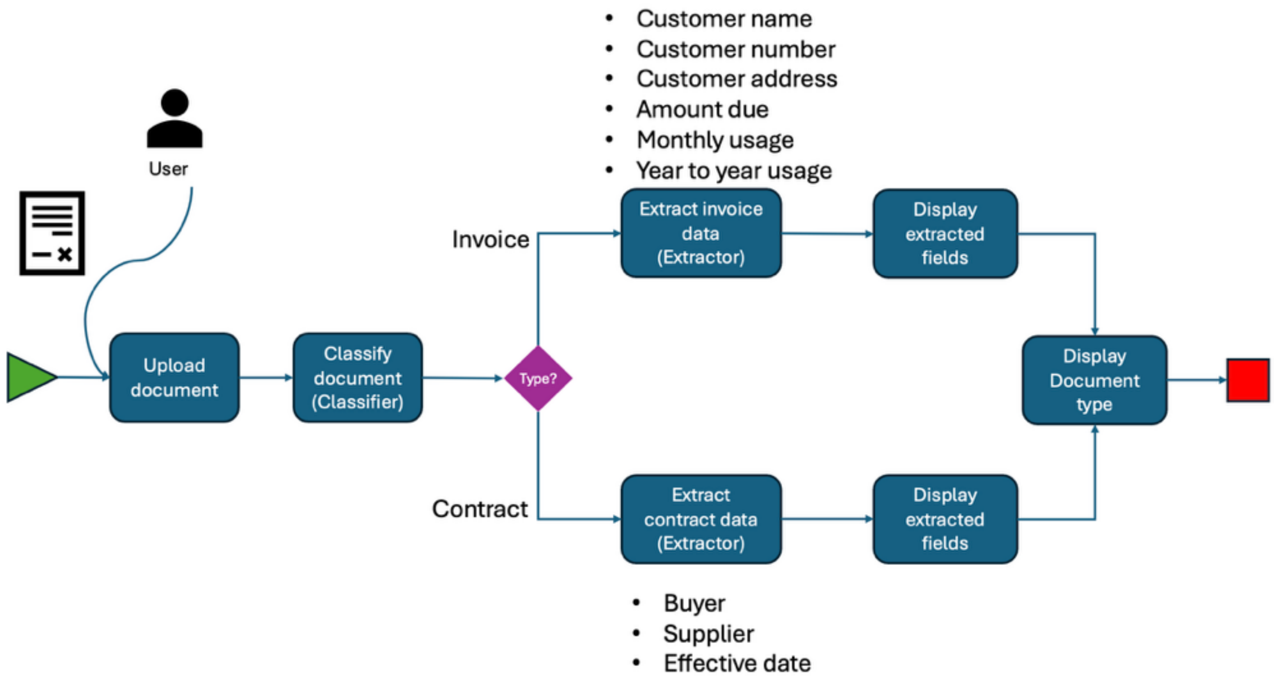
Solution: Watsonx Orchestrate

The firm implements Watsonx Orchestrate to automate the extraction of key data from contracts and invoices, streamlining the process and reducing errors.

- Automated Extraction: Watsonx Orchestrate automatically extracts relevant data from documents, eliminating manual entry.
- Workflow Integration: The extracted data is seamlessly integrated into the firm's systems (e.g., billing and accounting), speeding up the process.
- Confidence Review: If the extraction confidence is low, a review step ensures data accuracy without delays.

The process begins with configuring the document classifier and extractor tools to identify the type of document being ingested. Based on the classification, the system then determines which key-value pairs to extract. Finally, the flow concludes with a summary of the main fields retrieved by the agent during the extraction process, completing the end-to-end user journey.

The diagram below presents the workflow supporting this use case:



This lab guide will guide you through the different steps to create this flow.

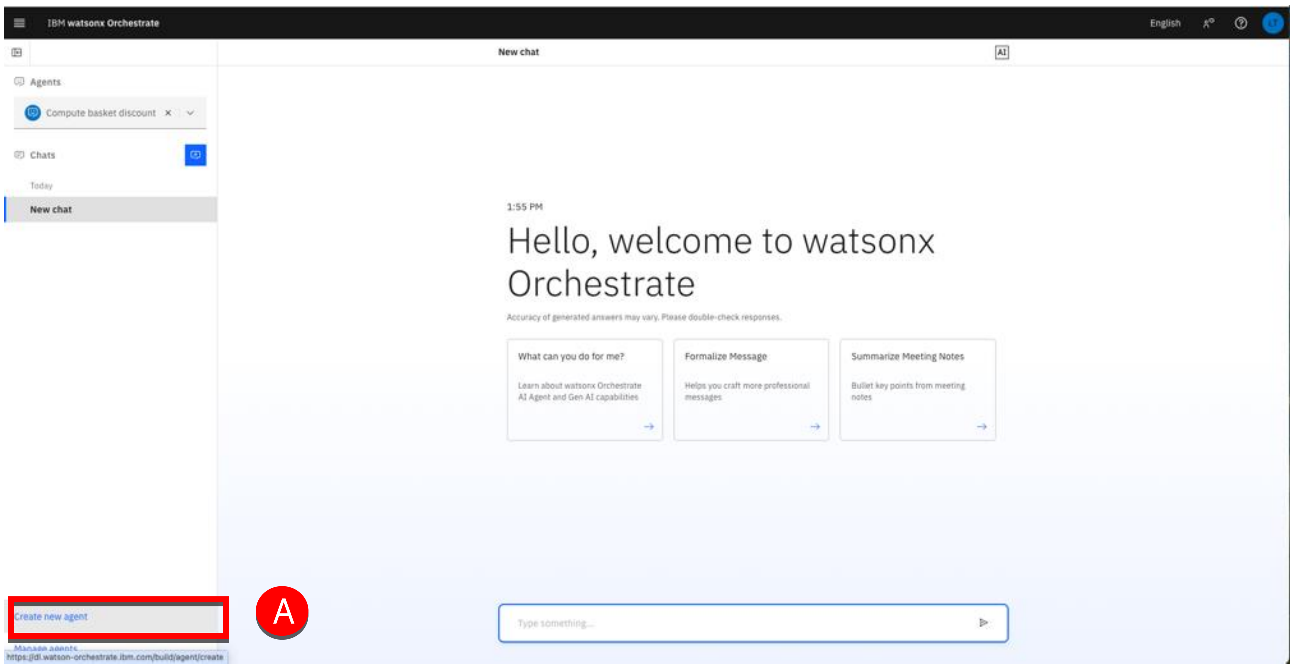
Let's get started.

Create an Agent

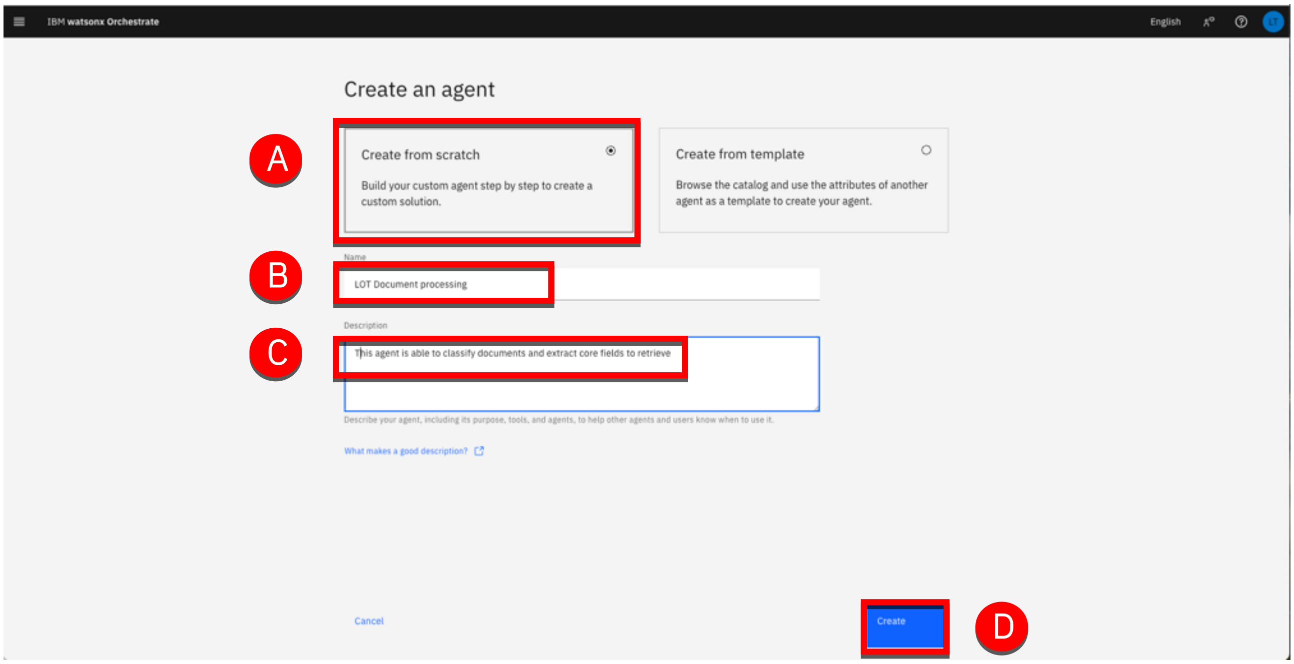
In this section, you will create a document processing agent. This agent will implement the process previously described using the watsonx Orchestrate flow capabilities, as well as a Document Classifier and Document extractors.

1. Launch your watsonx Orchestrate SaaS instance.

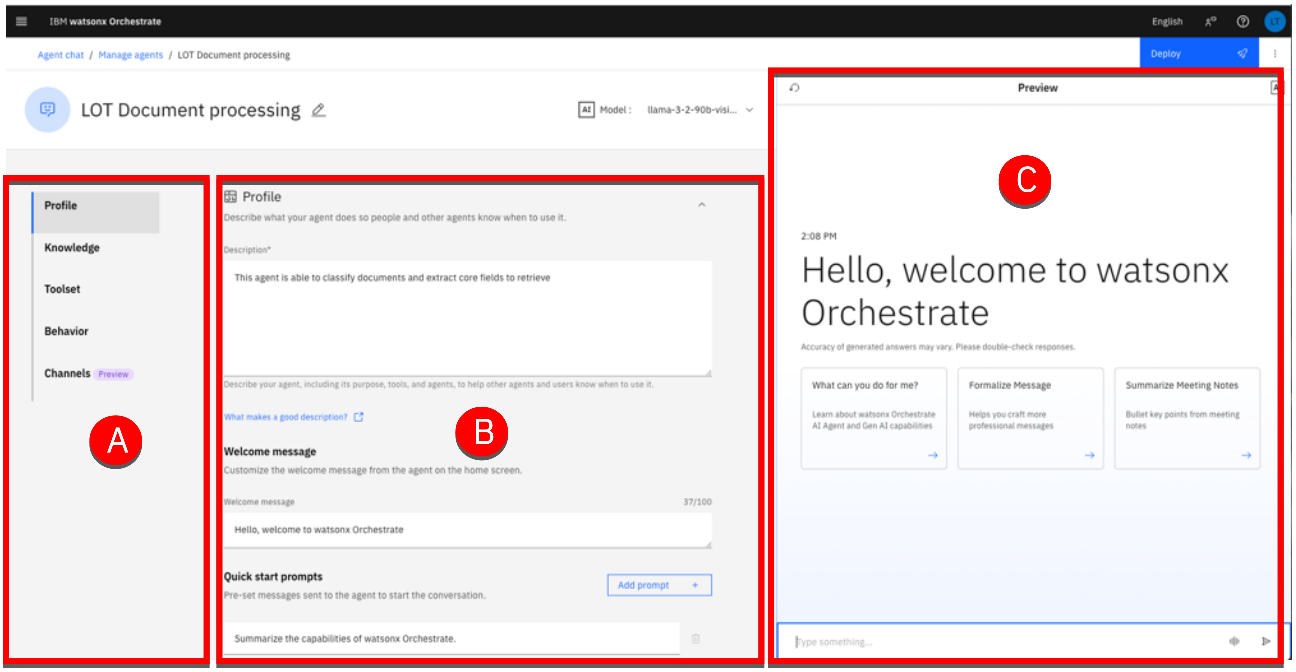
2. Click **Create new Agent** (A).



3. Select **Create from scratch** (A). Type "`[YOUR INITIALS] Document processing`" in the **Name** field (B). Type "`This agent is able to classify documents and extract core fields to retrieve`" in the **Description** field (C). Click **Create** (D).



4. Note the different sections of the Agent builder tool.



The Agent Builder opens; the screen is divided into three key areas:

- (A) Navigation Panel – move between different sections of the Agent Builder.
- (B) Configuration – set up and customize the core functionality of your agent.
- (C) Preview – preview and test your agent.

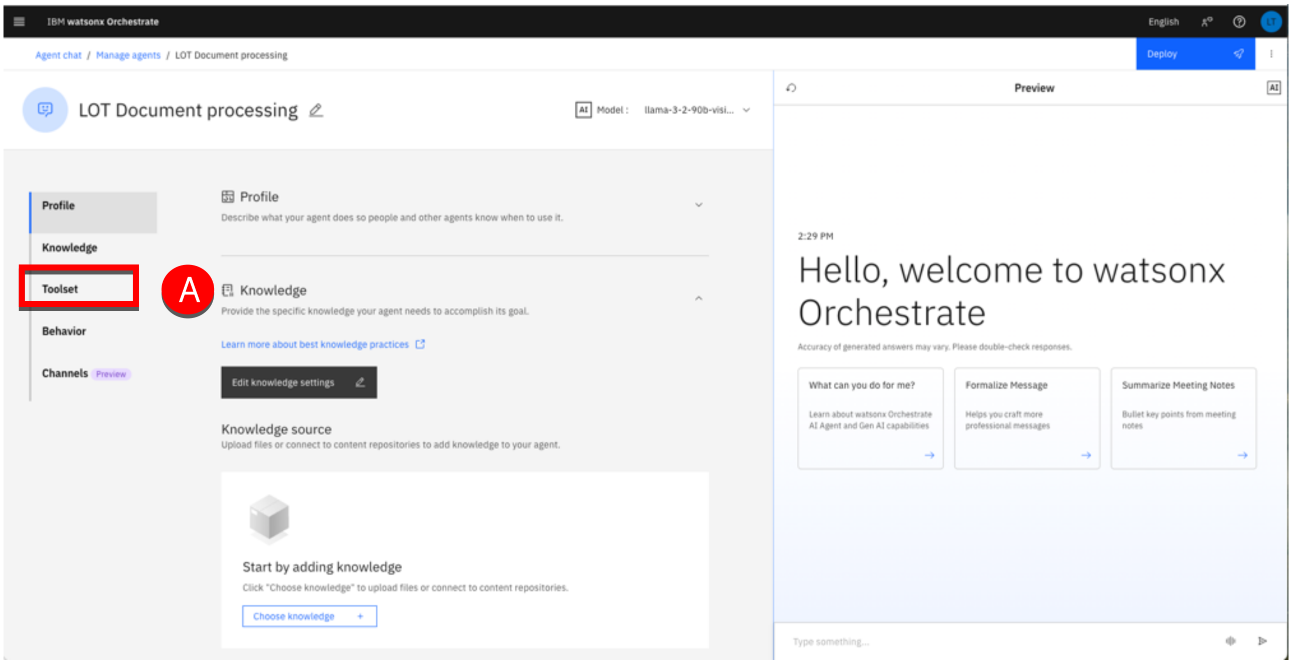
Below is a description of each section:

- Profile: Define your agent's purpose, usage scenarios, and interaction style. Describe what the agent does, when it should be used (especially in multi-agent configurations), and choose its style—Default or ReAct—to guide how it interprets requests, plans and uses tools.
- Knowledge: This is where you equip your agent with knowledge by uploading files or connecting to conversational search platforms such as Milvus, Elasticsearch, or custom-built sources. This ensures the agent can generate accurate, contextual responses by drawing on relevant content.
- Toolset: Provide your agent with tools to perform tasks. Tools can be added from the Catalog, imported from OpenAPI specification files and MCP servers, or built with custom flows. Tools extend the agent's capabilities, enabling it to automate actions such as retrieving data or sending emails.
- Behavior: Here, you define how the agent interacts with users, formats data and how it handles requests. Add rules and instructions to shape its tone, response style, and overall behavior during interactions.
- Channels: This is where you connect your agent to communication platforms like Slack or embed the agent in a website.

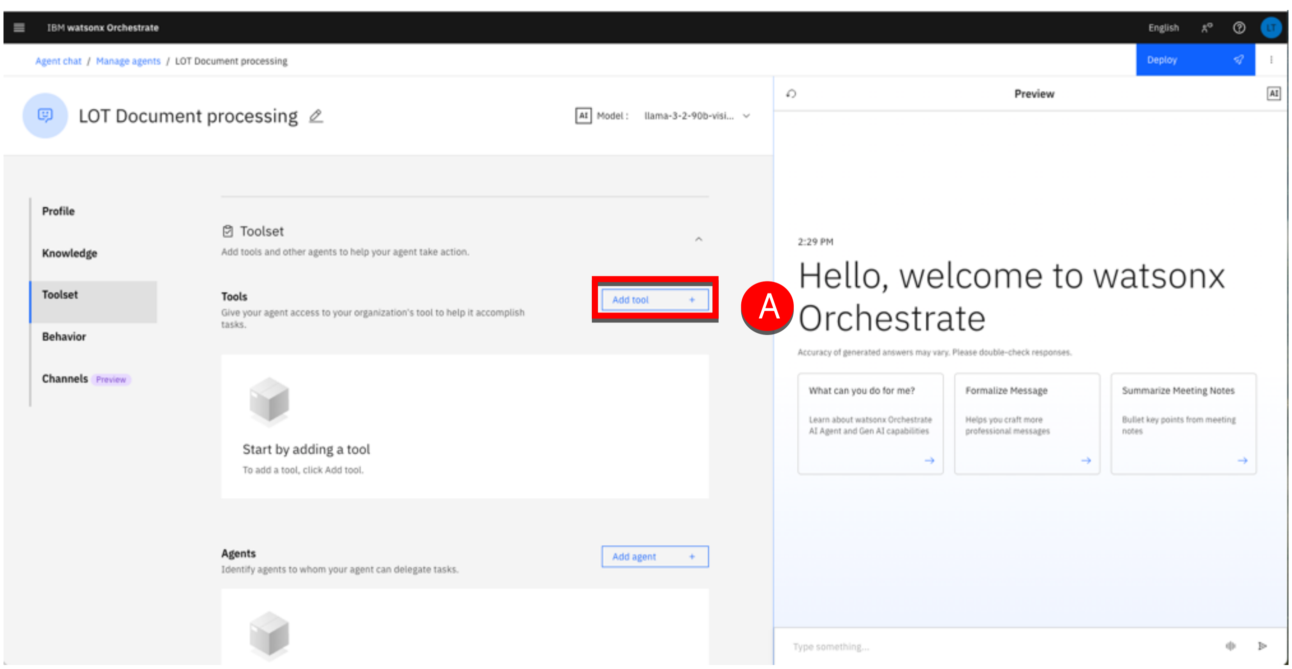
Add tools

In this section you will add a tool to your agent. This tool will contain the process described in the introduction of this lab. You will start by creating the tool and implementing the tool behavior as a flow.

1. Click **Toolset** (A).

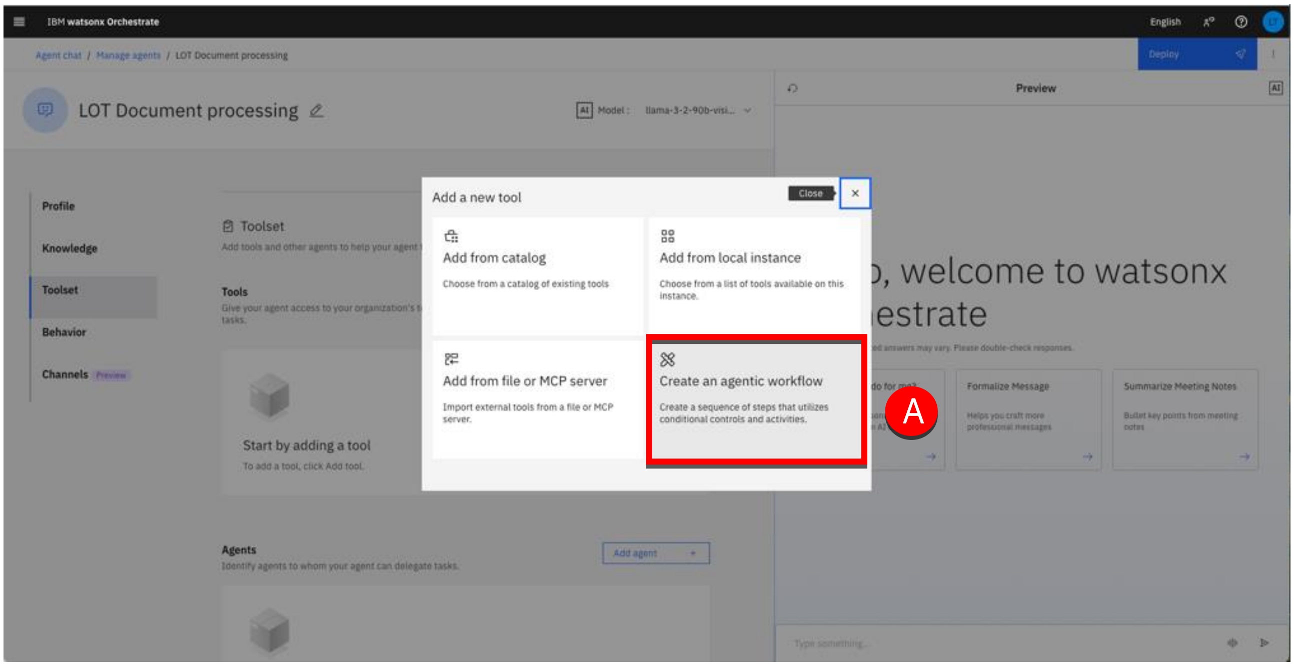


2. Click **Add tool** (A).

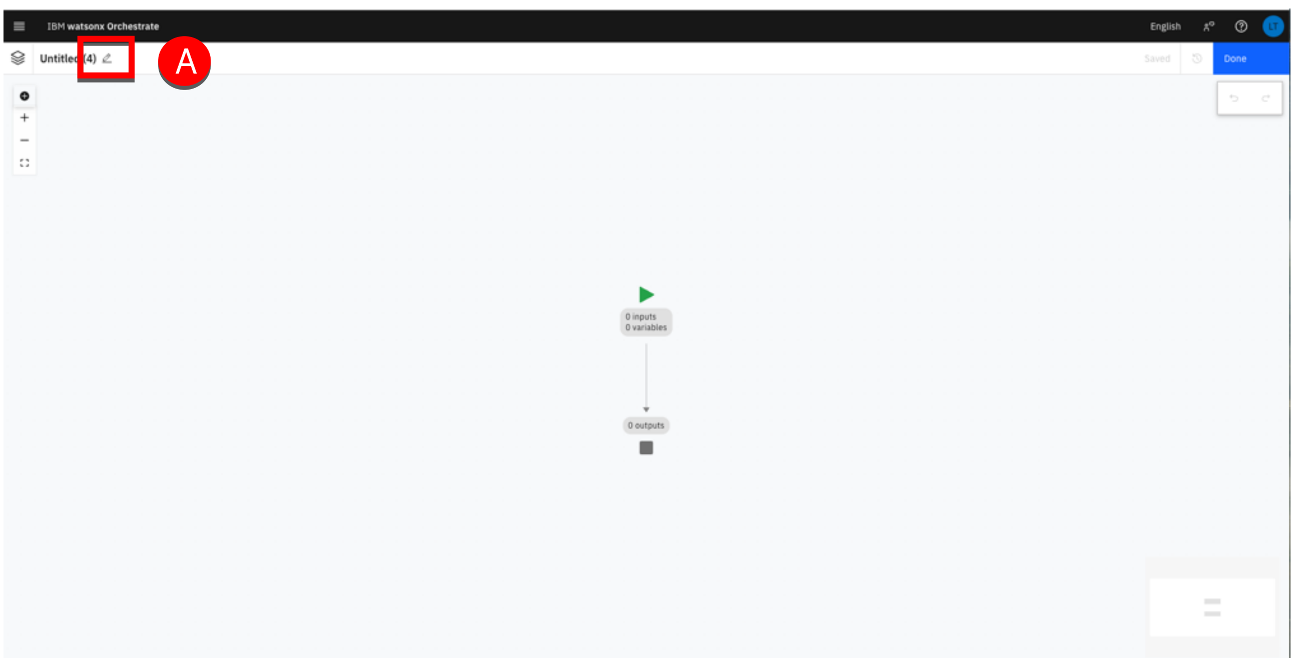


Create the workflow

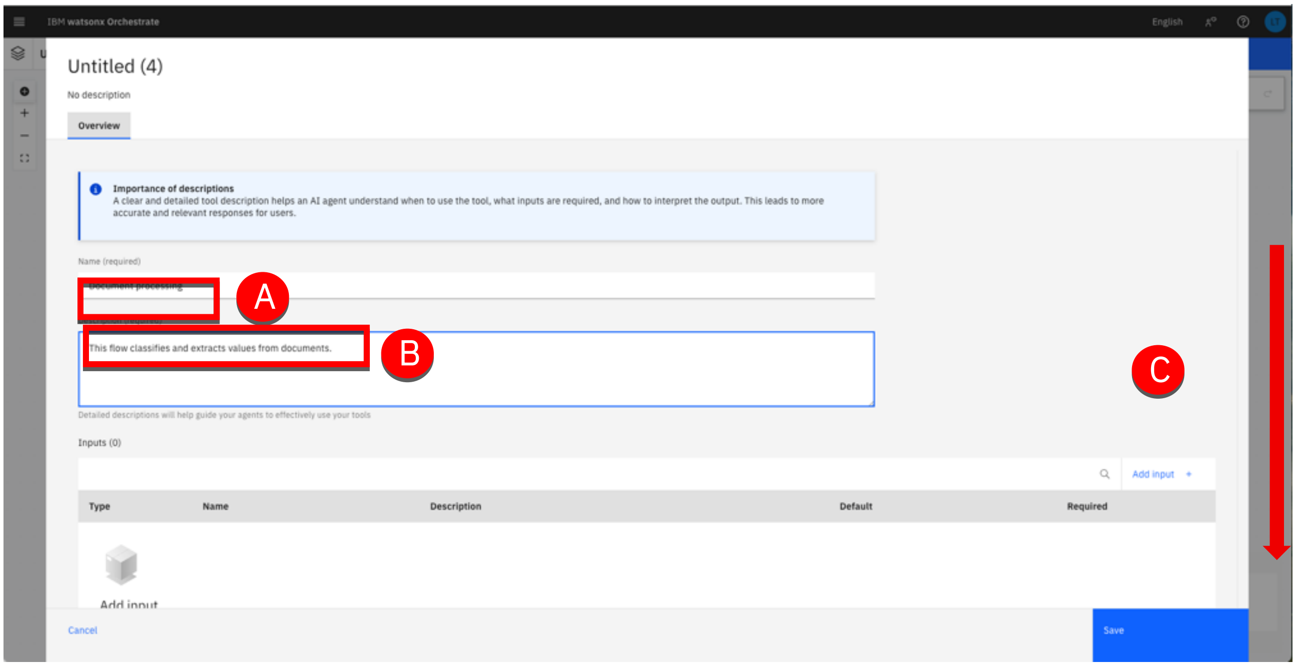
1. Click **Create an agentic workflow** (A).



2. Click the pencil icon (A) to rename the workflow.



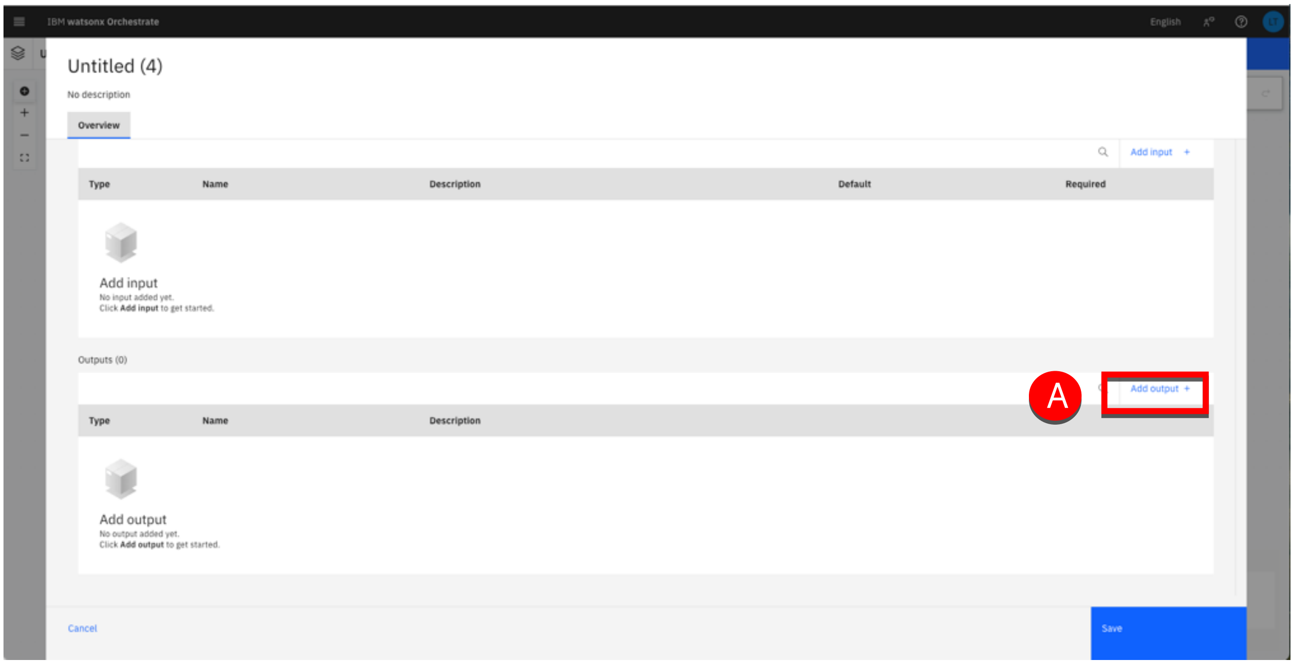
3. Type 'Document processing' in the **Name** (A). Type "This flow classifies and extracts values from documents." In the **Description** (B). Scroll down to the **Output** section (C).



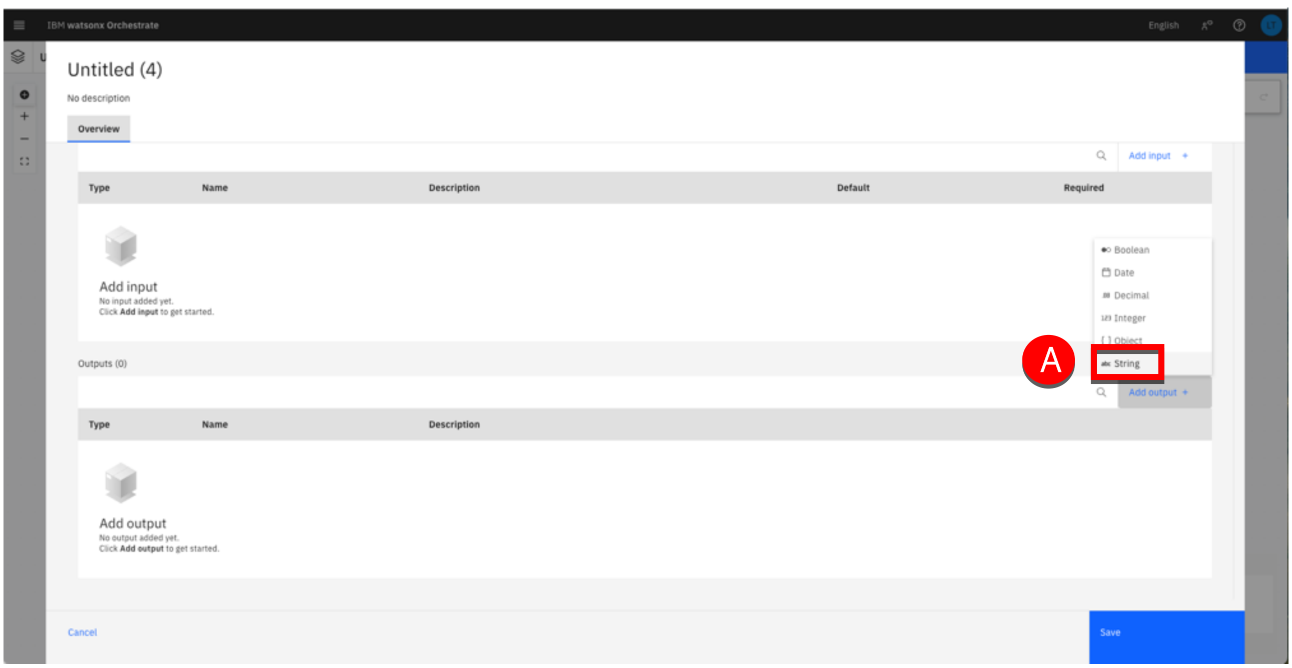
Define the process output

Scrolling down, you'll find the Input and Output sections, which apply to the entire flow. In this case, no inputs are required since they are already handled within the flow itself through the user activity file upload. For the output, instead of using a text output node, the flow can return the classification result, defined as a string.

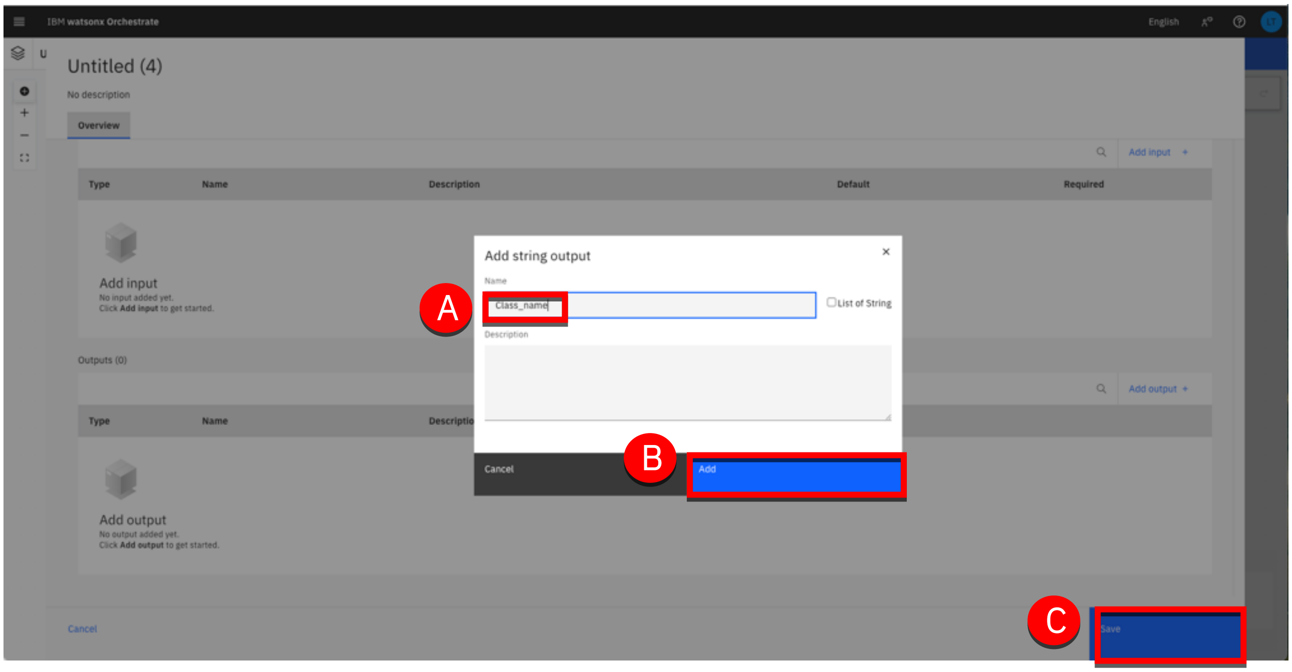
1. Click **Add output** (A).



2. Select **String** (A).



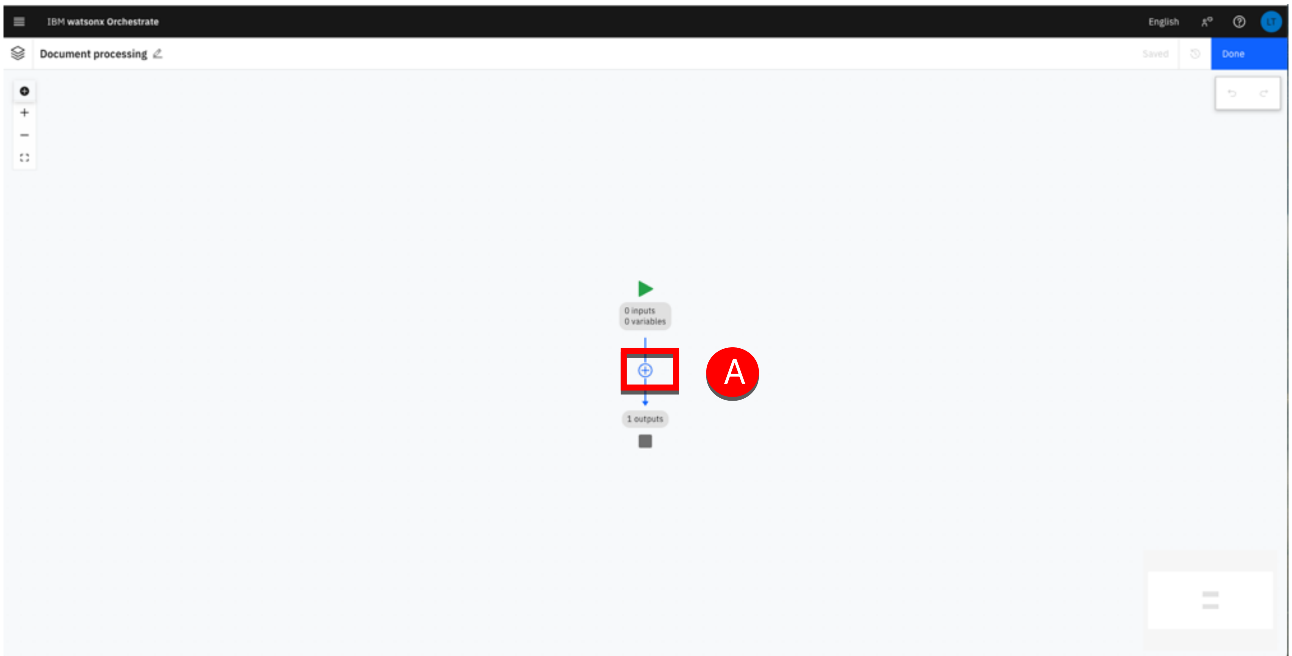
3. Type "`Class_name`" in the **Name** field (A). Click **Add** (B). Then click **Save** (C).



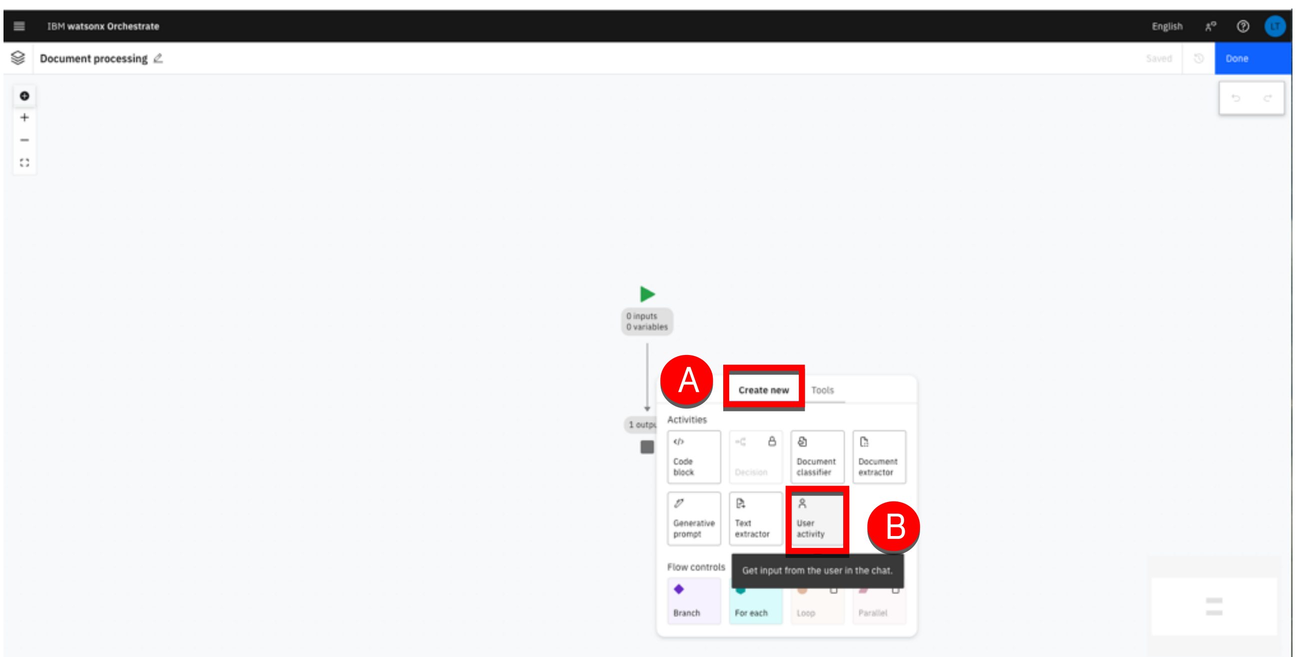
Design the workflow diagram

In this section, you will add the different workflow activities including the document classifier and the two extractors.

1. Move the cursor over the link until the + icon appears. Click the + to add an activity (A). The first activity of the process consists in uploading the document to process. It will be implemented as a user activity.

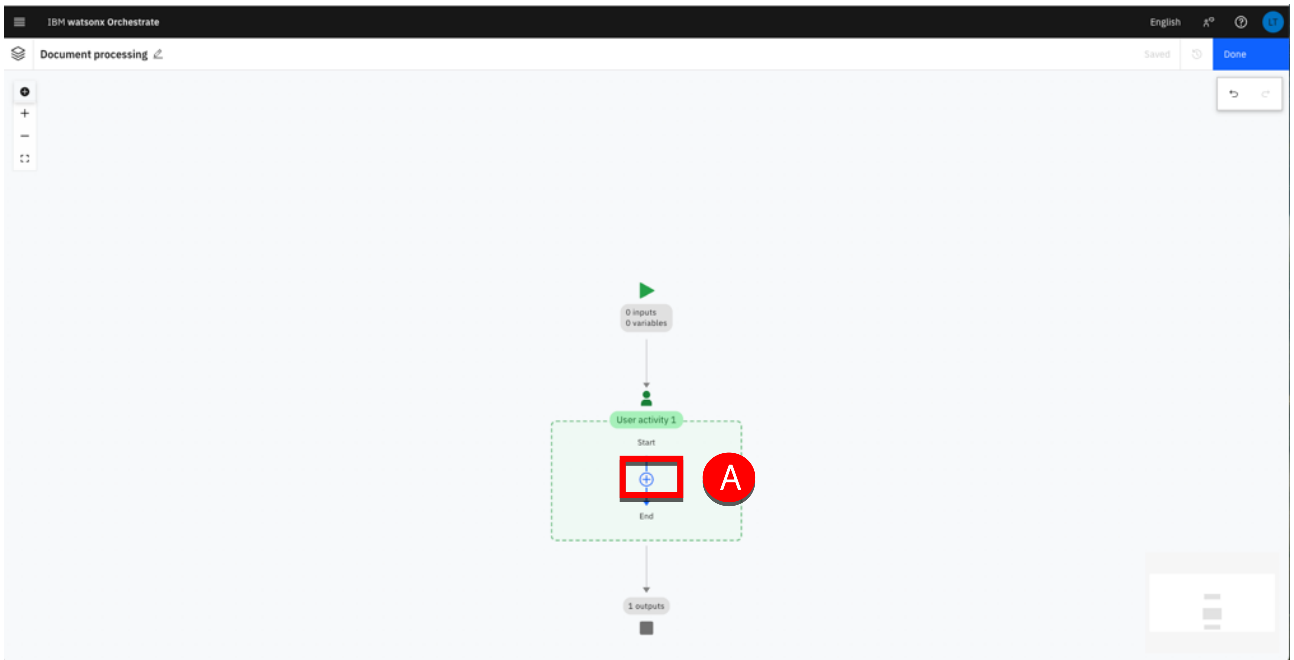


2. Select **Create new** (A) and then select **User activity** (B).

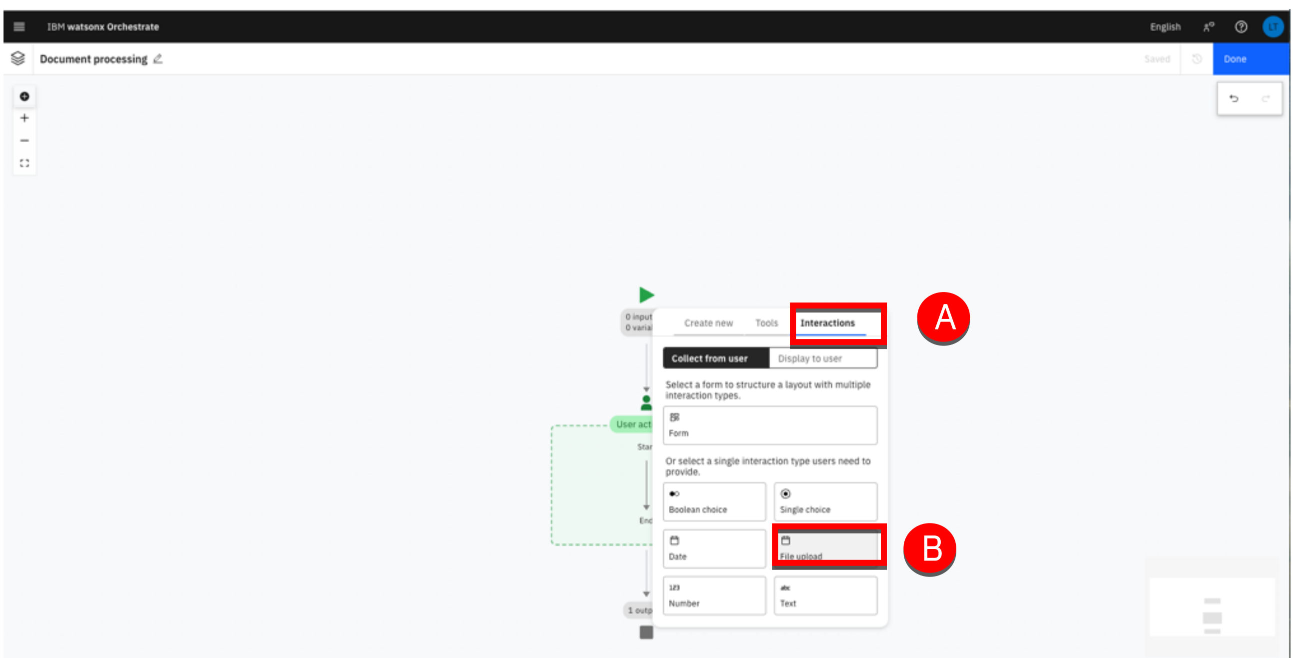


Note: A green box appears, indicating the user activity. This will contain the series of activities that the user must perform whilst interacting with the agent.

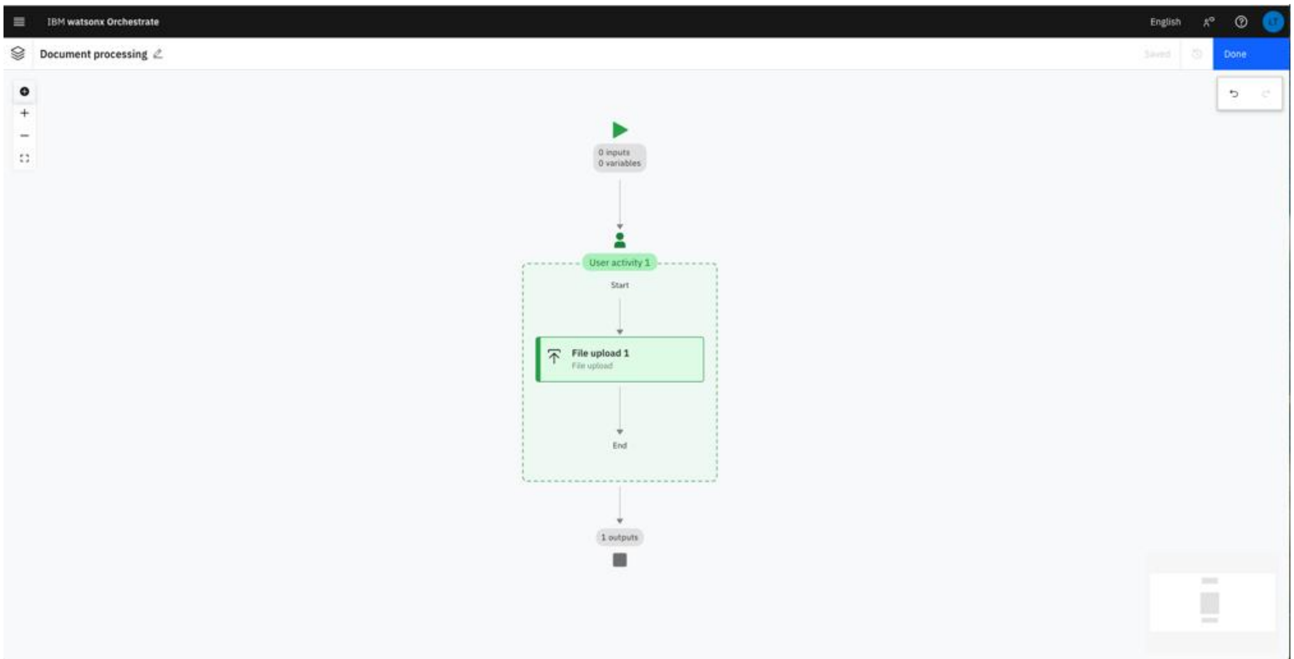
3. Move the cursor into the user activity and click + (A) to add the activities the user will have to perform.



4. Select the **Interactions** tab (B) then click **File upload** (C).



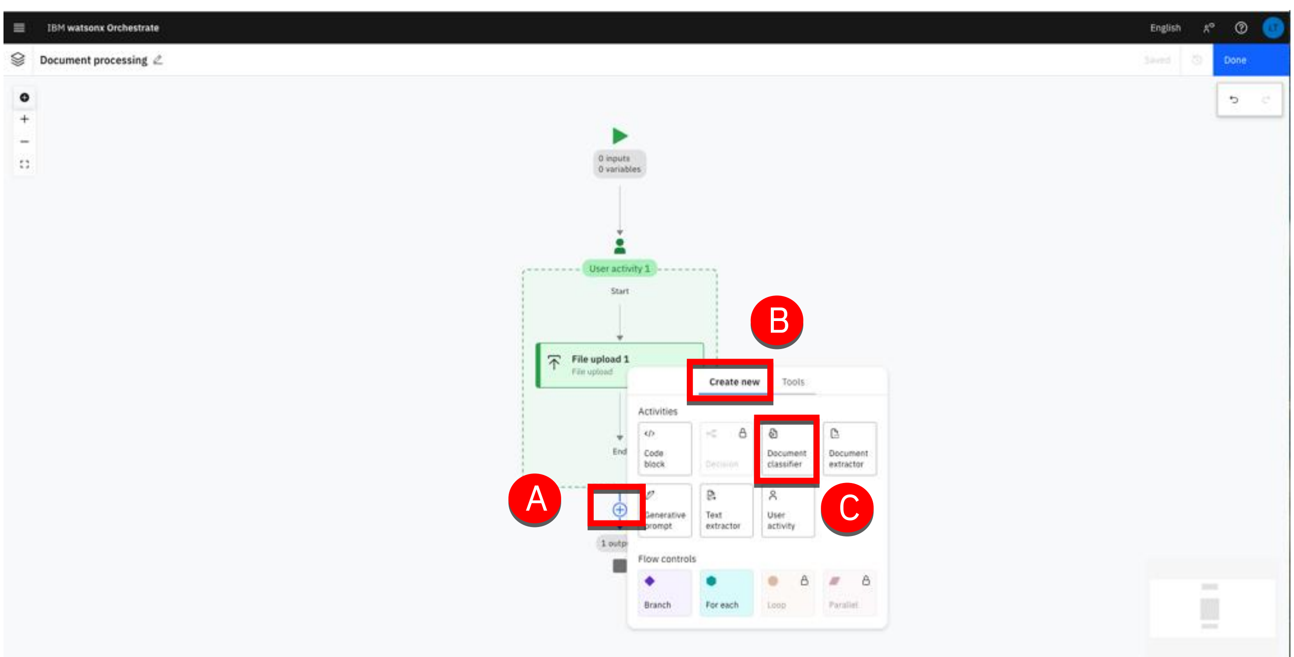
The first activity of the process is now defined. The next step is to classify the document the user will upload. To do this, you will add a watsonx Orchestrate Document Classifier.



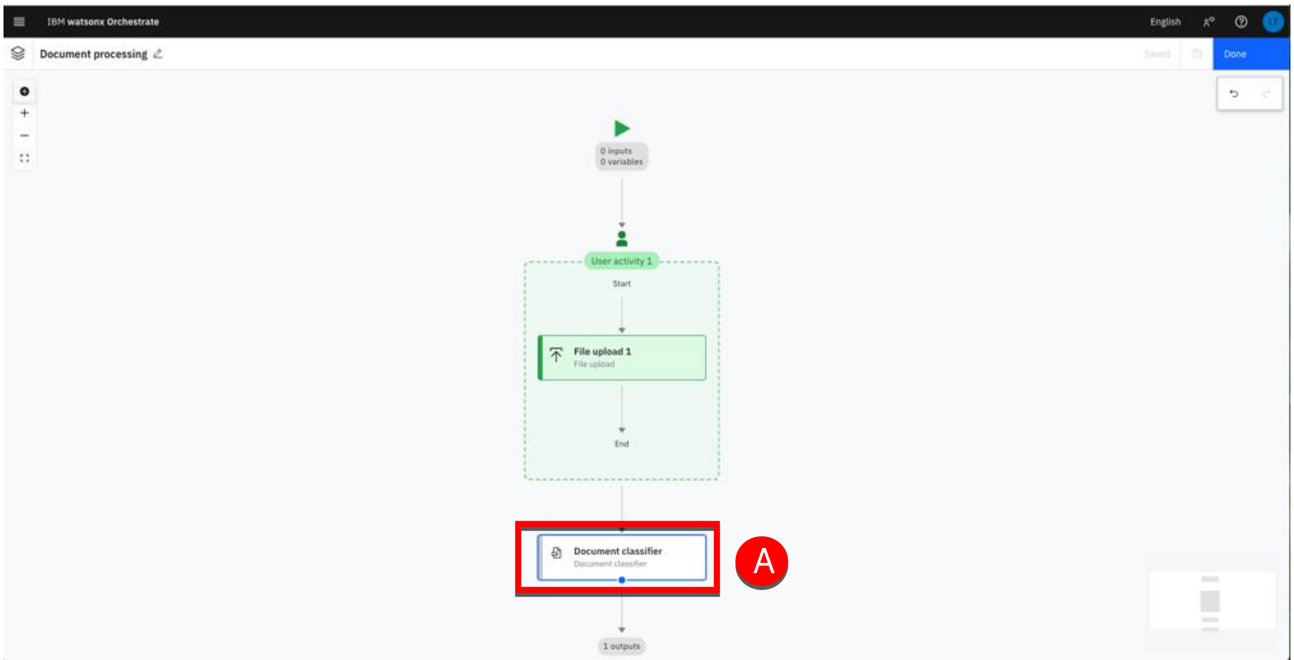
Add a document classifier

In this section, you will add a document classifier to distinguish Invoice documents from the Contract ones.

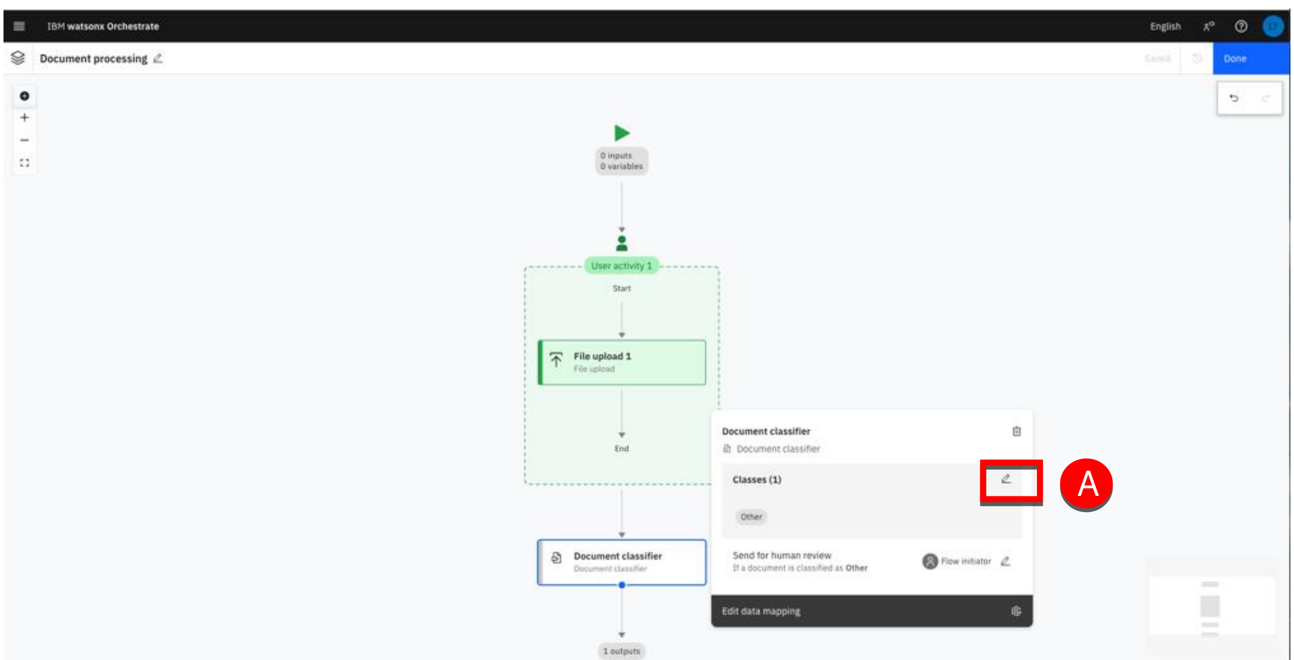
1. Move the cursor over the last link of the process and click + (A). Select **Create new** (B) then click **Document classifier** (C).



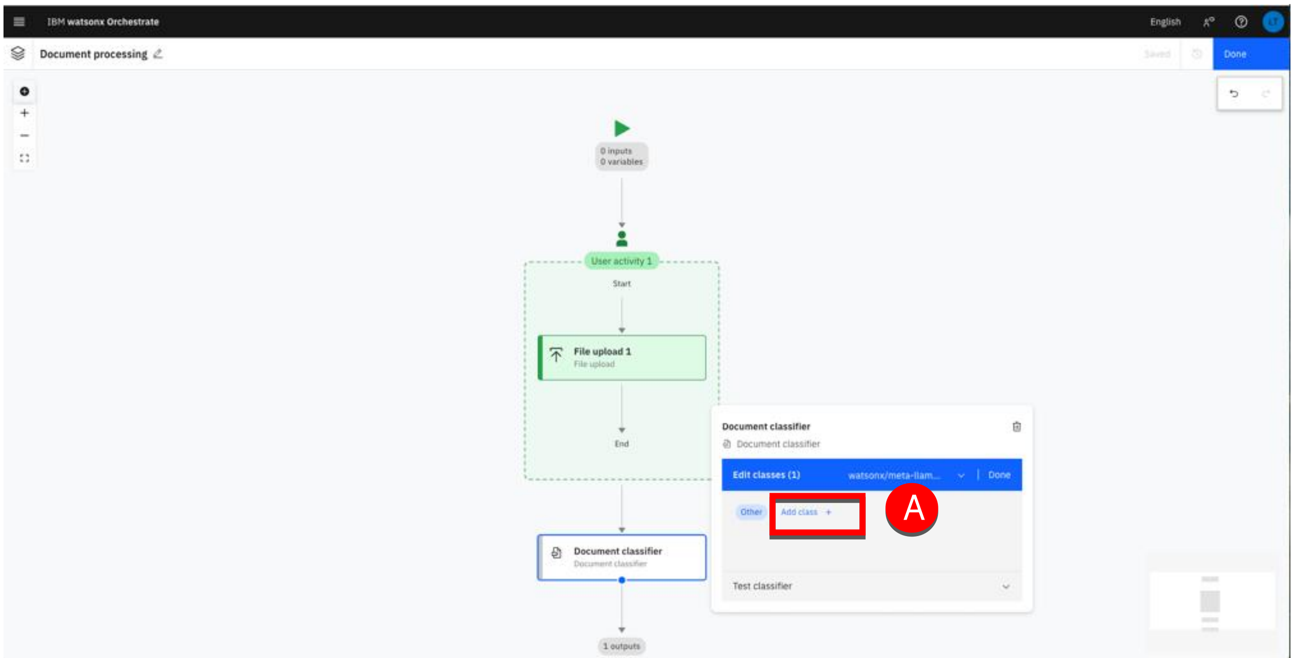
2. Click the **Document classifier** node (A).



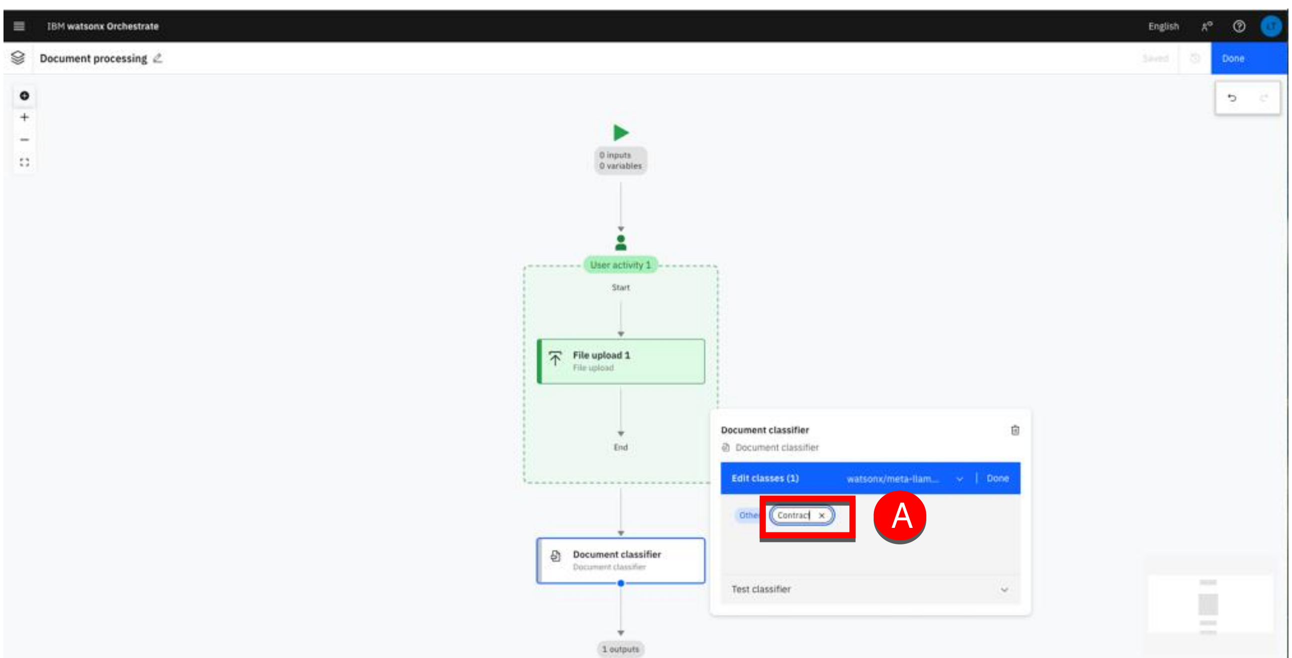
3. Click the edit icon (A).



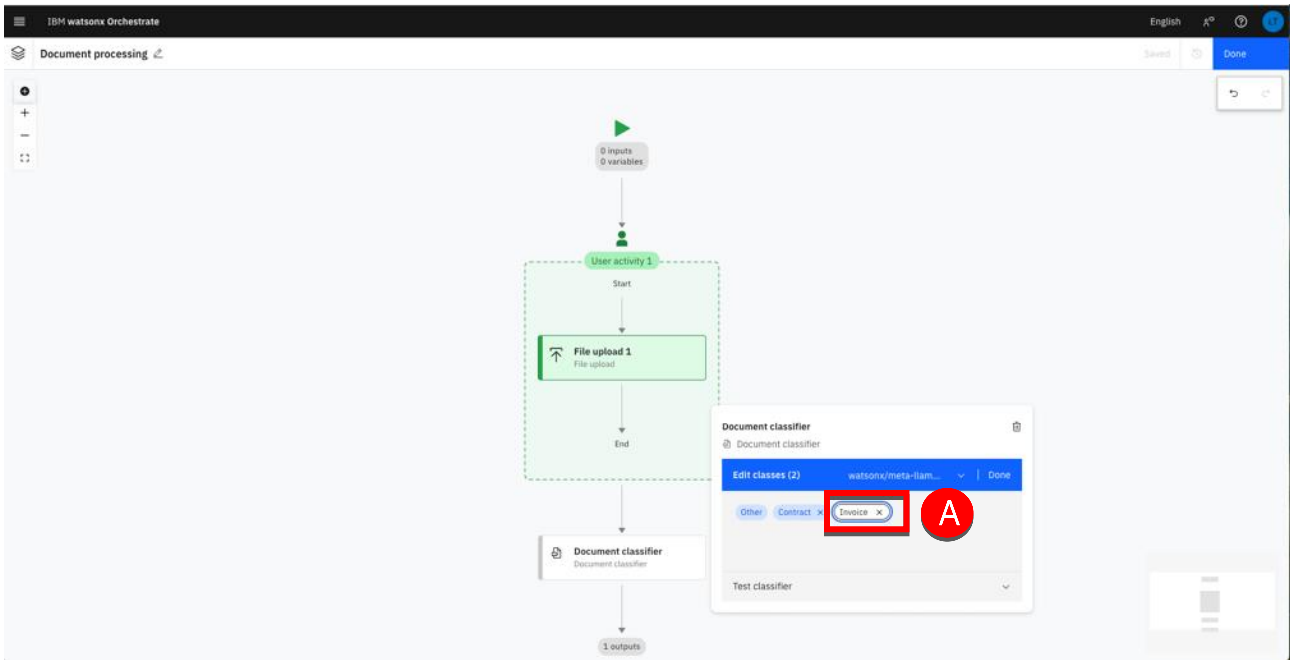
4. Click Add class (A)



5. Type ‘`Contract`’ (A), then Enter.

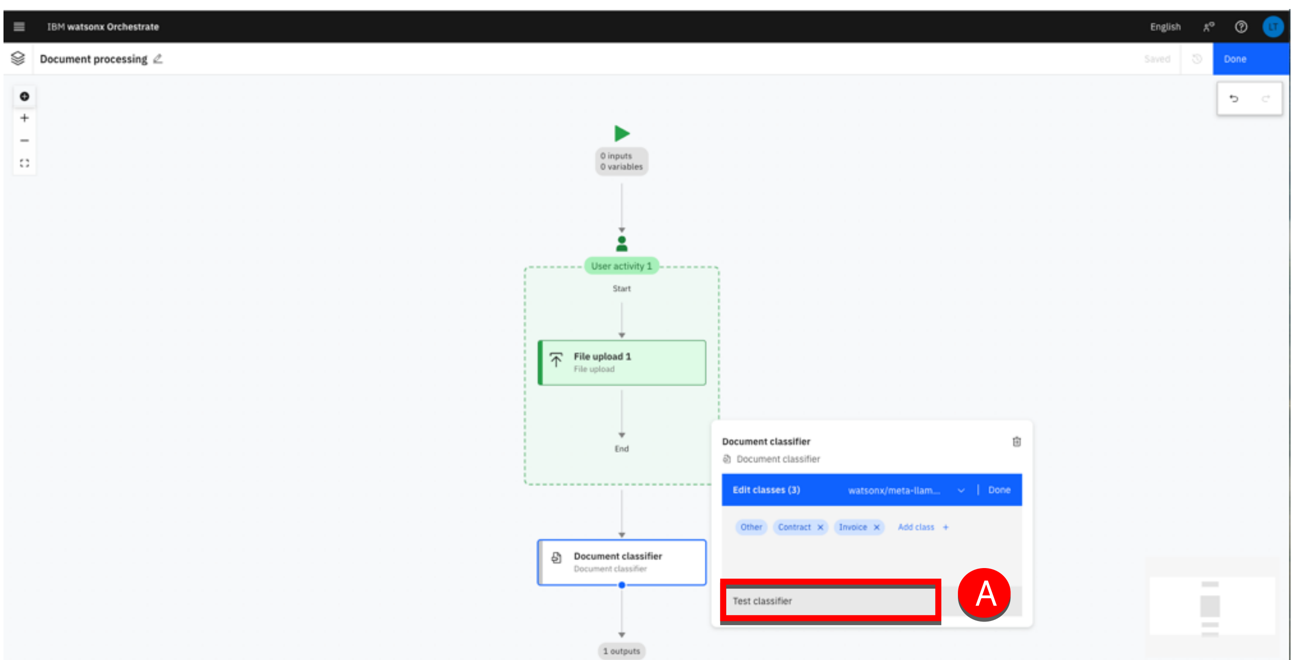


6. Type '`Invoice`' into the new field (A), then TAB out of the field.

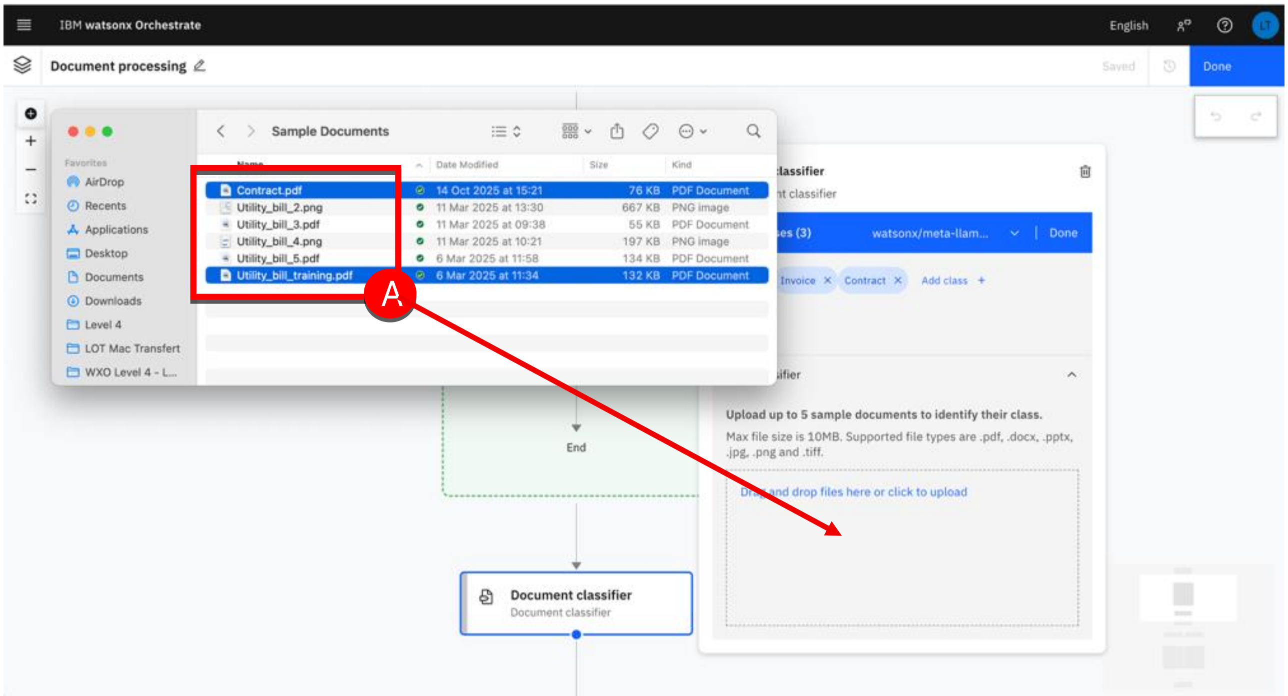


You are now ready to test your document classifier.

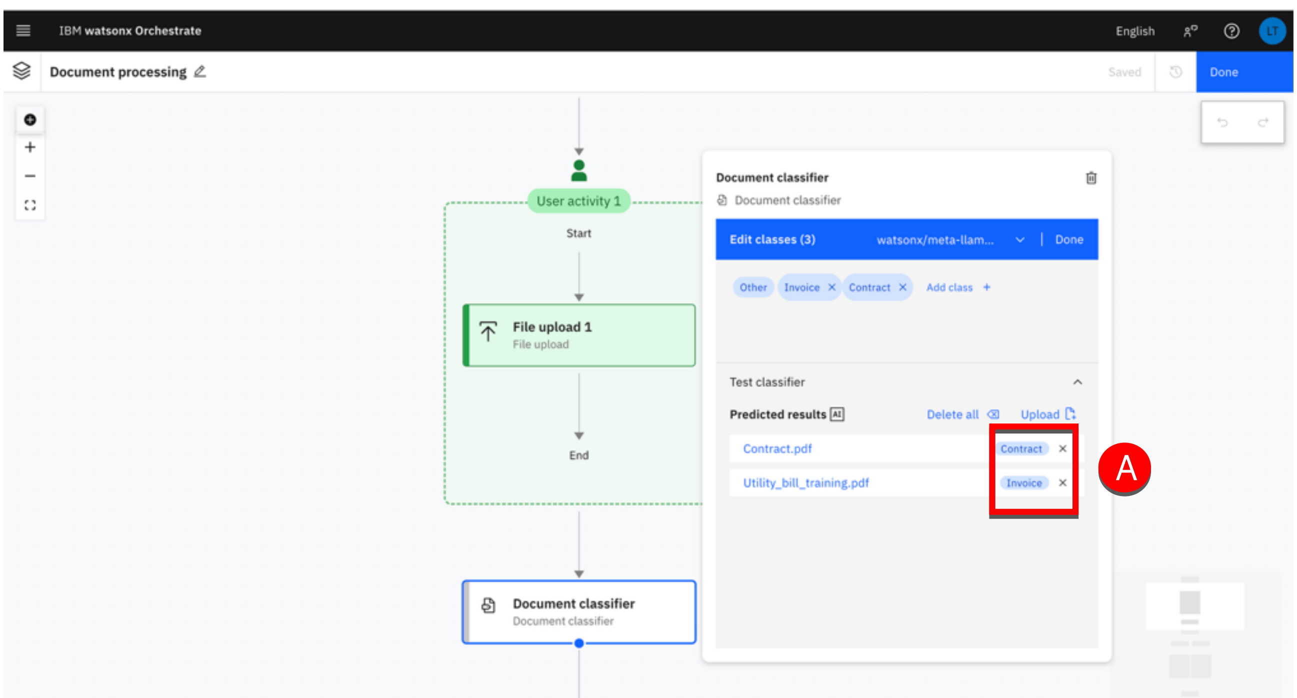
7. Click **Test classifier** (A).



8. Drag and drop the two sample files, **Utility_bill_training.pdf** and **Contract.pdf** from your local machine location (A). Note: ensure you select both files before dropping in the upload area, if you upload a single document you will need to click the **Upload** icon to upload the second document.

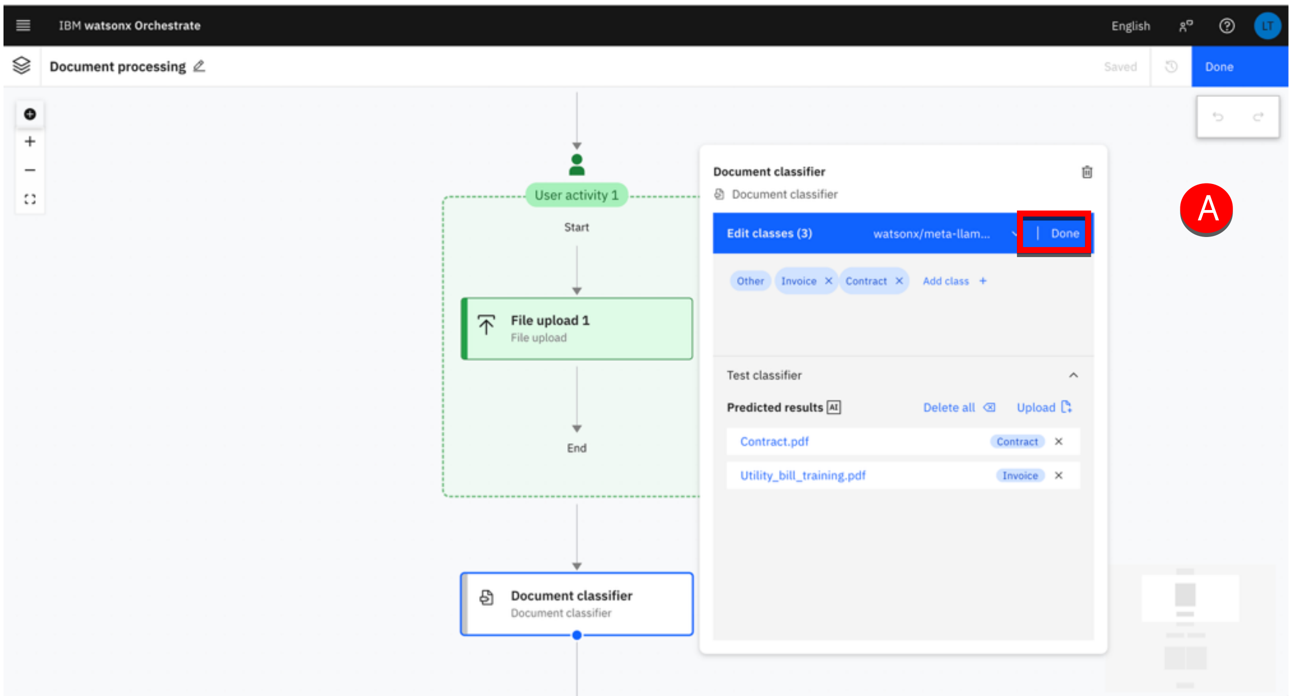


9. Wait for the documents to be analyzed and classified (A).



Note. The two documents have been successfully classified by the LLM based on their content. Contract.pdf has been classified as a Contract and Invoice.pdf has been classified as an Invoice. If your documents last too long to be classified, you may have to delete them and import them back again.

10. Click **Done** (A).

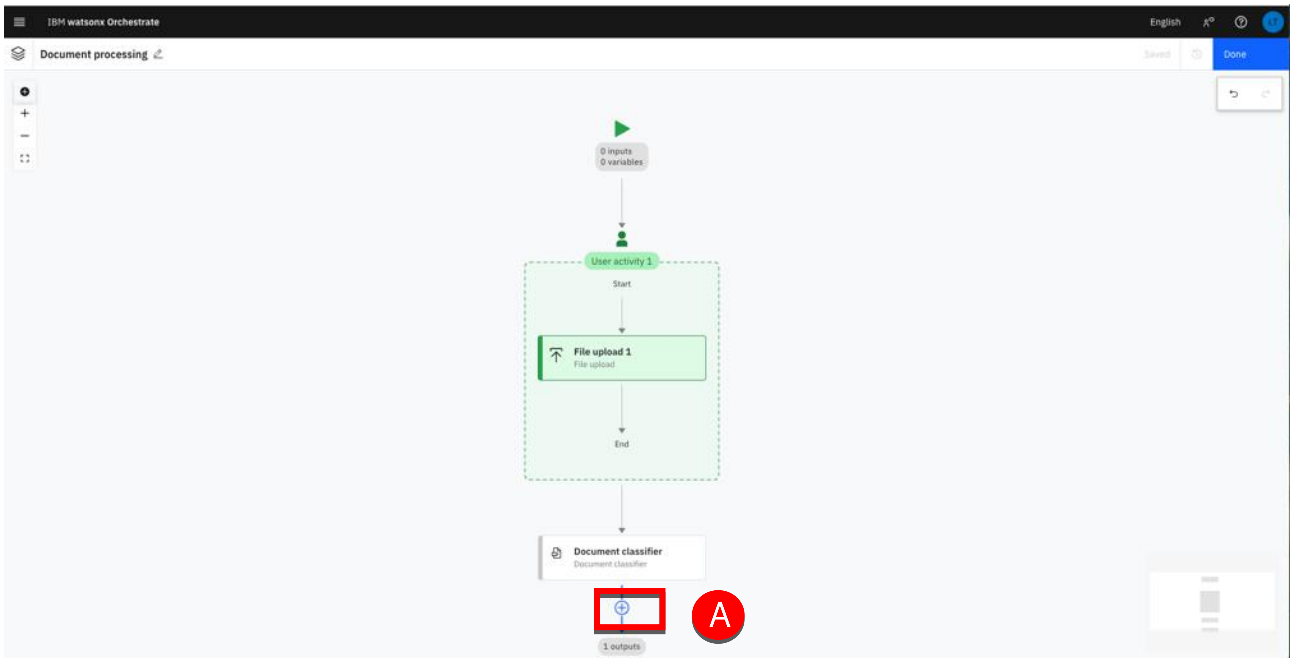


Note: You can click the 'Upload' icon seen above to test out the classifier with more document types and verify its accuracy.

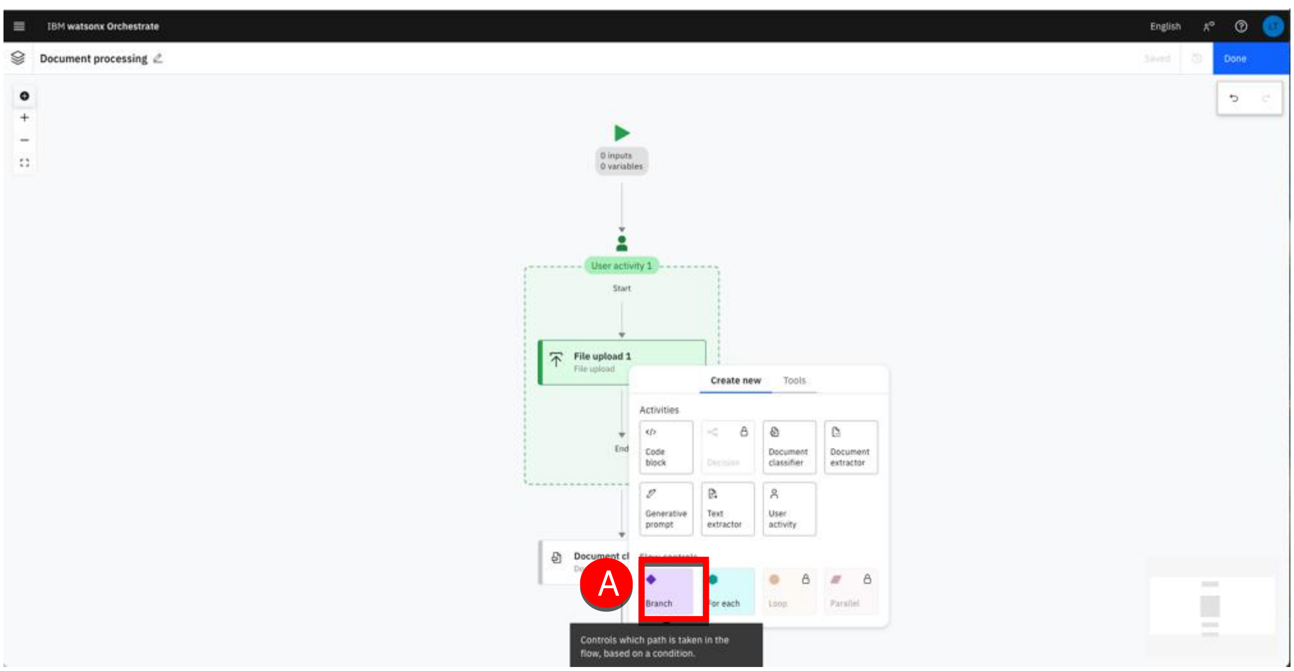
The next step is to route the process on the right extractor path depending on the document type. To do this, you will add a branch node responsible for the triage.

Add a Branch node

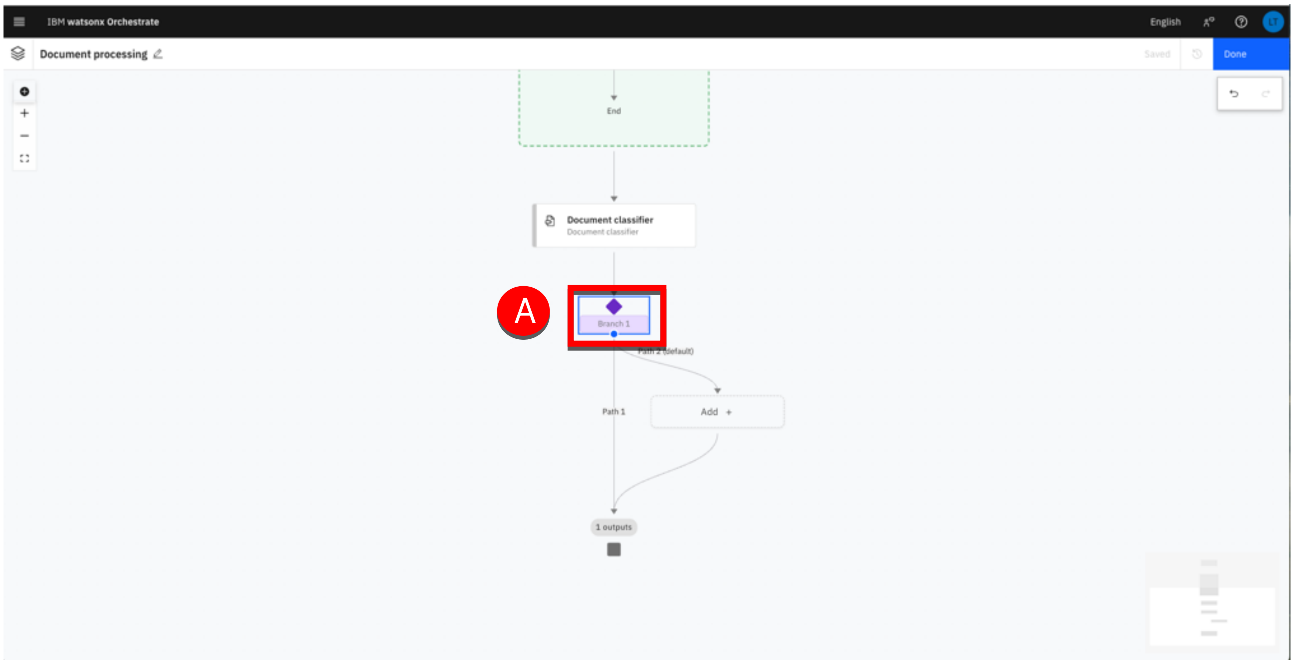
1. Move the cursor over the last link of the diagram then click + (A).



2. Click **Branch** (A).

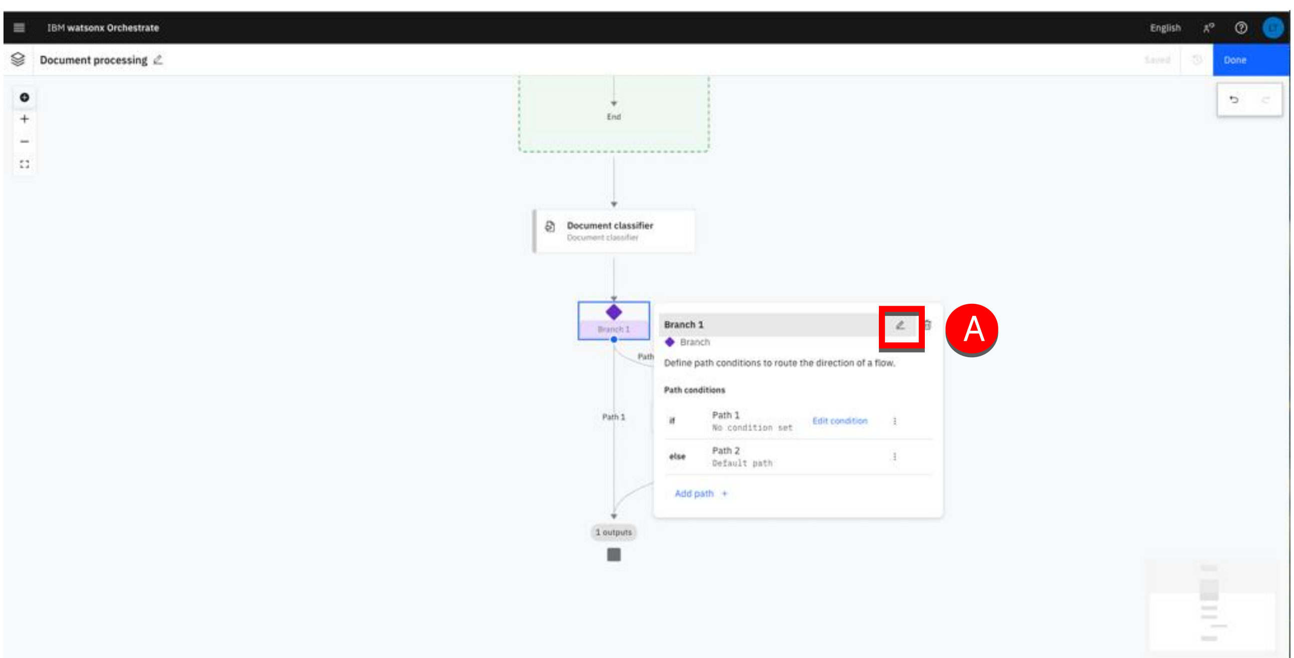


3. Click the **Branch 1** node (A).

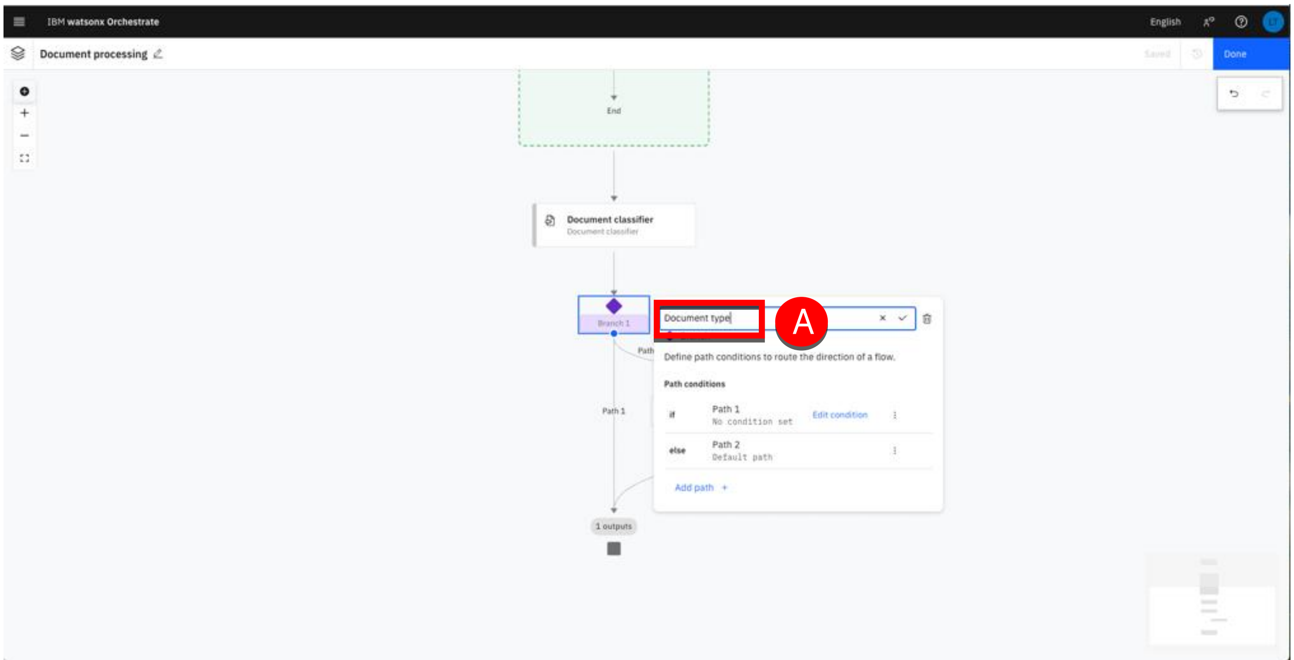


Note: When creating a branch from scratch, two paths are generated by default, but additional paths can be added as needed.

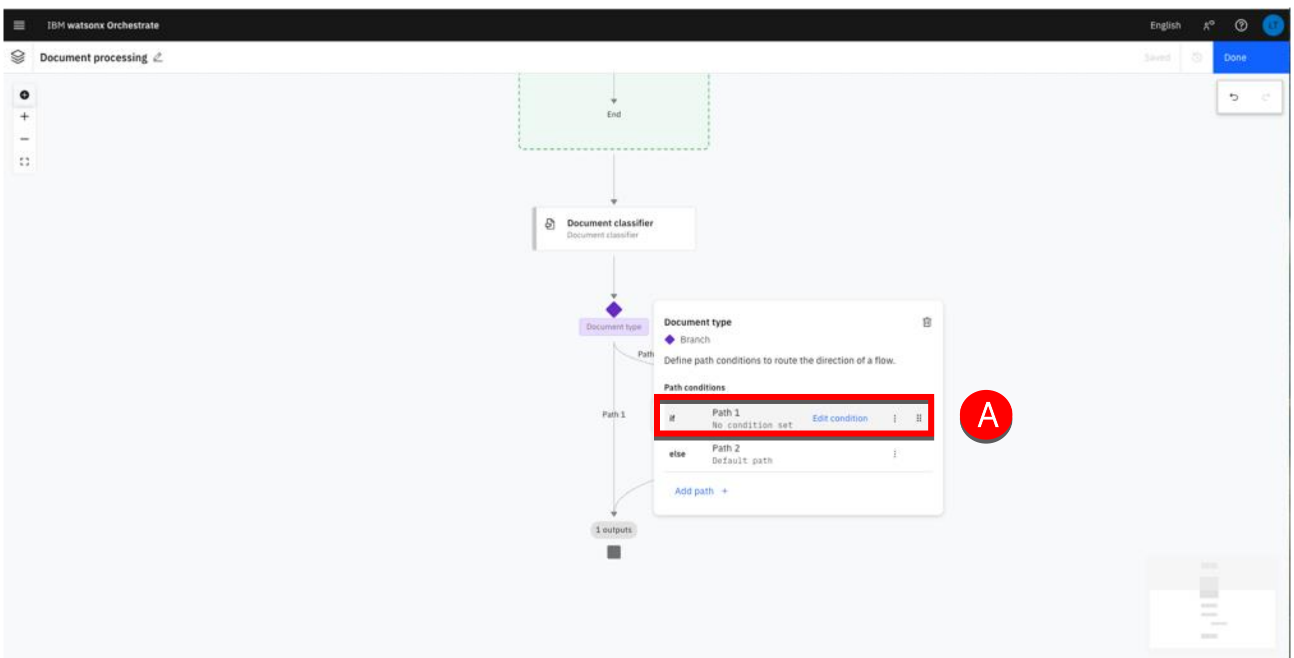
4. Move the cursor over the **Branch 1** name then click the edit icon (A)



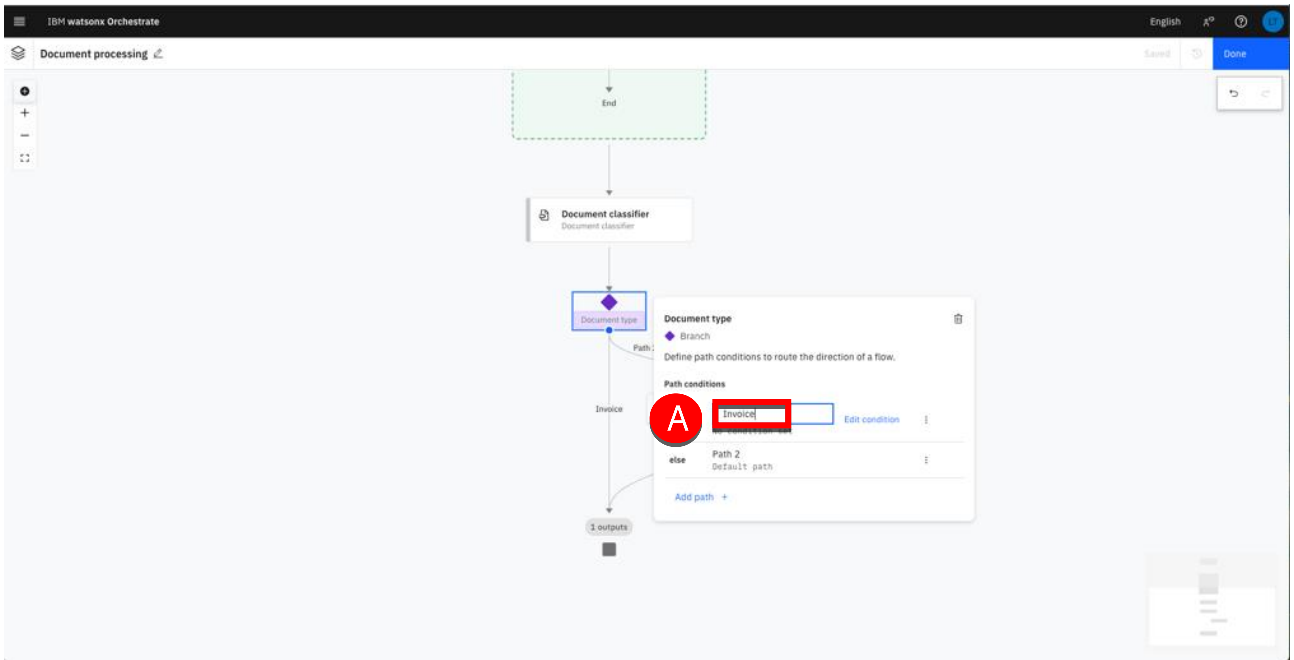
5. Type ‘`Document type`’ in the node name field (A) then hit Enter.



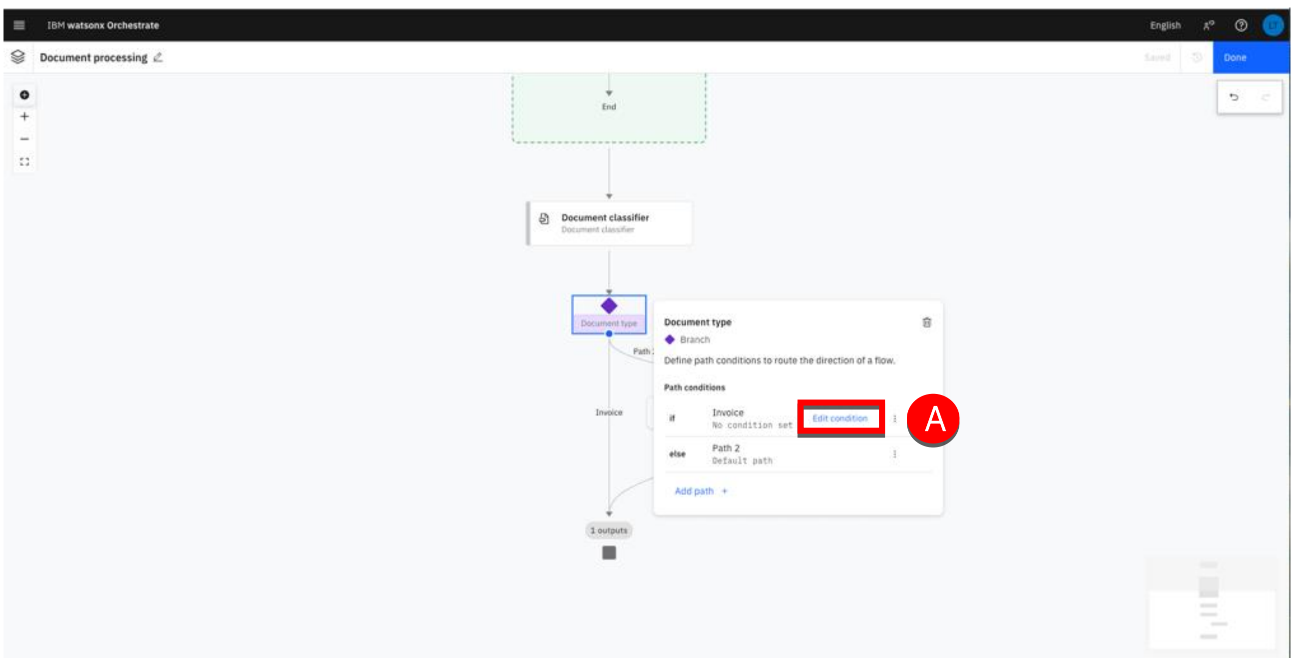
6. Click the **Path 1** row (A)



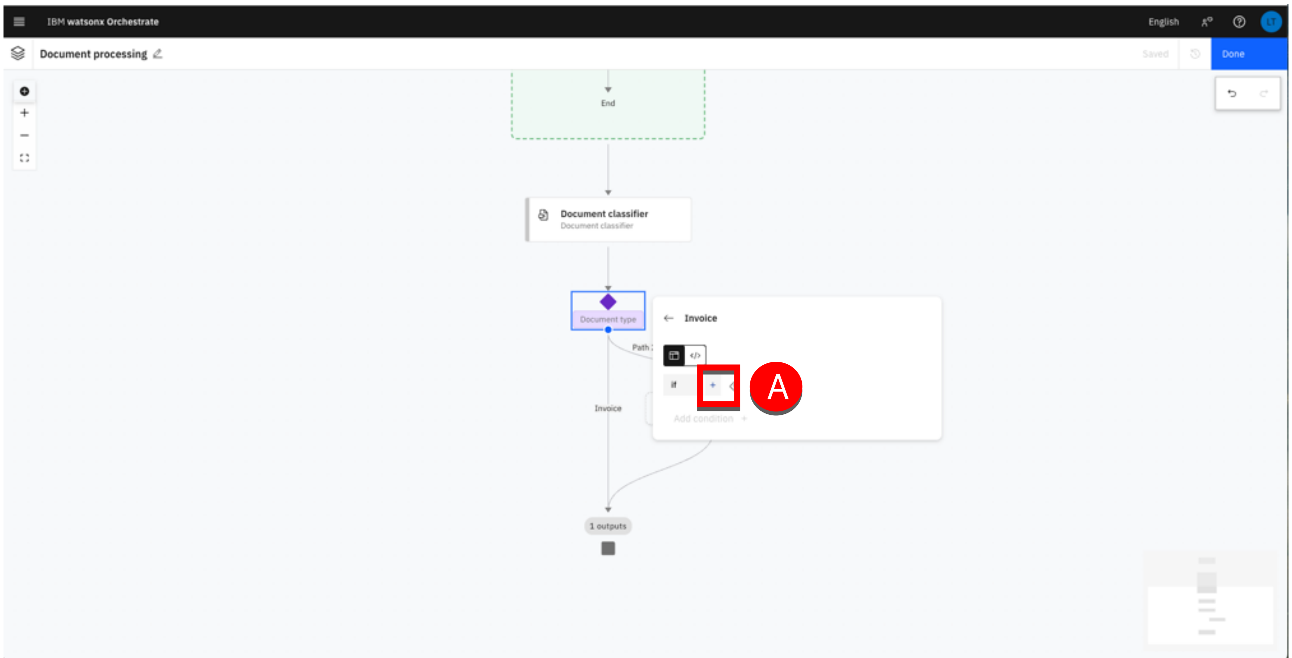
7. Type 'Invoice' (A).



8. Click **Edit condition** (A) to edit the condition and select invoice documents.

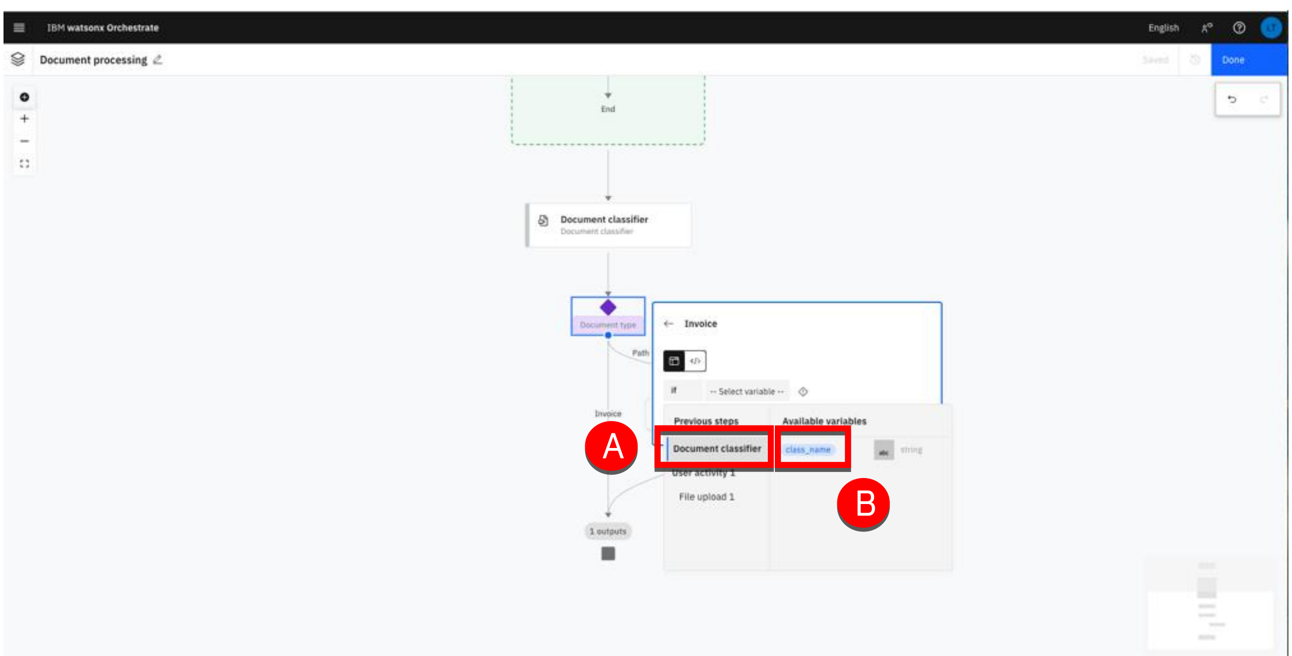


9. Click the + icon (A) to add conditions.

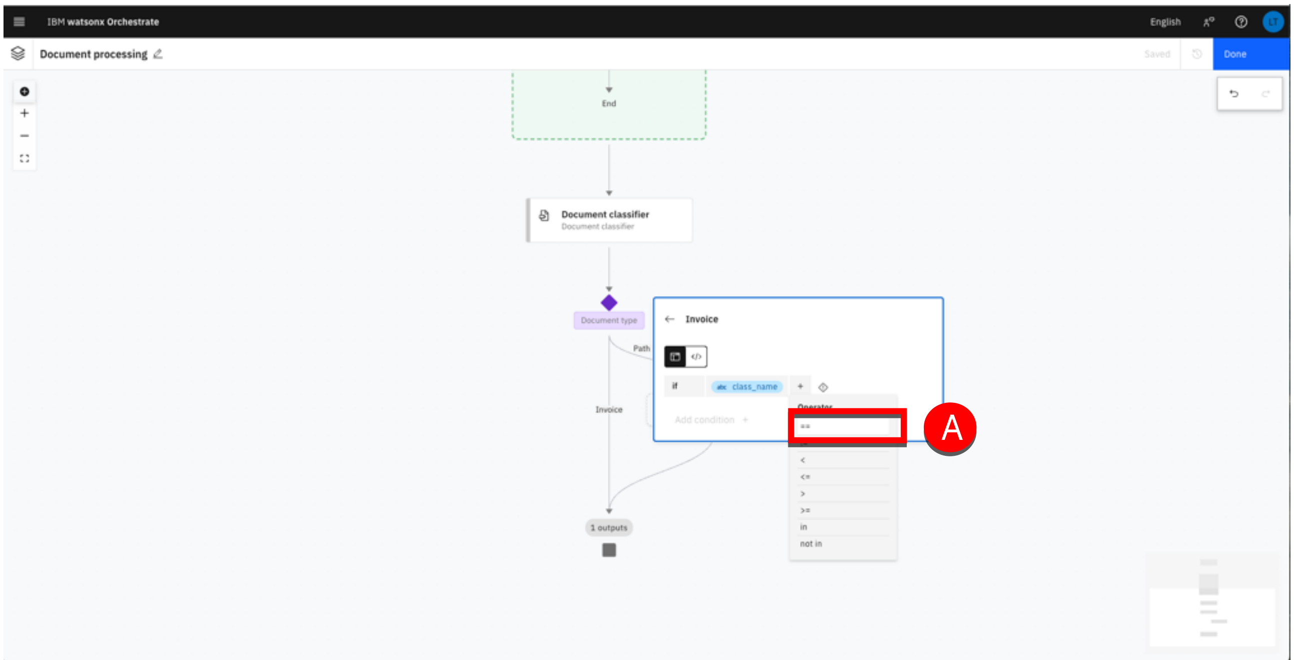


You will use the result of the Document classifier step as the variable to evaluate for the routing.

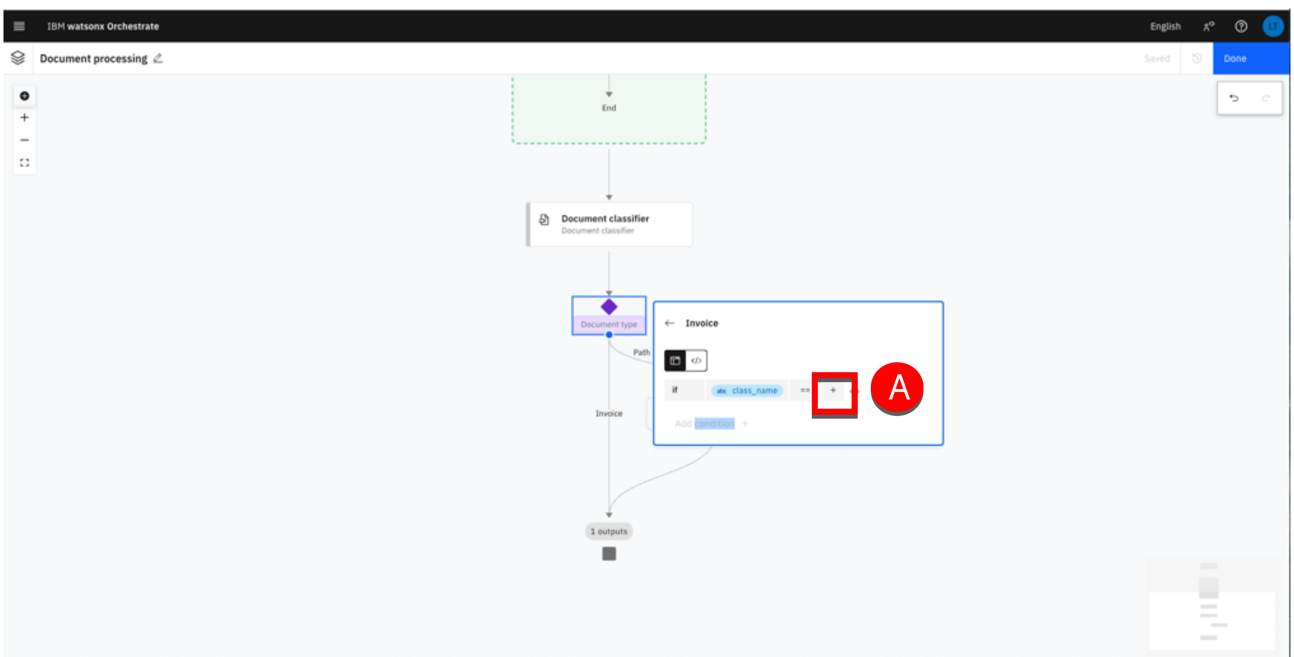
10. Click **Document classifier** (A) and then click **class_name** (B).



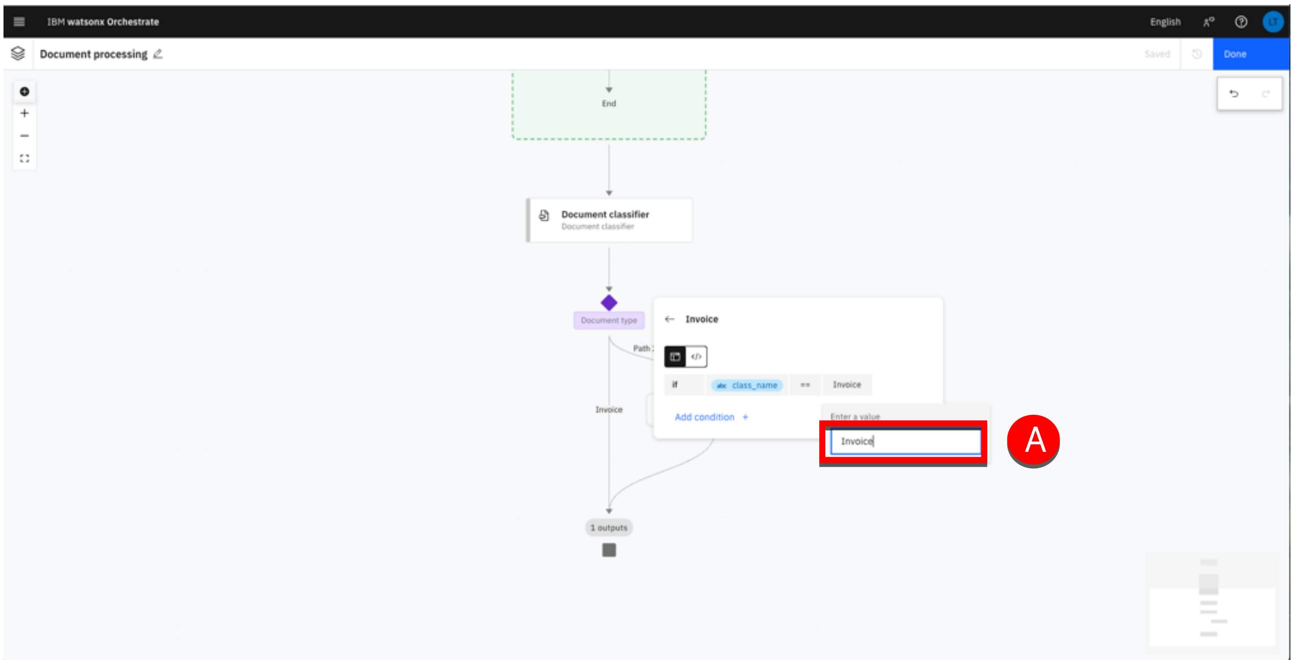
11. Select the '==' operator (A).



12. Click + to specify the value to check (A).



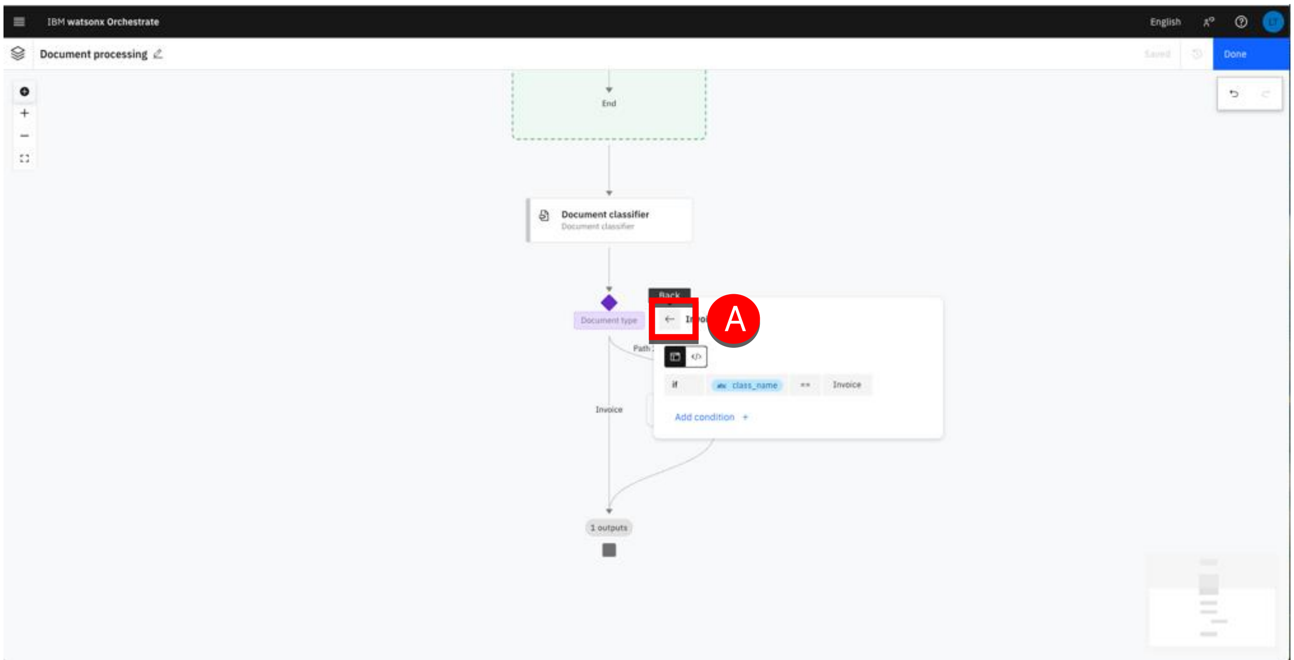
13. Type ‘`Invoice`’ (A) then hit Enter.



The process will be routed in this branch if the document type is Invoice.

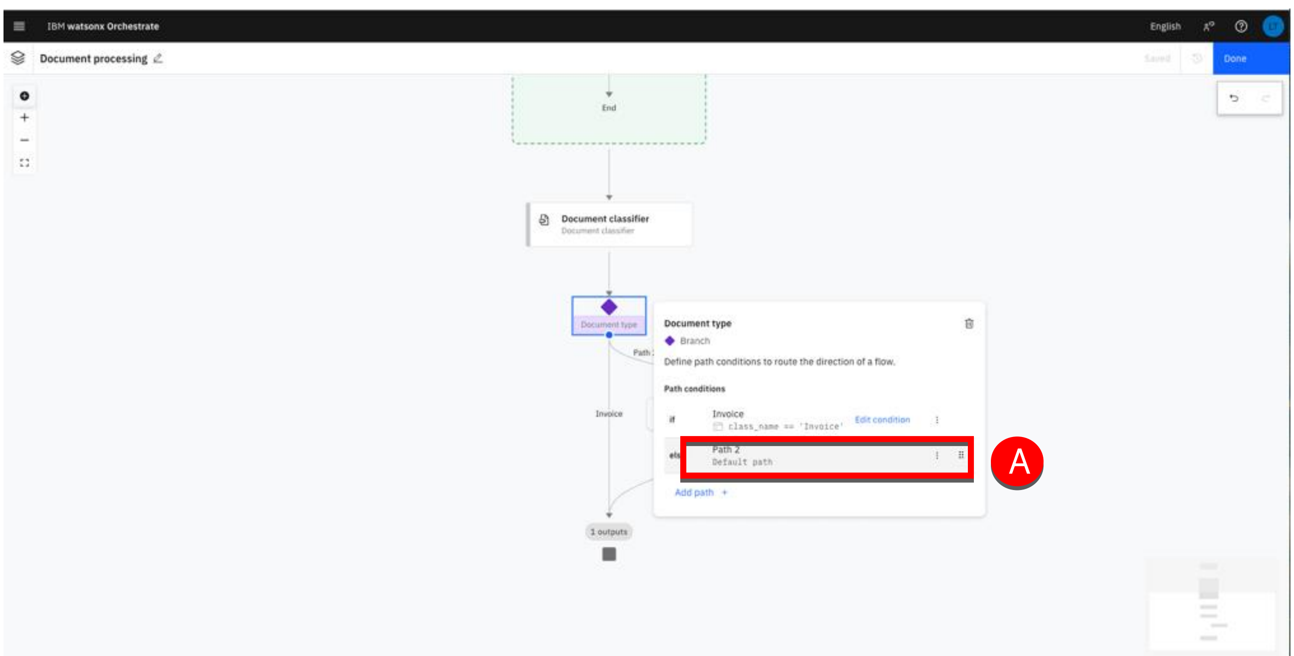
> Note: Conditions can also be edited with the expression editor (</> icon). The equivalent expression to enter using the expression editor will then be: **flow["Document classifier"].output.class_name == "Contract"**

14. Click the **Back** icon (A).

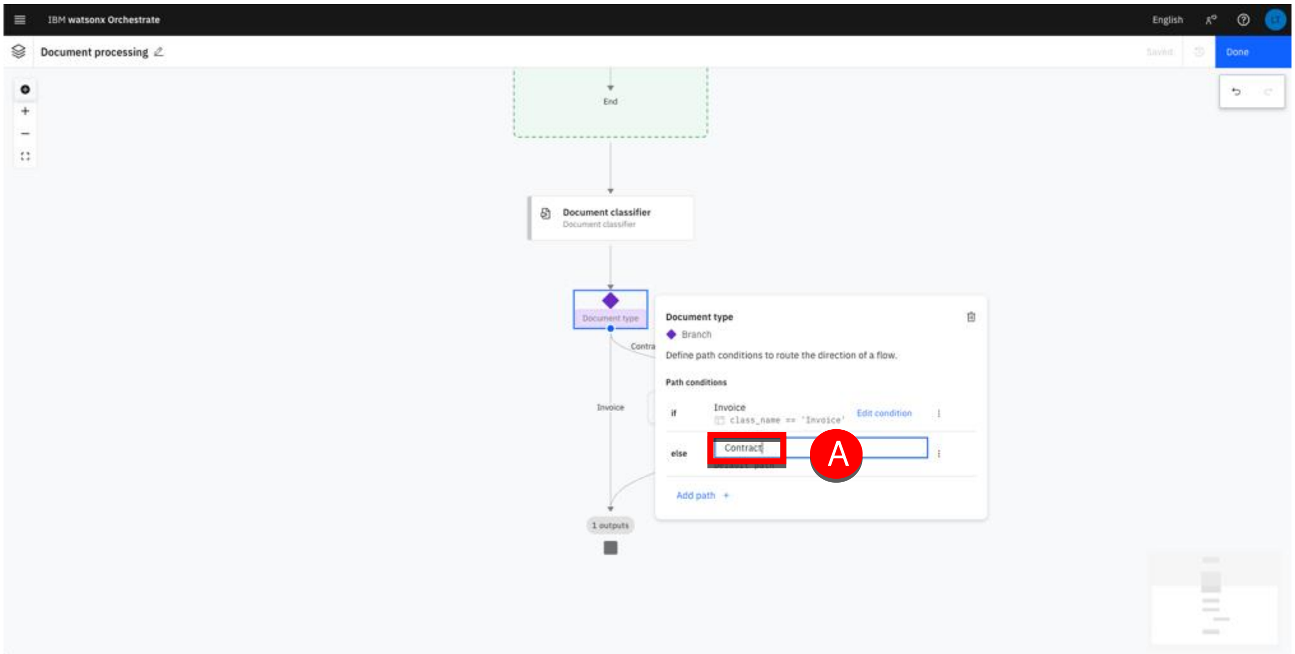


Let's just rename the second path Contract. As you will just have 2 types of documents, any document that is not an invoice (i.e. contracts) will be routed to this branch.

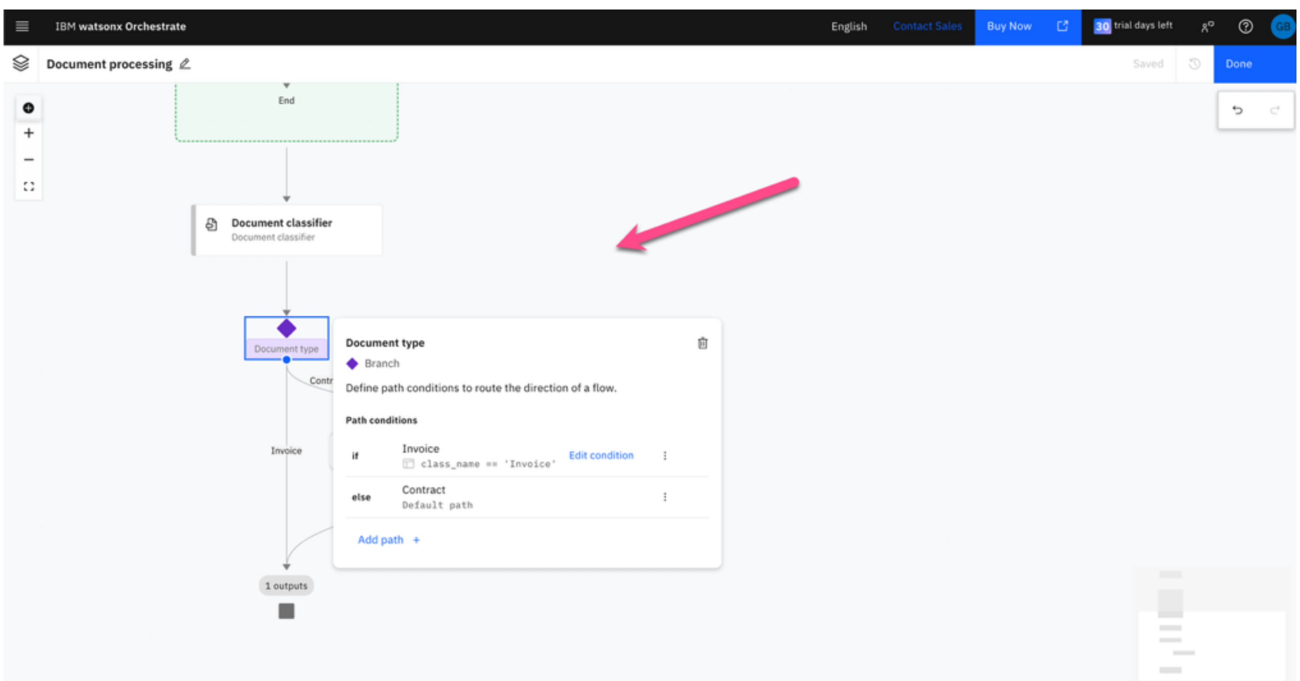
15. Click the **Path 2** row (A).



16. Type ‘`Contract`’ (A) then hit Enter.



17. Click anywhere on the canvas to close the **Document type** branch panel.



You are now ready to add the 2 document extractors, one for each document type.

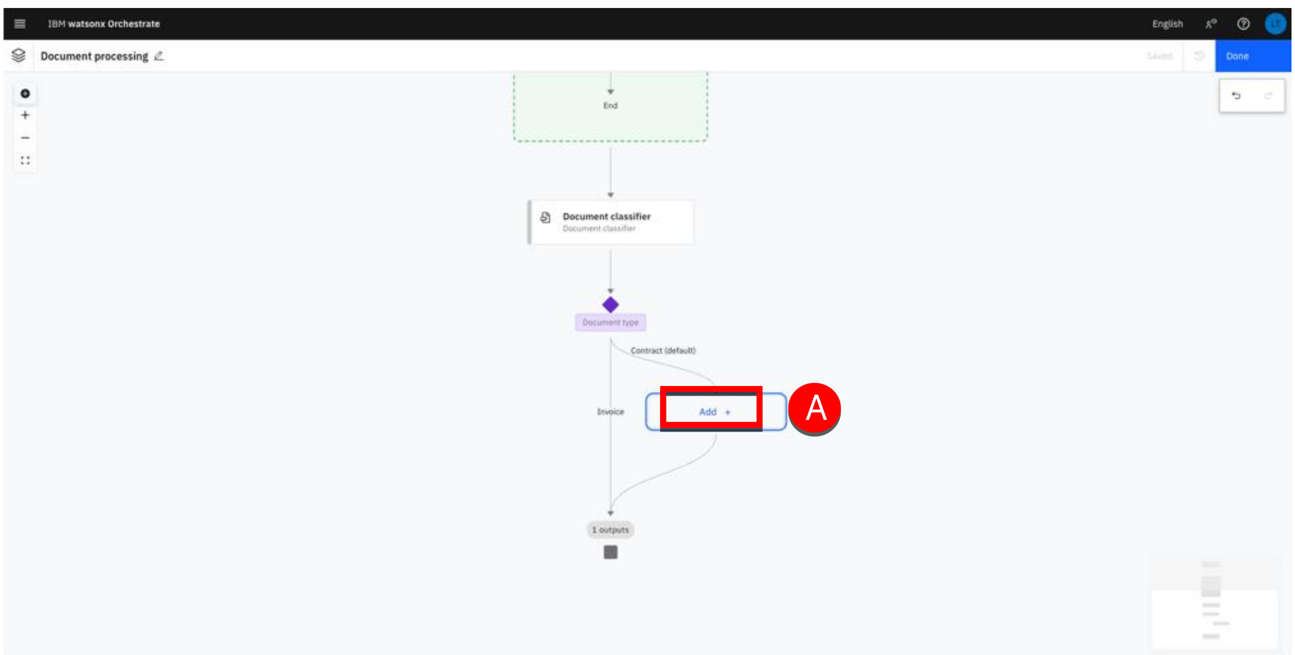
Add a document extractor

In this section, you will create two document extractors. Each document extractor will be responsible for extracting specific data from each document class. You will start with the Contract type of documents for which you will extract the following fields:

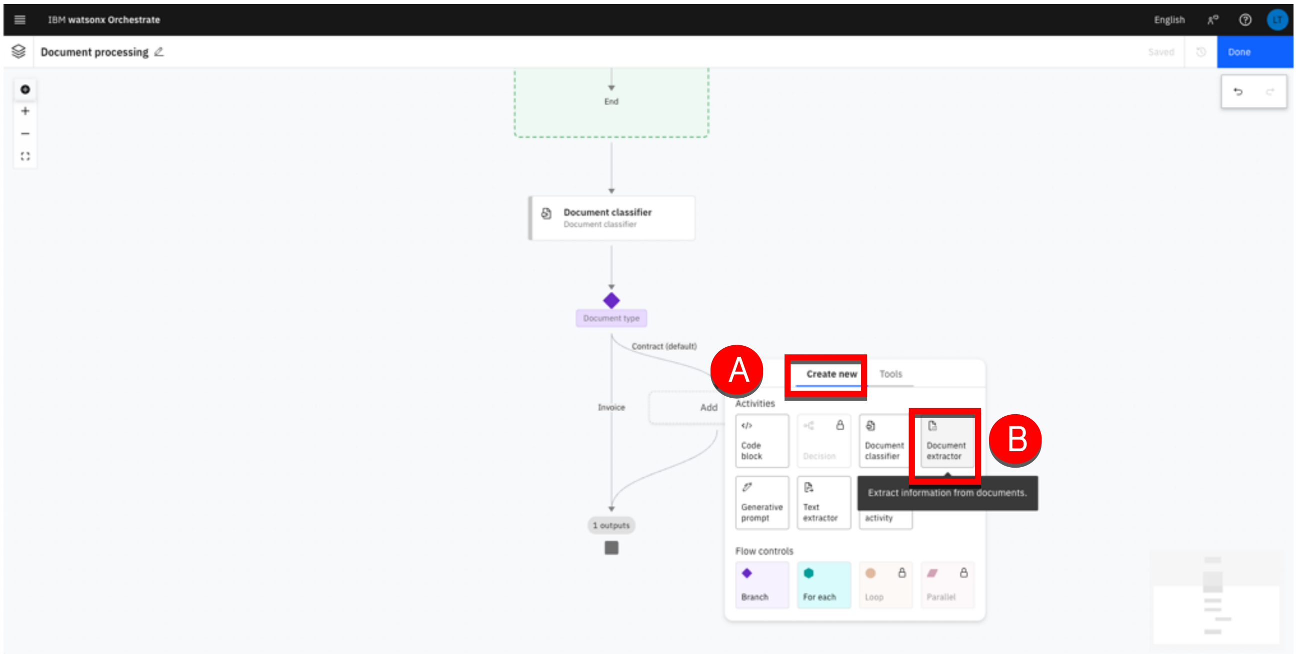
- Buyer
- Supplier
- Effective date

Note: During this section, extractors may return different values from the one shown in this guide. LLM are evolving and may produce slightly different results. In many situations, you will have to refresh the interface manually closing/re-opening or clicking on the interface to see the extracted values updated after a few seconds (minutes depending on your LLM shared usage).

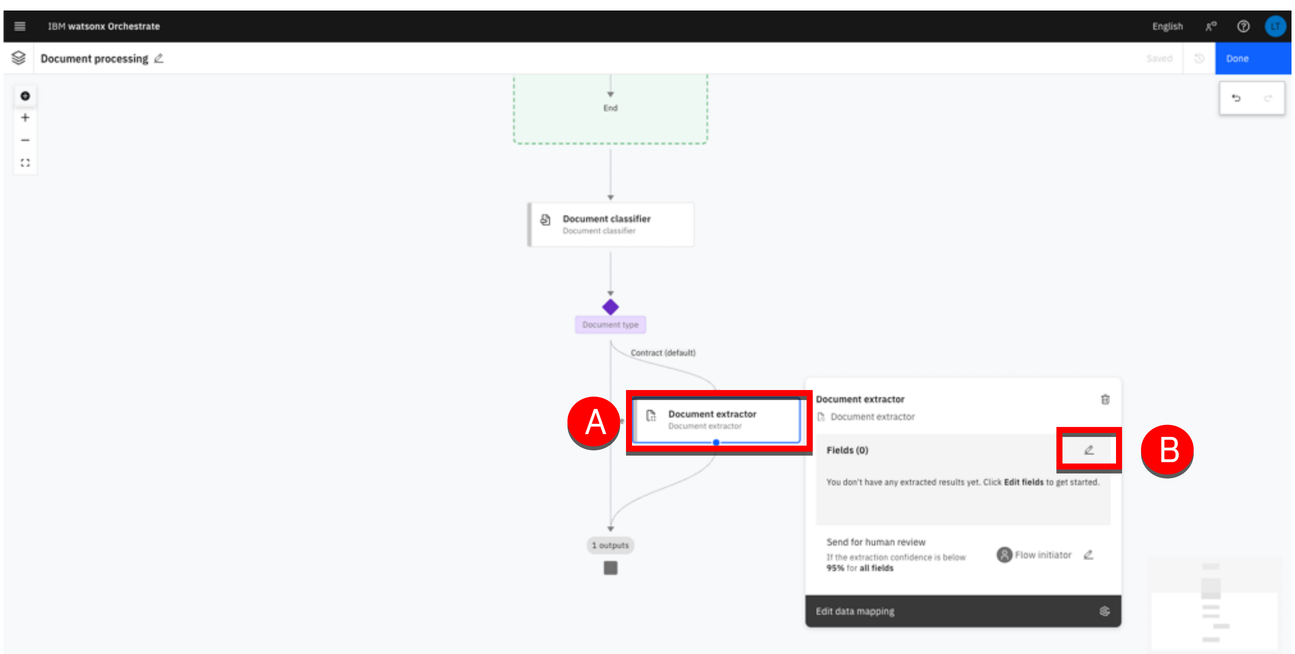
1. Click **Add +** (A).



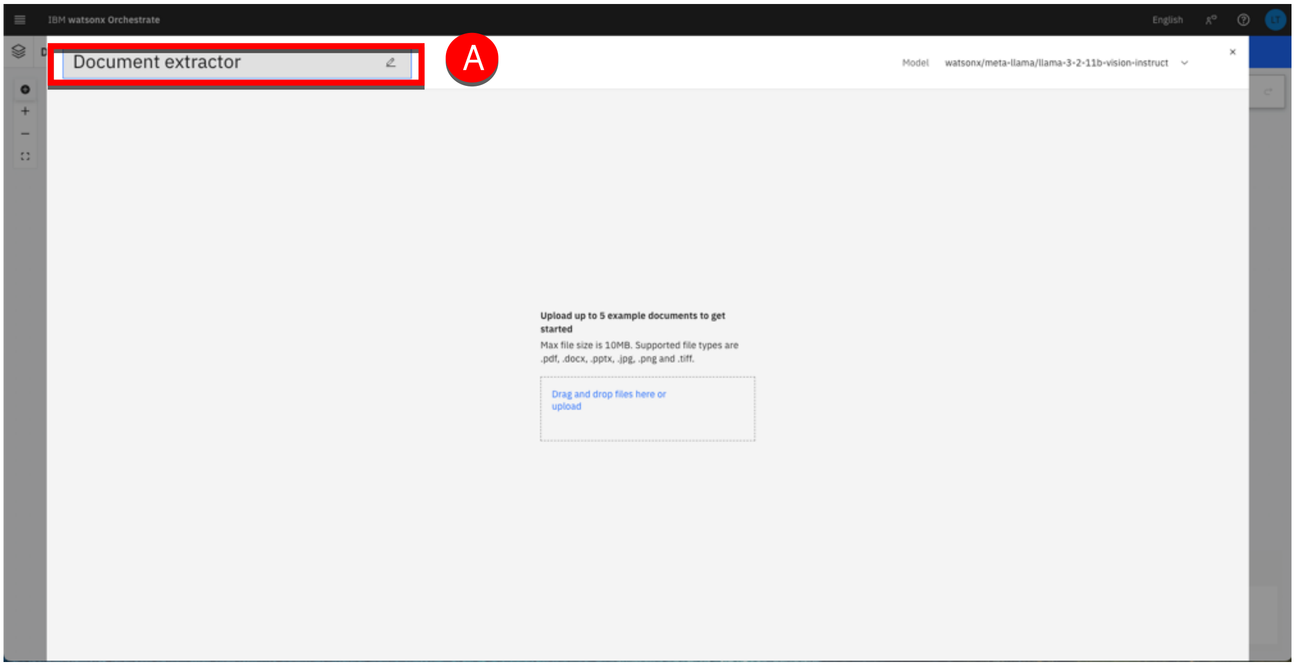
2. Select **Create new** (A) then click **Document extractor** (B).



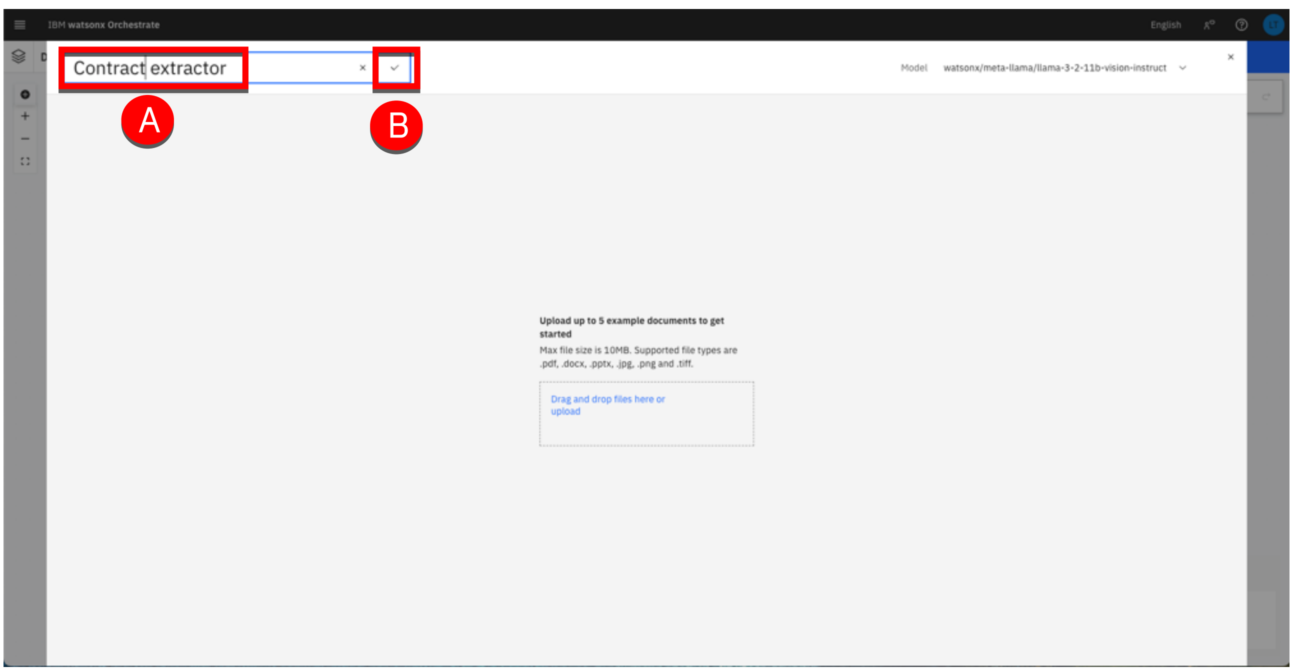
3. Click the **Document extractor** node (A) then click the edit icon (B).



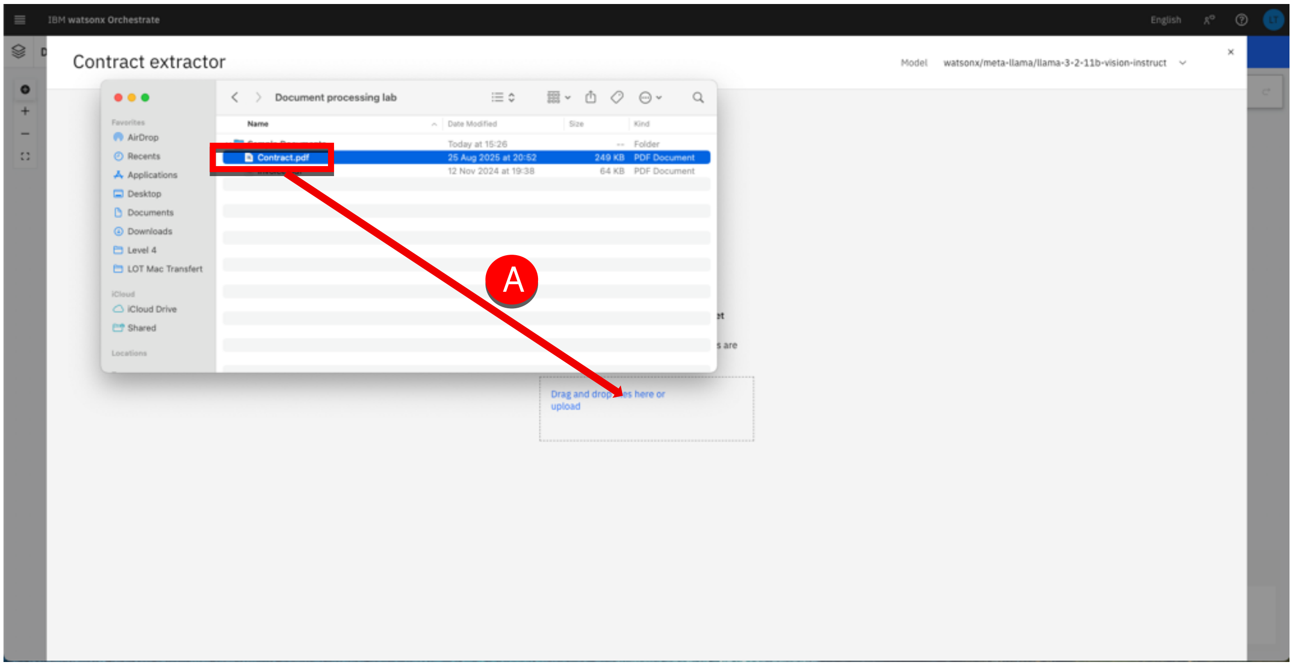
4. Click the **Document extractor** name (A) to edit it.



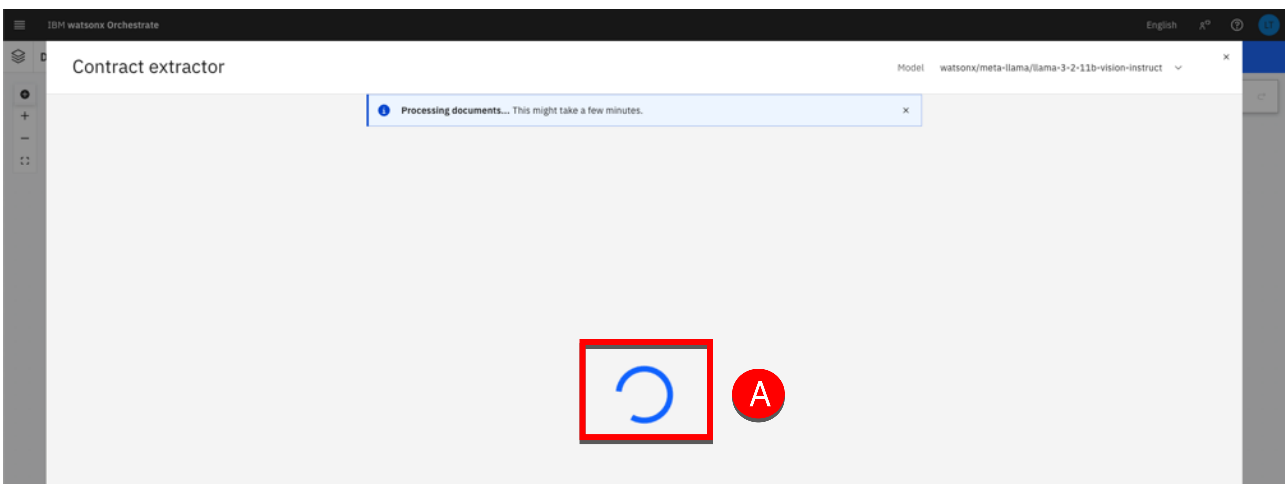
5. Type '`Contract extractor`' (A) as a new name then click the save icon (B).



6. Drag and drop the **Contract.pdf** file (A) from your computer to the dropping area.



7. Wait for the document to be processed.



8. Once uploaded, the sample document is displayed, and the page is divided into the following key segments:

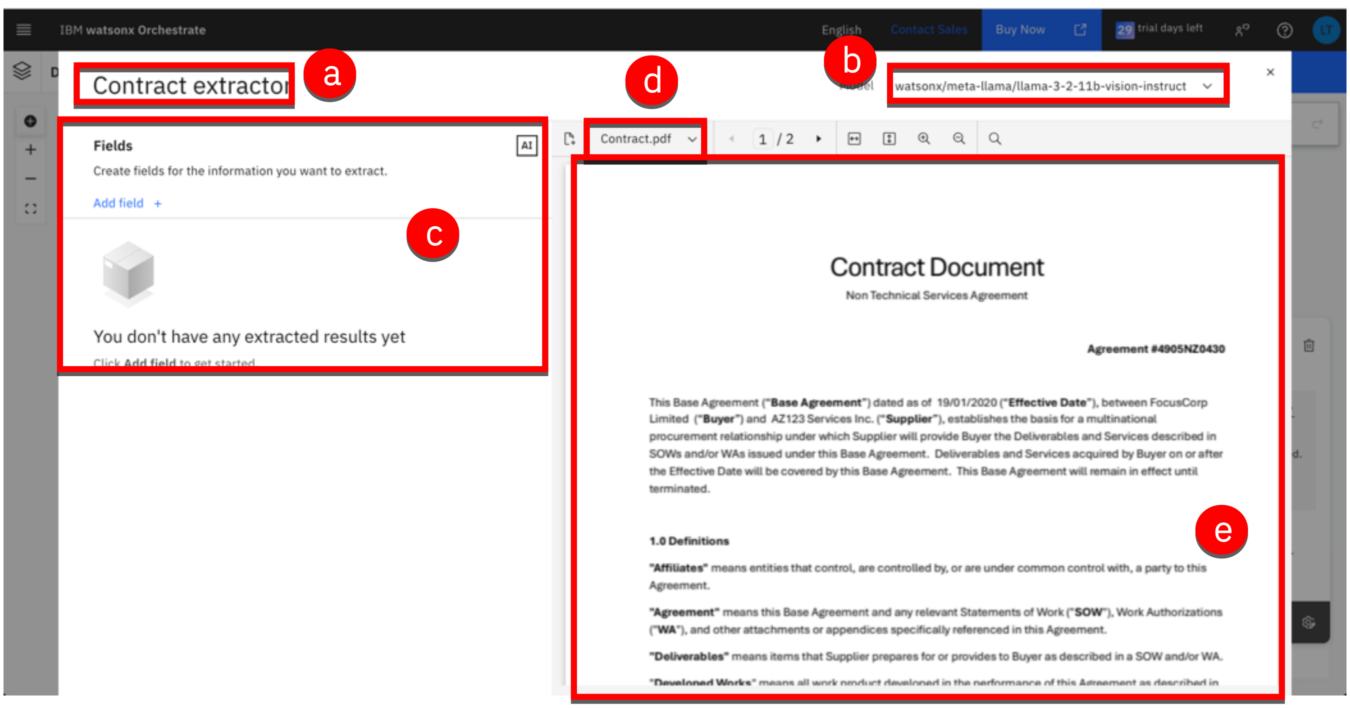
   a. Activity Name – Displays the name of the document extractor activity. Rename it to reflect the sample document you've uploaded.

   b. LLM Model Selection – Choose your preferred LLM model from the drop-down menu. You can switch models at any time while prompting them for document extraction.

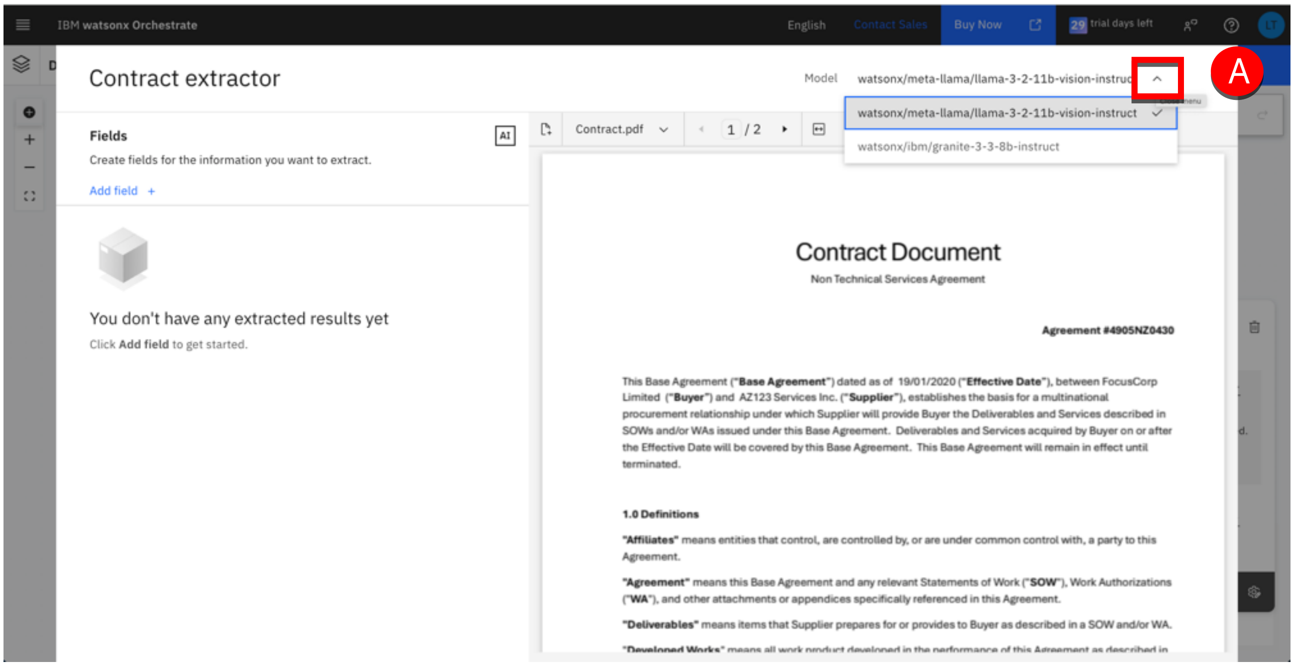
   c. Field Definitions – Define the list of fields you want to extract from the sample document. The selected LLM will retrieve the corresponding values.

   d. Document Navigation & Upload – Use the drop-down to navigate between uploaded documents. You can upload up to five additional documents for prompting and testing.

   e. Document Viewer: Shows the uploaded document with the specified fields for extraction highlighted.

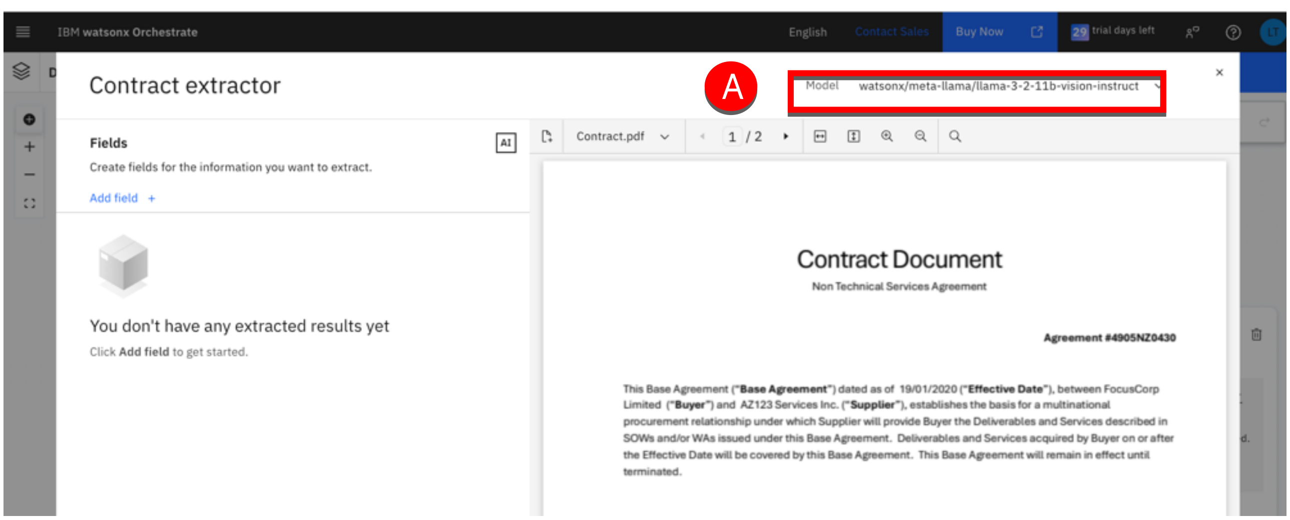


9. Click the model expand icon (A)



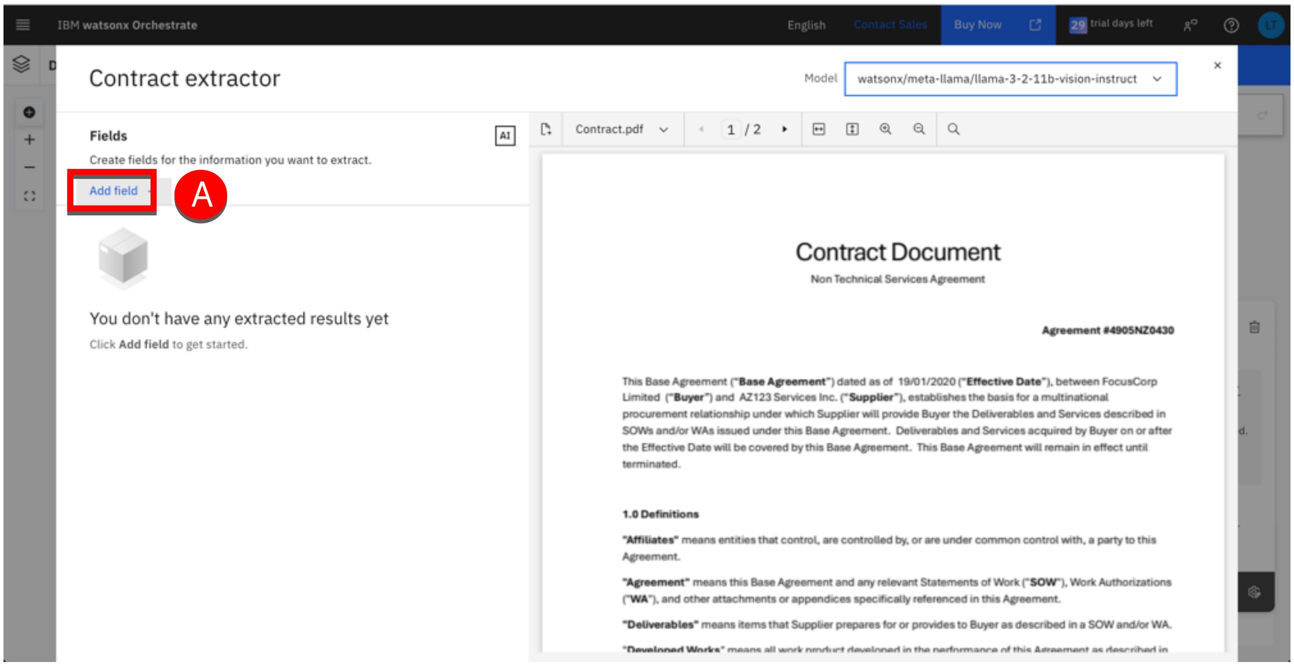
Note: You can change the LLM model by clicking on the ‘Model’ drop-down. Here you’ll see all the available models that you can select. You can identify the current model that is being used, as you’ll see a tick icon.
You can change the model at any time while you test to see which one is most accurate at extracting fields

10. Keep the meta-lama model (A).

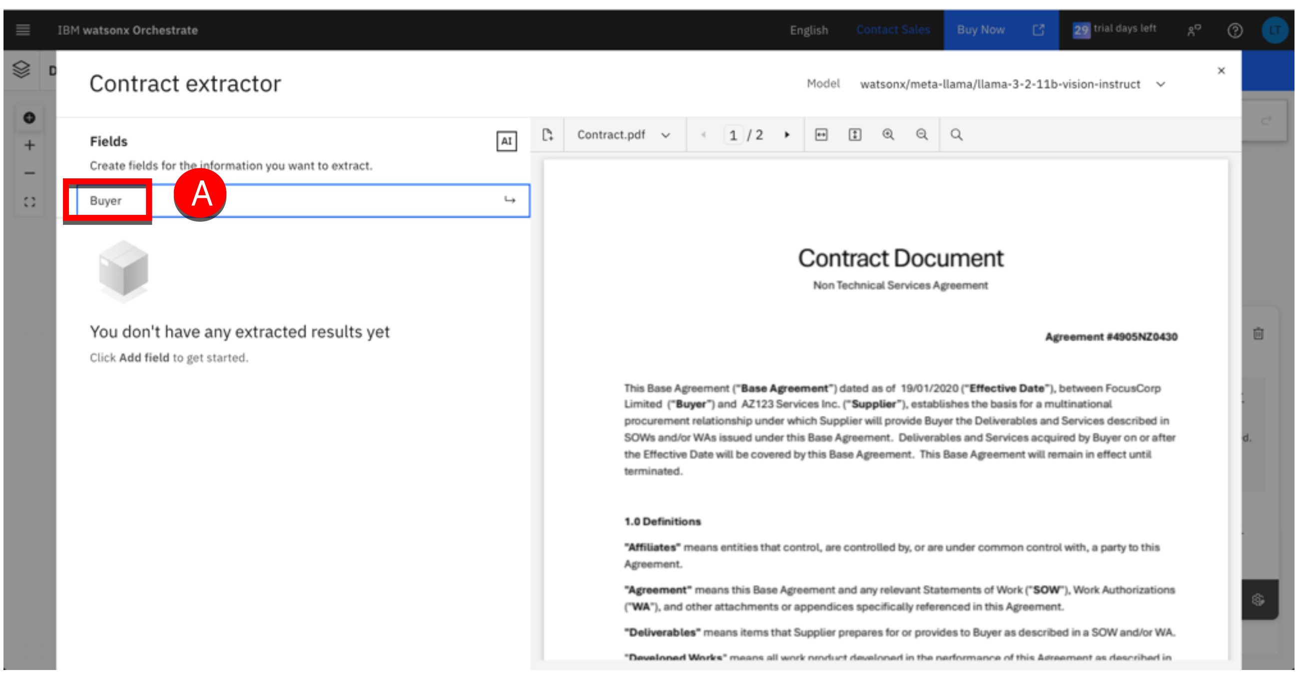


Let’s now define the fields to extract.

11. Click **Add field** (A).



12. Type `Buyer` (A) and hit Enter.

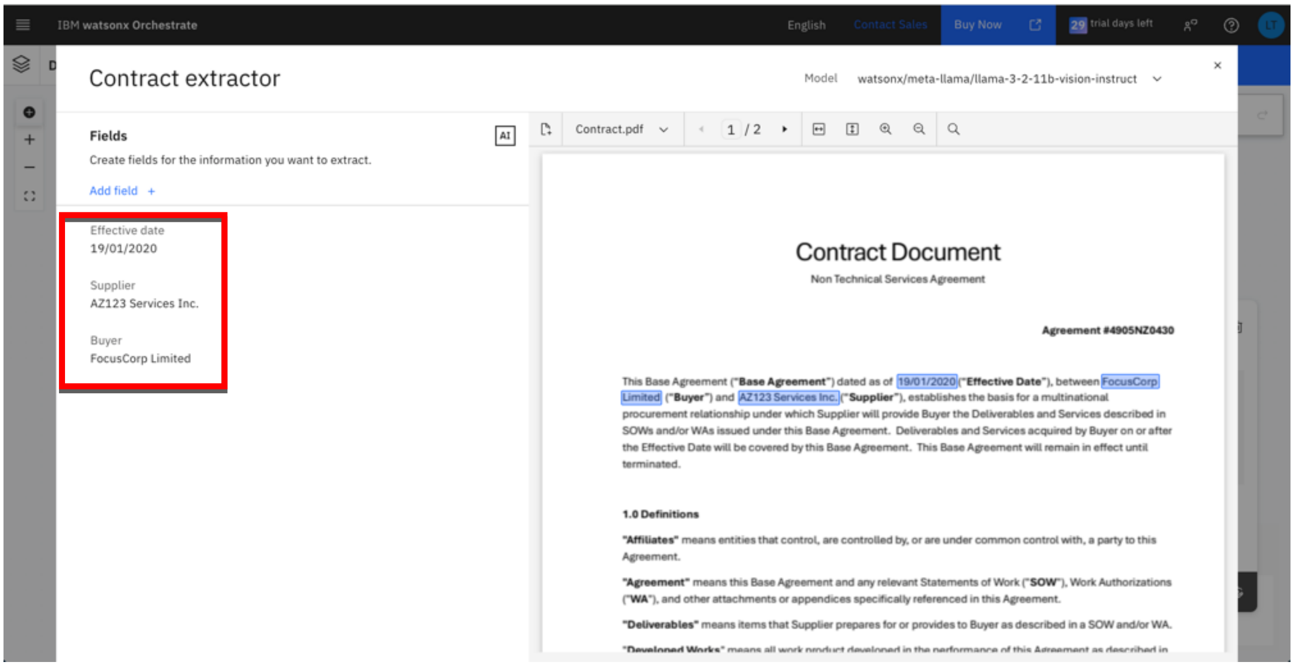


Note: The LLM will retrieve a value associated with the key you define here, which will auto-highlight in the document preview.

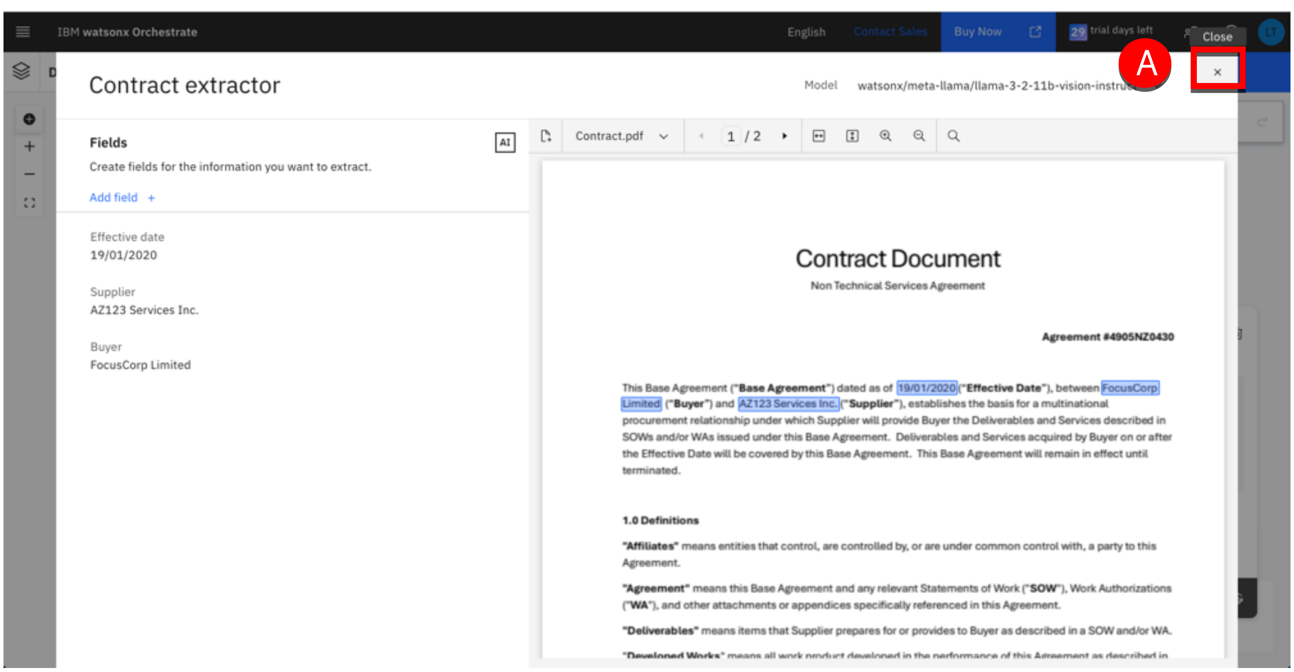
13. In the same way, add the 2 following fields:

   - Supplier
   - Effective date

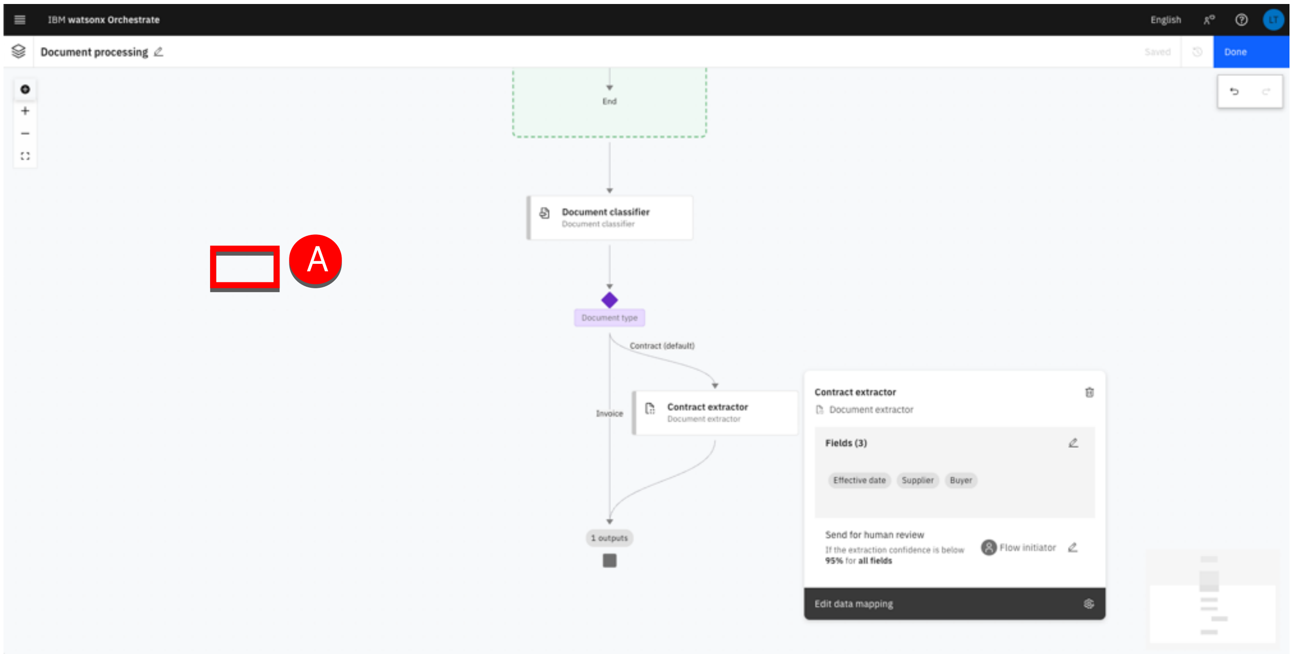
   Your screen should look like:



14. Click x (A) to close the extractor.



15. Click in the background to close the **Contract extractor** property view (A).

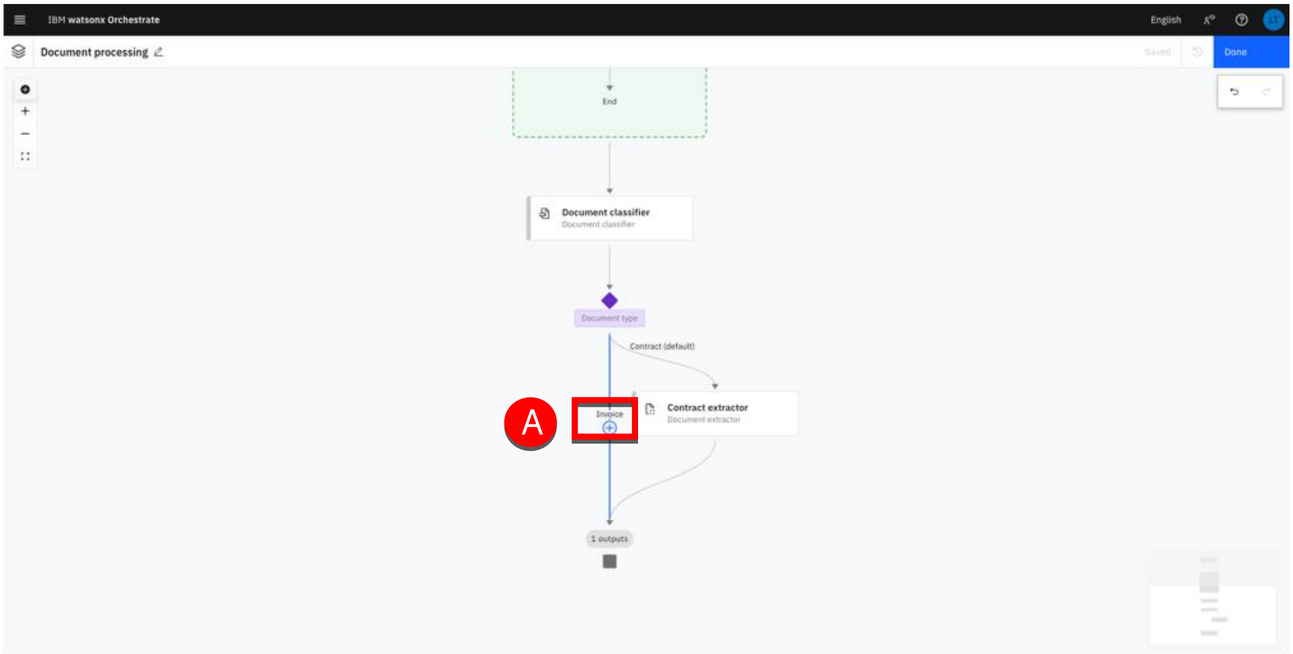


Next, you will create the Invoice extractor in the corresponding branch of the process. This extractor will be more complex and harder to implement. You will learn how to train the model to extract values from different kind of documents (PDF files or images).

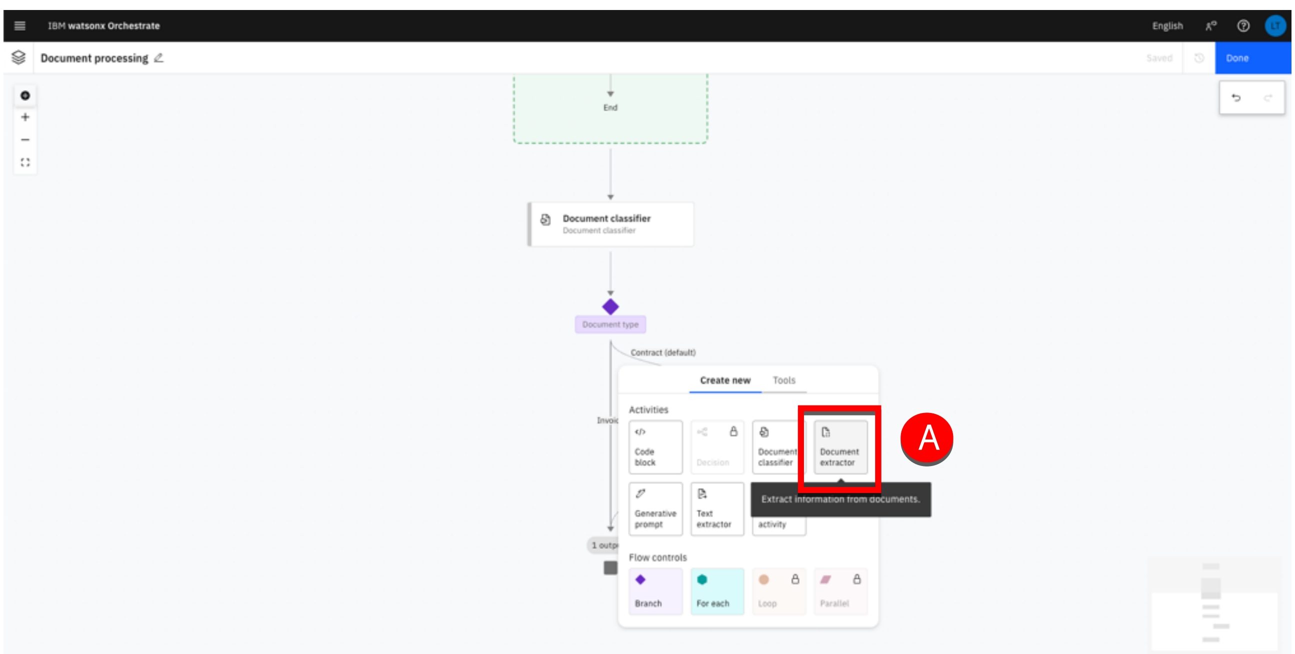
For the invoice documents, you will extract the following fields:

- Customer name
- Customer number
- Customer address
- Amount due
- Due date
- Monthly usage
- Year to year usage

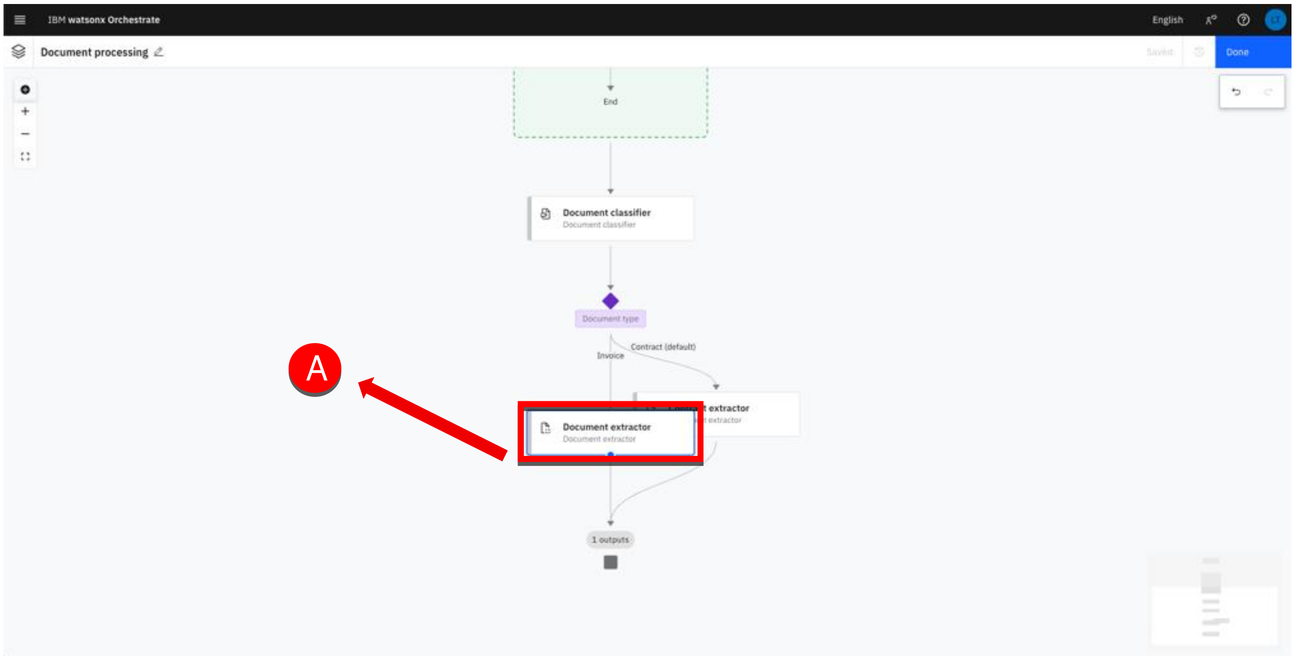
16. Move the mouse over the **Invoice** branch and click **+** (A)



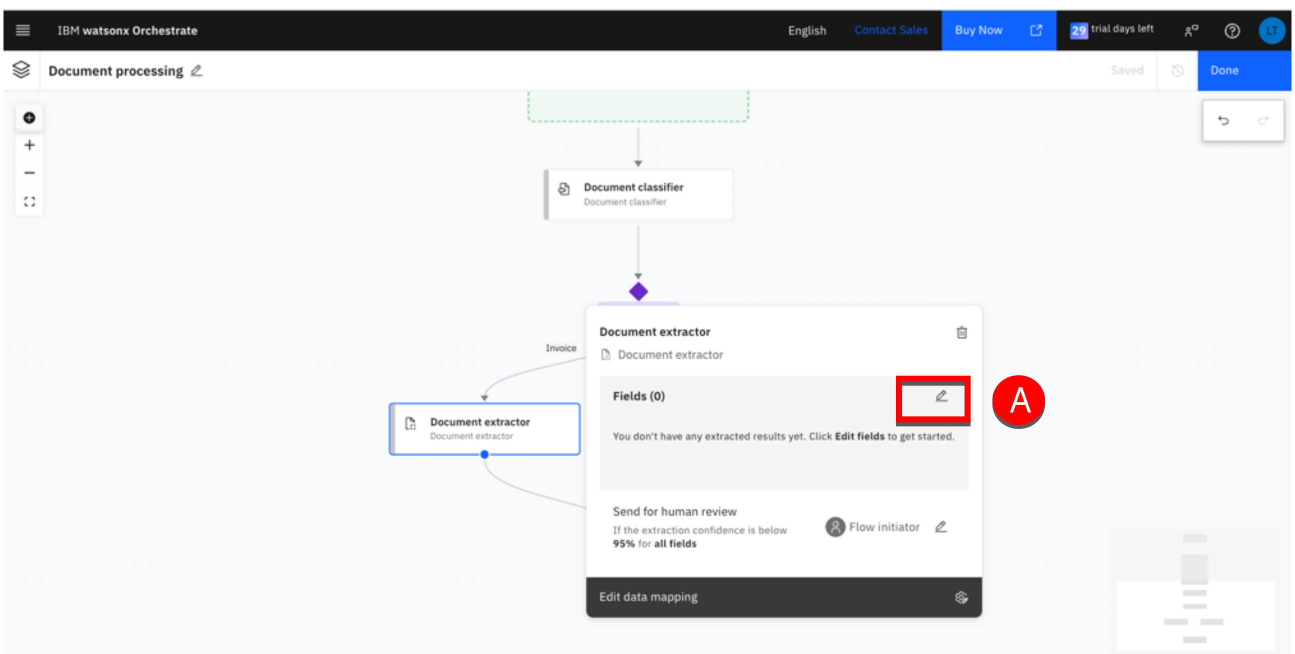
17. Select **Document extractor** (A).



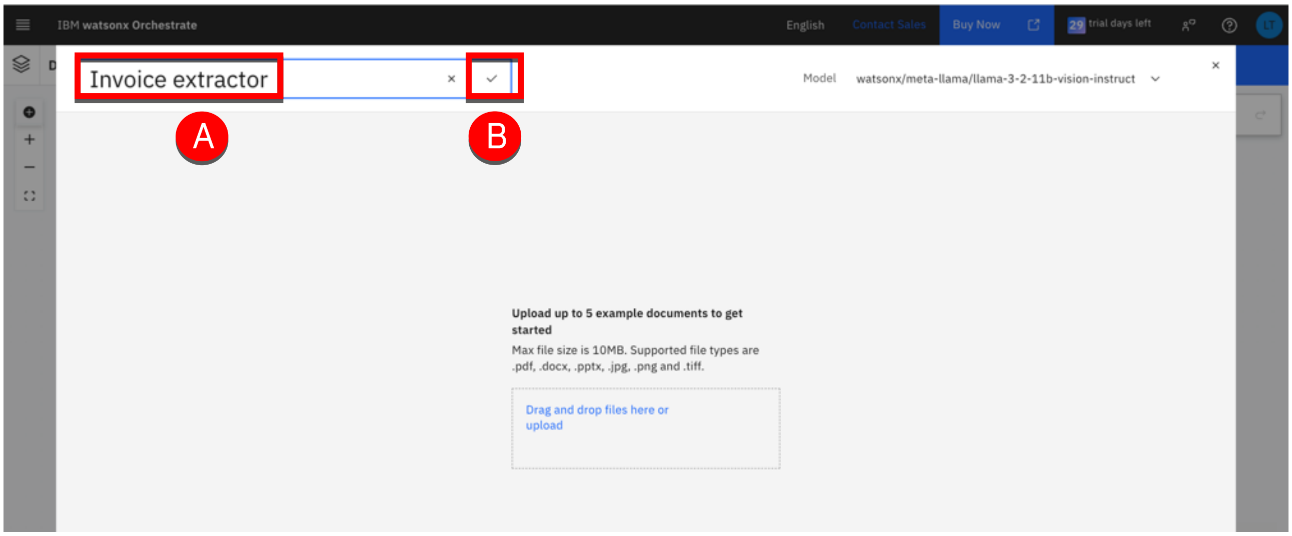
18. Click the **Document extractor** node a move it for a clearer layout (A).



19. Click the edit icon (A)

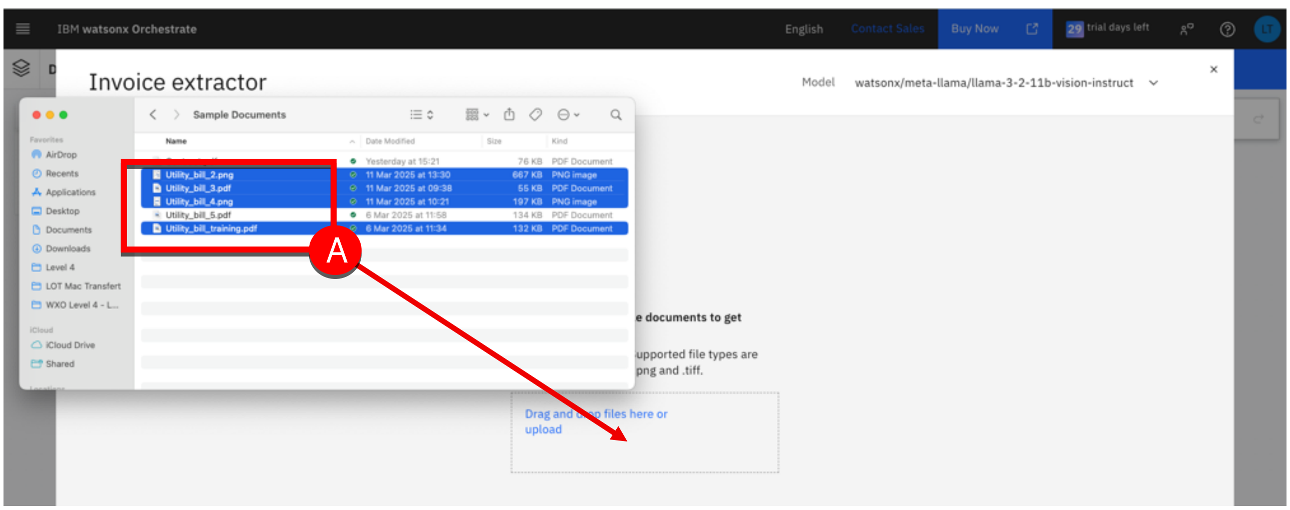


20. Type ‘`Invoice extractor`’ in the name field (A). Click the save icon (B).

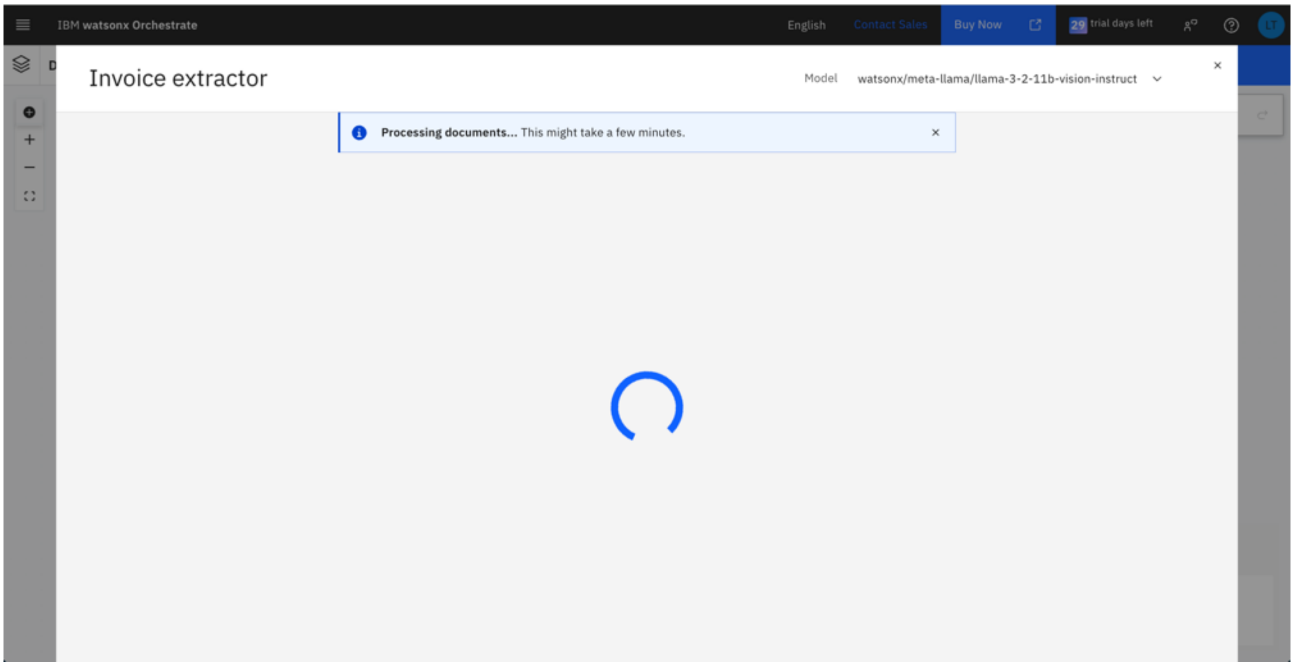


21. Drag and drop the four following files in the upload area (A)

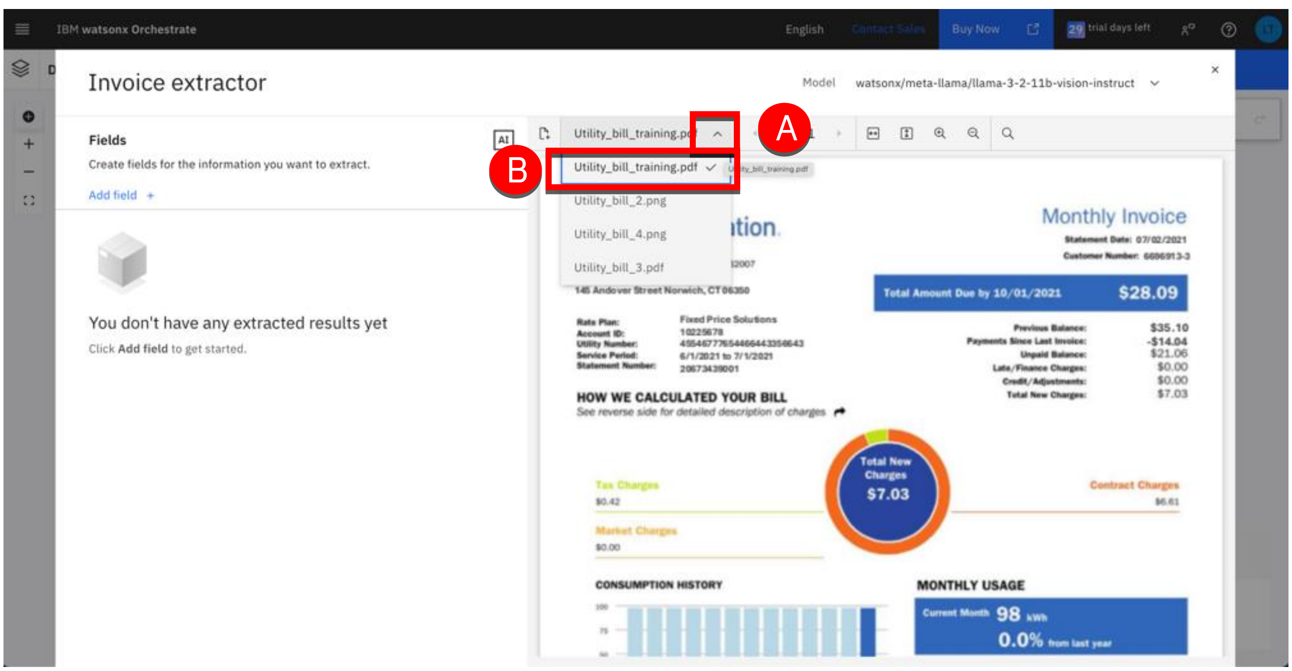
   - Utility_bill_training.pdf
   - Utility_bill_2.png
   - Utility_bill_3.pdf
   - Utility_bill_4.png



22. Wait for the 4 files to be processed.



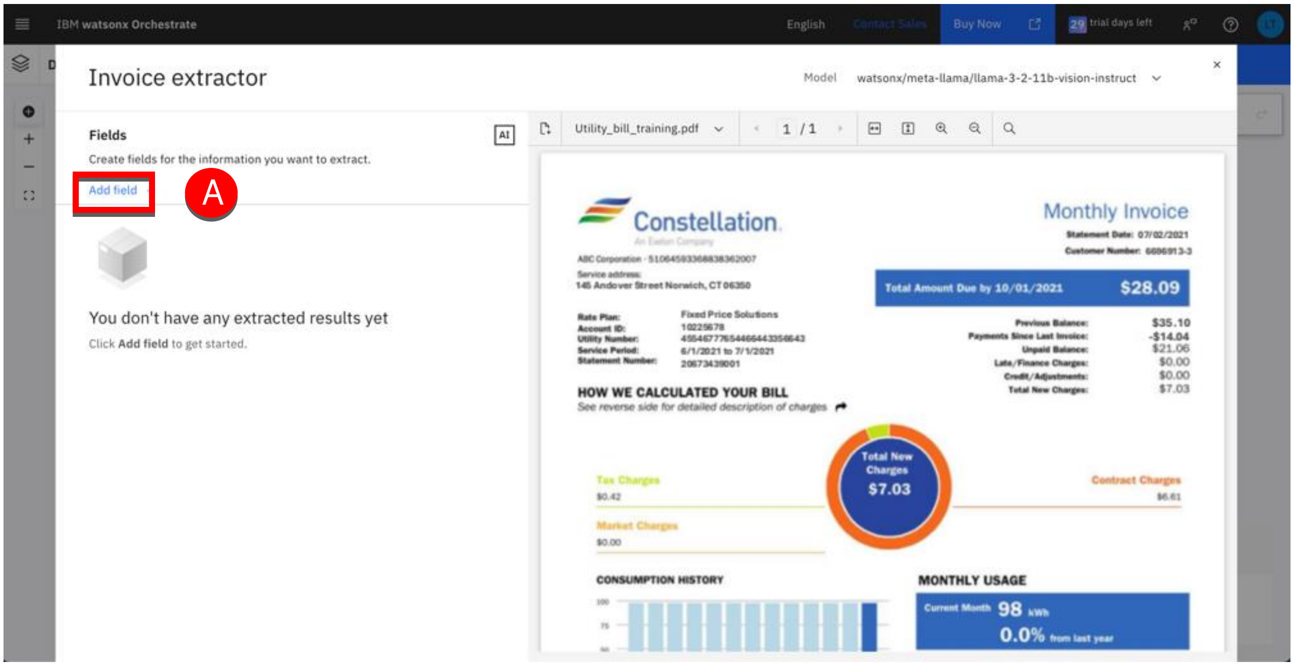
23. Expand the file selector (A) and select the **Utility_bill_training.pdf** file (B).



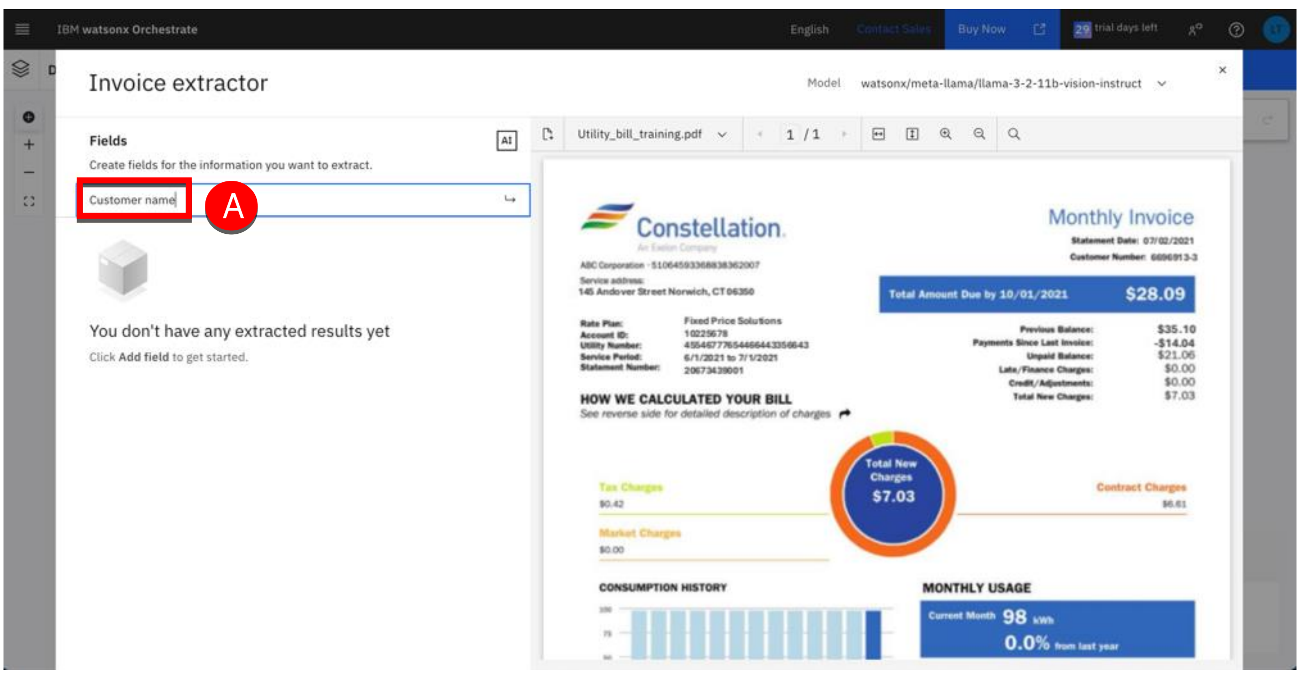
Define the fields to extract

You are now ready to specify the fields to be extracted.

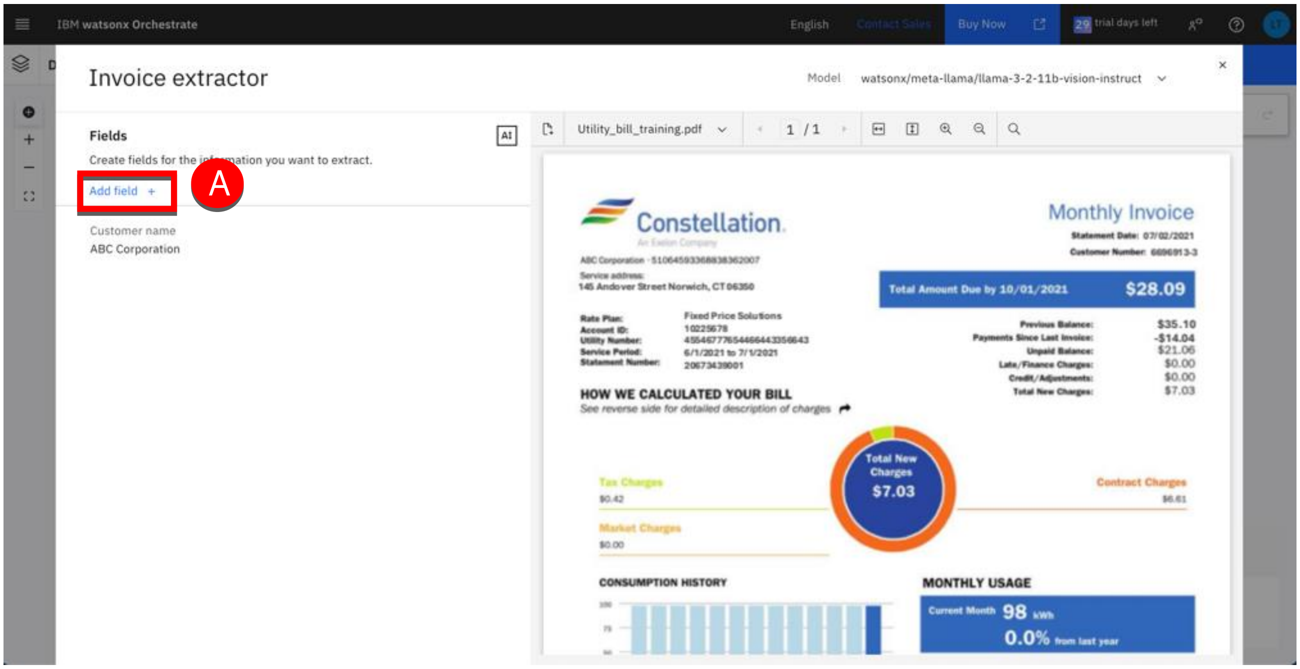
1. Click **Add field** (A).



2. Type ‘`Customer name`’ into the field name (A) then press Enter.

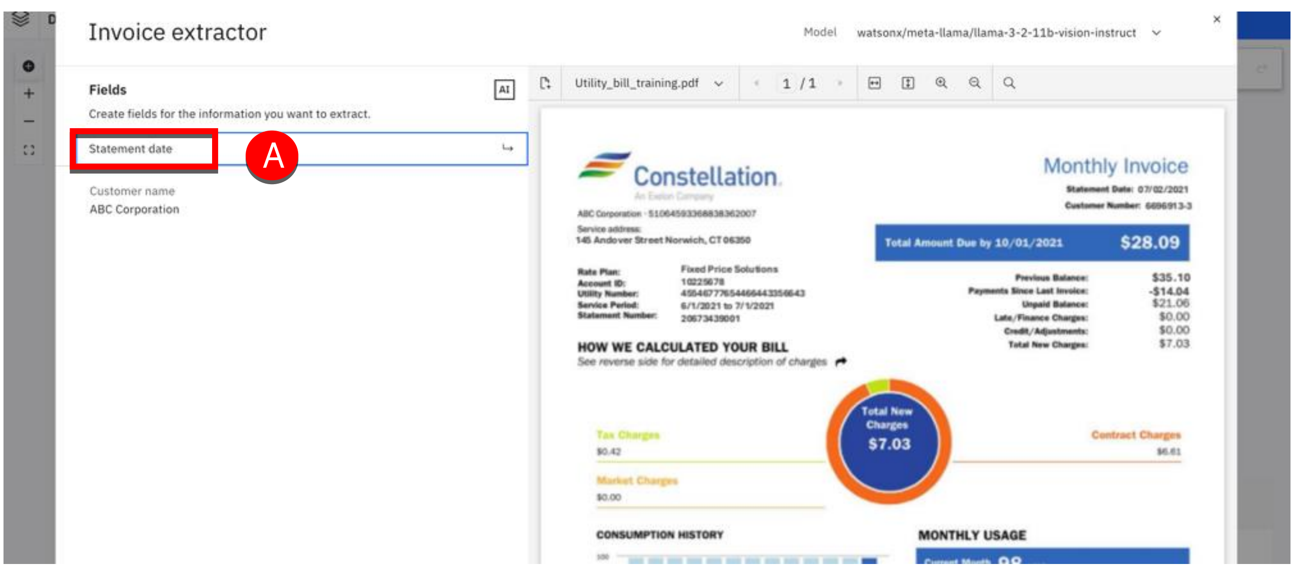


3. Click **Add field** (A).



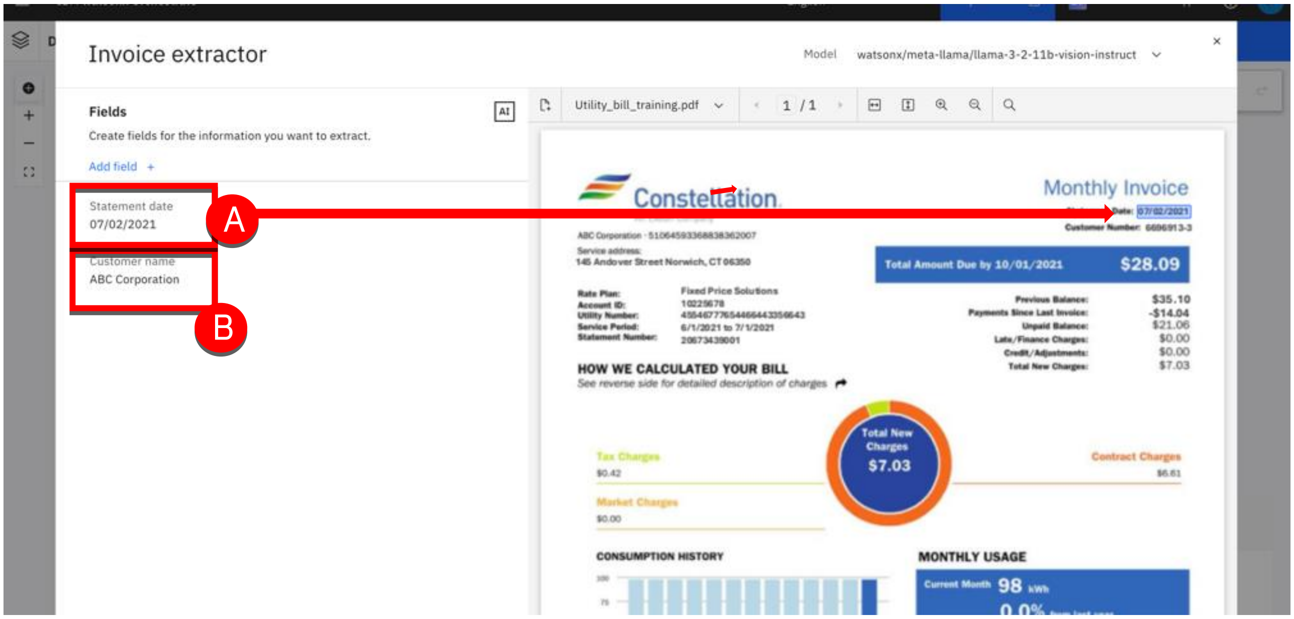
Note: The extractor may not recognize the field directly. We will see in the next step how to train the extractor.

4. Enter '`Statement date`' into the field name (A) then press Enter.



5. Review the extractor results:

- The **Statement date** was recognized (A).
- The **Customer name** was recognized (B).

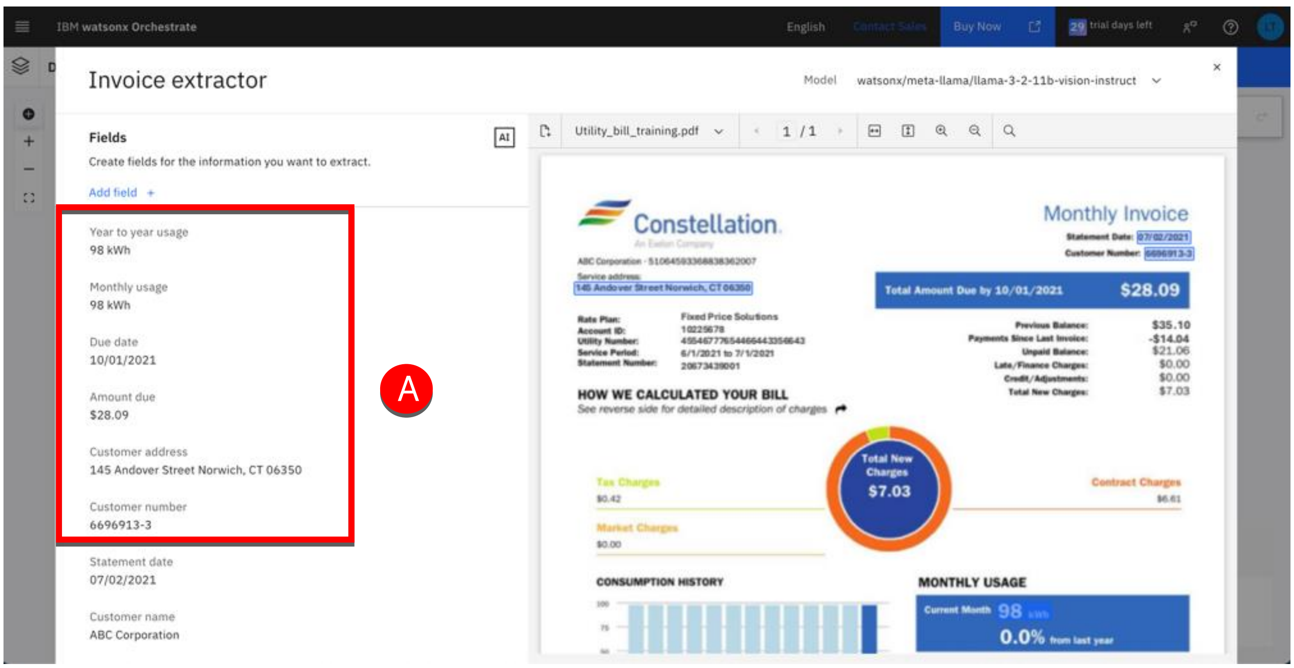


Note: The recognition results you see may vary from these screenshots depending on the maturity of the LLM model and its ability to recognize utility bill fields as LLM models are evolving rapidly.

You will add additional fields described in the next section before training the model to extract the missing fields.

6. Repeat the steps above to add the following fields (A):

- `Customer number`
- `Customer address`
- `Amount due`
- `Due date`
- `Monthly usage`
- `Year to year usage`

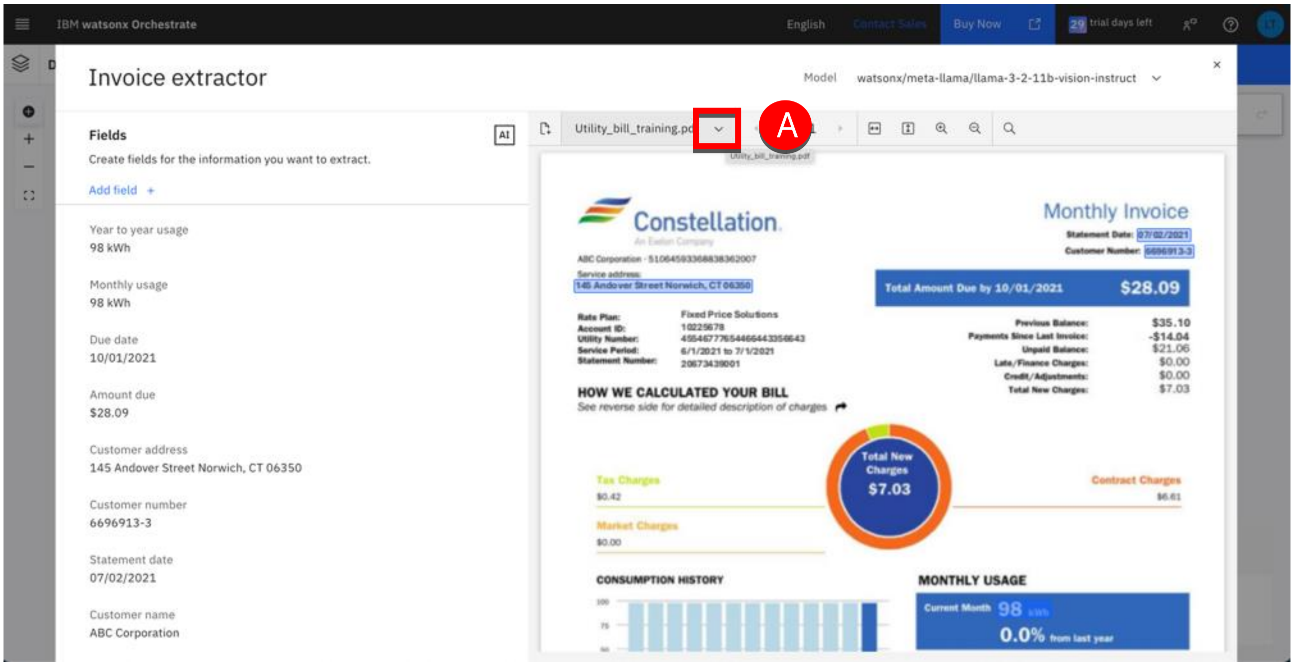


> **Note:** In this example, the extractor was able to correctly recognize all the fields. This may not be true in your situation. We will see in the next chapter how to improve the model.

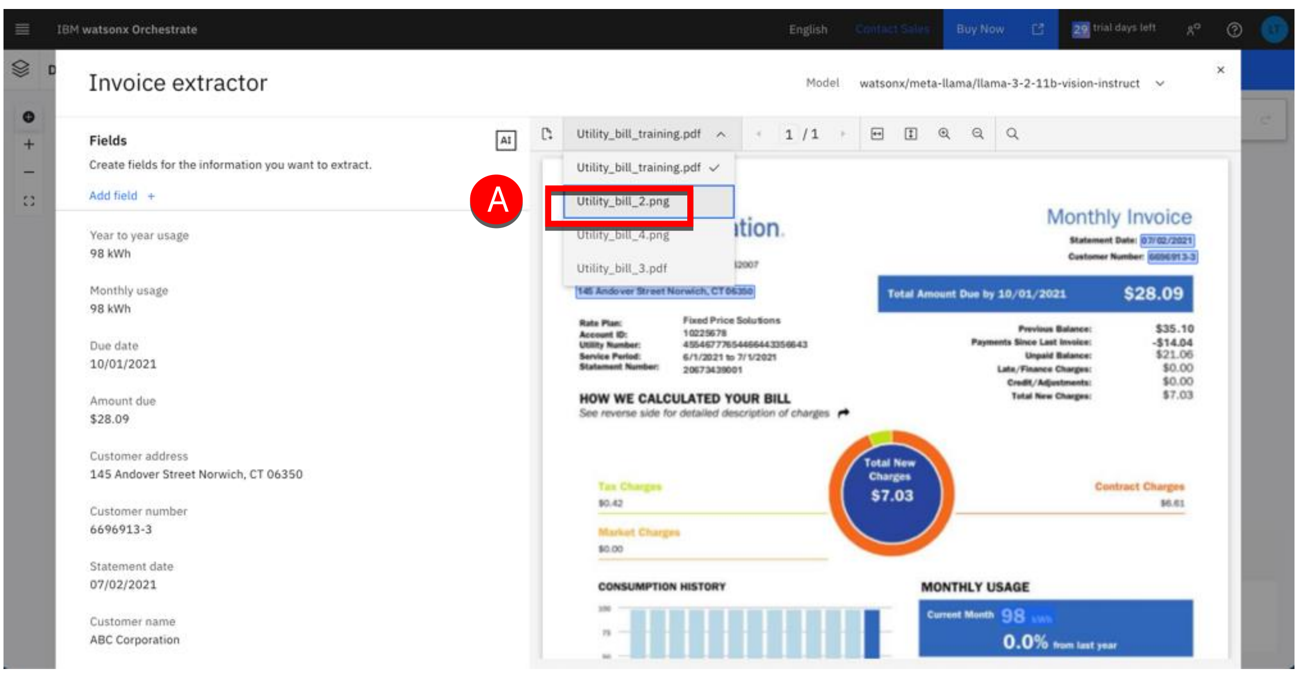
Test on an image file

In the previous step, the document extractor has been set-up using a pdf document. Scanned images can also be used in the same way. Let's use an image of the same document populated with different data.

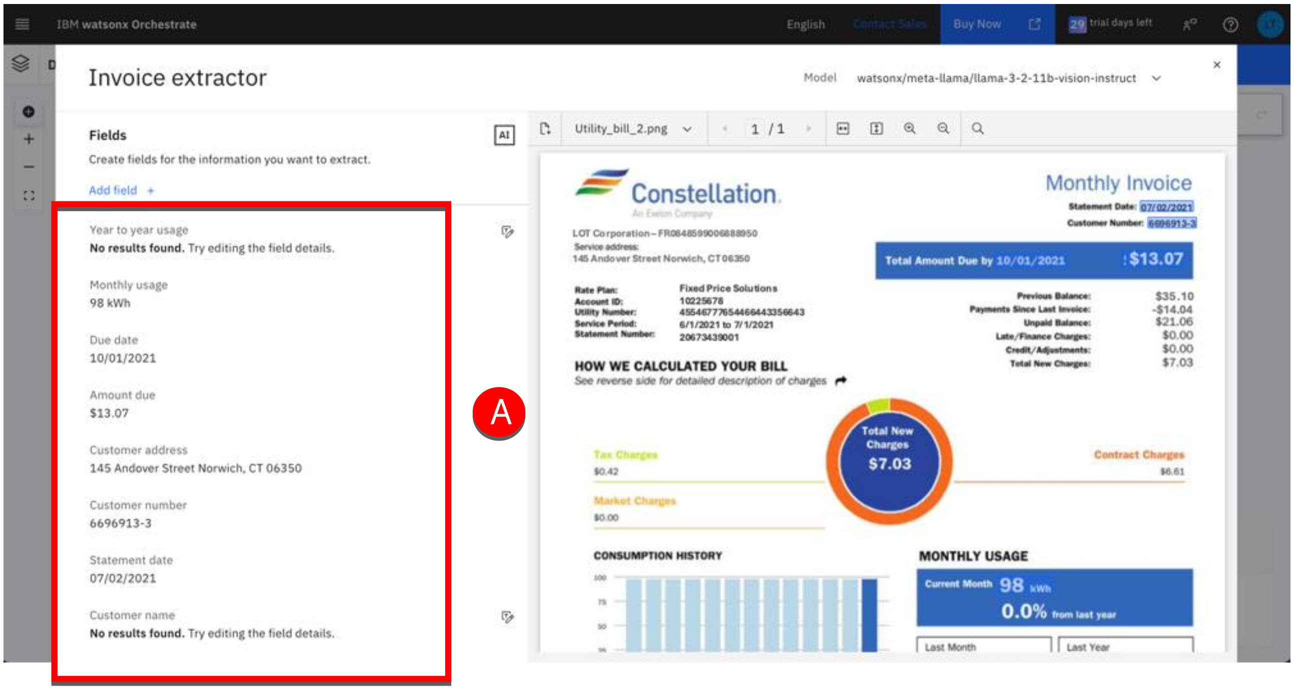
7. Click the manage document icon (A).



8. Select **Utility bill 2.png** (A).



9. Note that some fields may not have been recognized with this image version of the document (A).

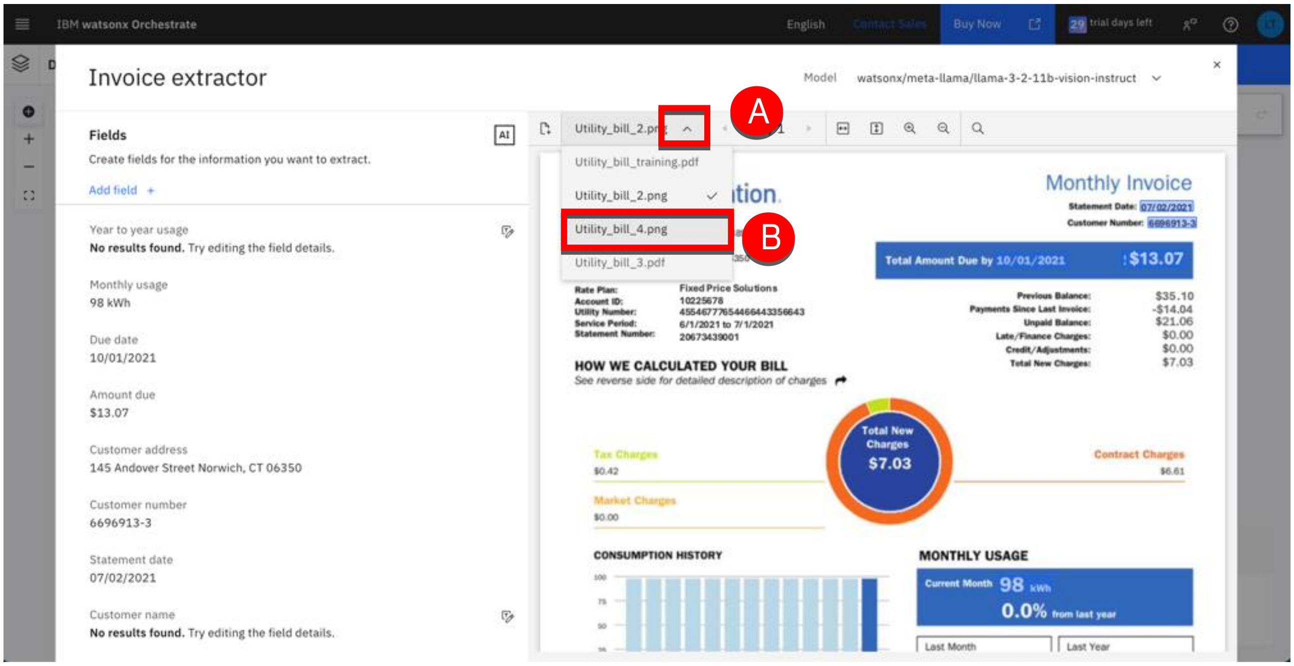


> **Note:** It may take a few seconds before the screen is refreshed with the extracted values. If all the fields are showing "No result found", refresh the view by switching to another document, then back again.
>
> **Note:** We will work to improve the LLM model in the next chapter.

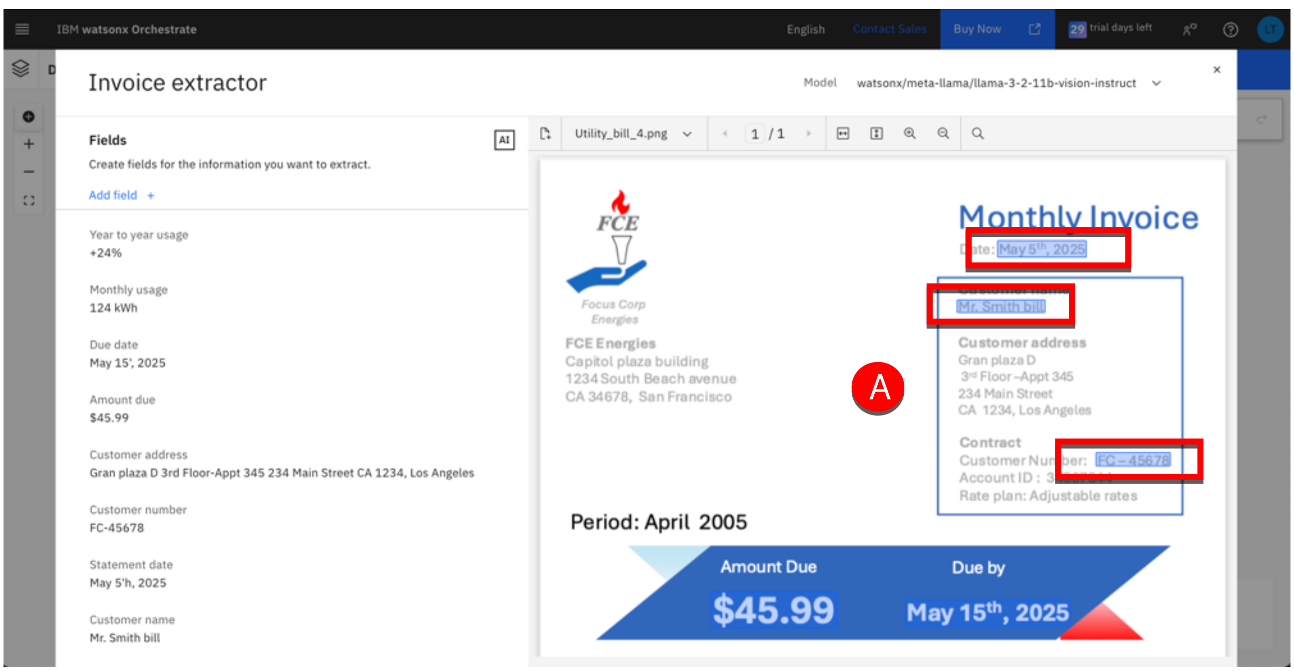
Test on a new document model

Not all invoices will have the same structure. In this section, you will test your document extractor on a utility invoice from a different energy company.

10. Click the expand icon to expand the document list (A) then select **Utility bill 4.png** (B).



11. Note that some fields may not have been correctly extracted with this new version of the document (A).



The document extractor may have been unable to recognize and extract all the fields with this new version of the document, and some errors may also be present. You will improve the results from the extractor in the next section by giving better descriptions of the fields and providing examples.

Note: Based on your LLM history and version, un-identified fields or errors may differ from the one presented in this lab. You may have to adapt the following script to correct errors encountered with your extractor.

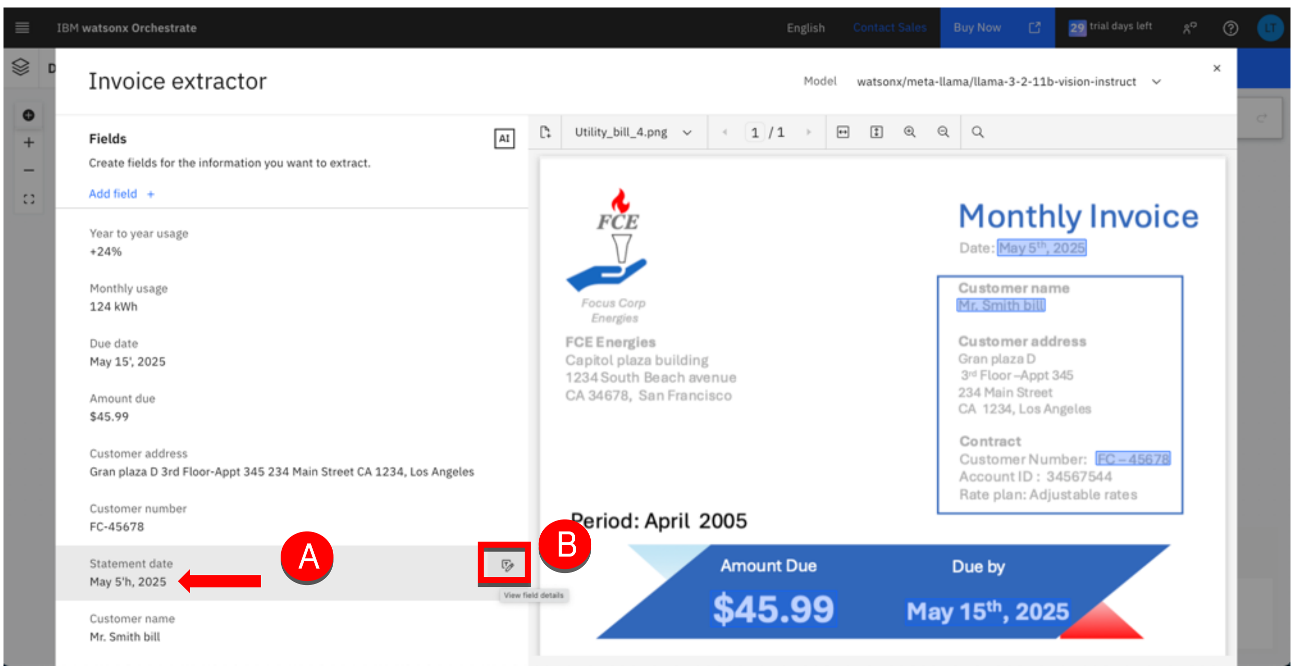
Improve your extractor

In this section, you will apply techniques to improve your results, helping the model to better identify and extract relevant data from your unstructured documents.

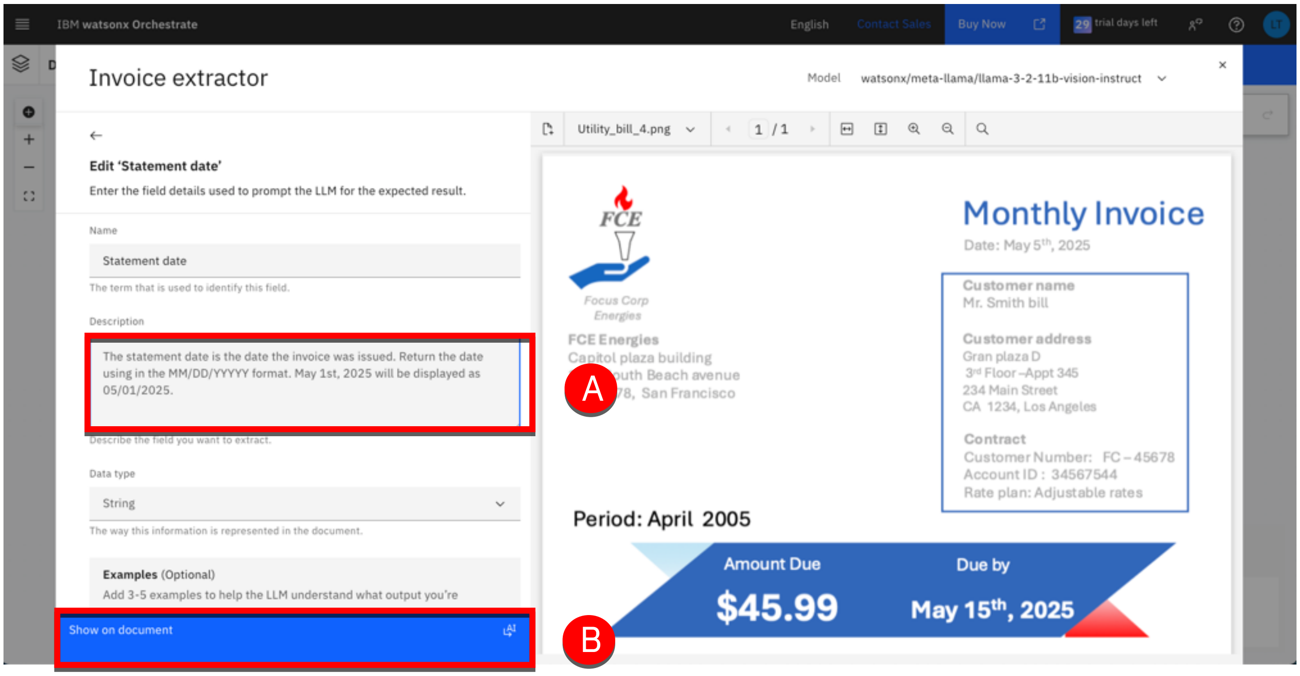
Add field descriptions

In our example (that may differ from yours), the Statement date is extracted but requires a reformatting. We would like all the dates to be formatted the same way with the mm/dd/yyyy format. Let's add a precise description of the field that will be used by the LLM to better extract the information in the format we want.

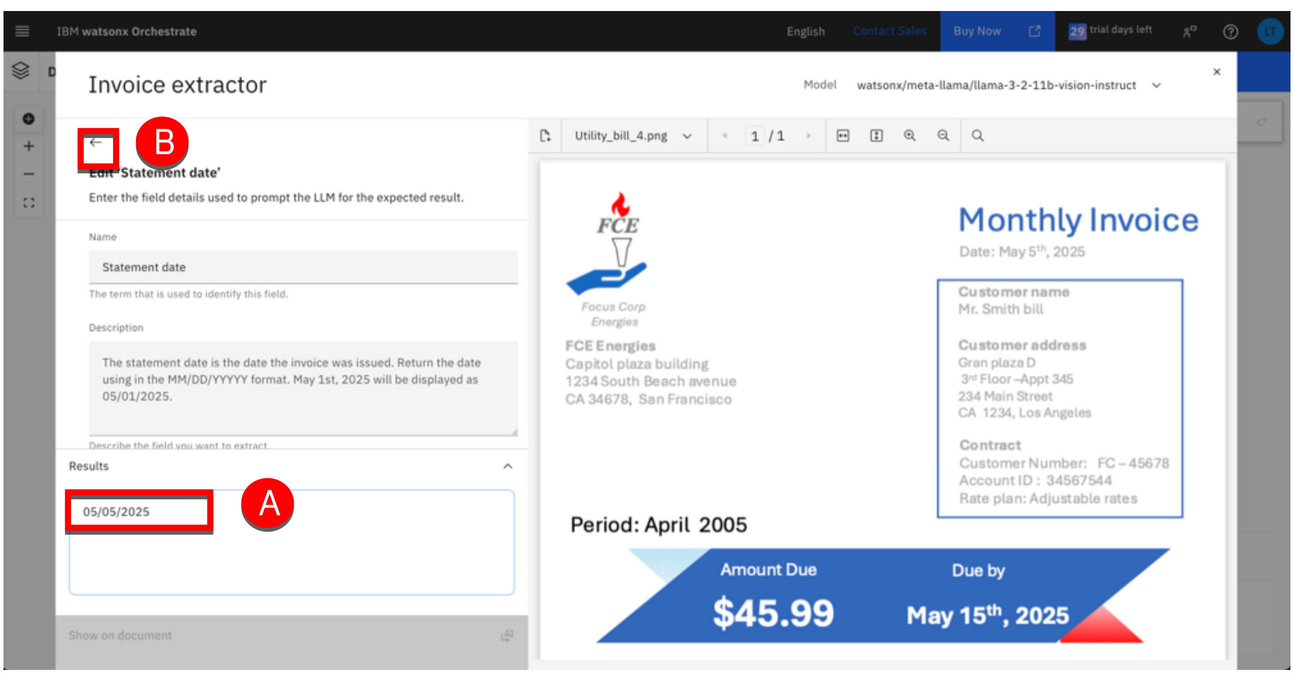
12. Identify a field displaying an error (A) then click the **View field details** icon (B).



13. Enter ‘`The statement date is the date the invoice was issued. Return the date using in the MM/DD/YYYYY format. May 1st, 2025 will be displayed as 05/01/2025.`’ into the **Description** field (A), then click **Show on document** (B).

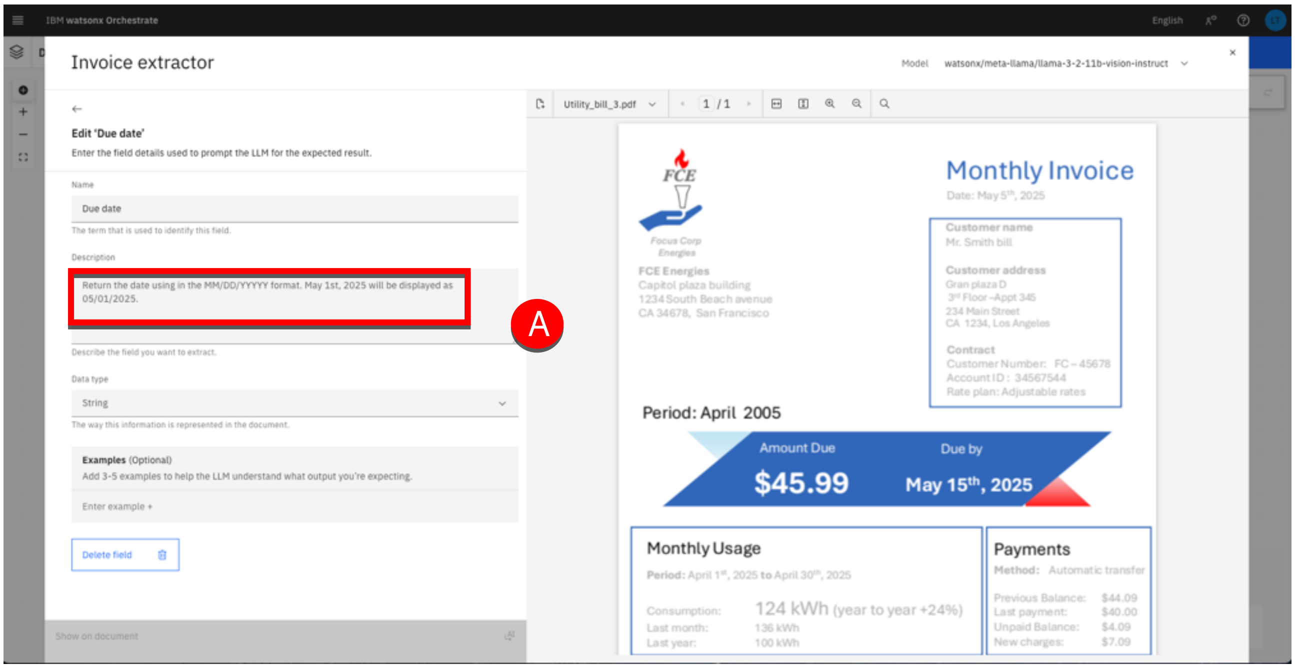


14. The field is now correctly identified and formatted the right way (A). Click the **<-** icon of the edit panel (B).



Note: If your extractor does not show the correctly formatted date, click the <- icon, then return back to the field details view to refresh the values.

15. Repeat the instruction definition (adapt it to the due date) so that the Due date is also displayed in the MM/DD/YYYY format (A).

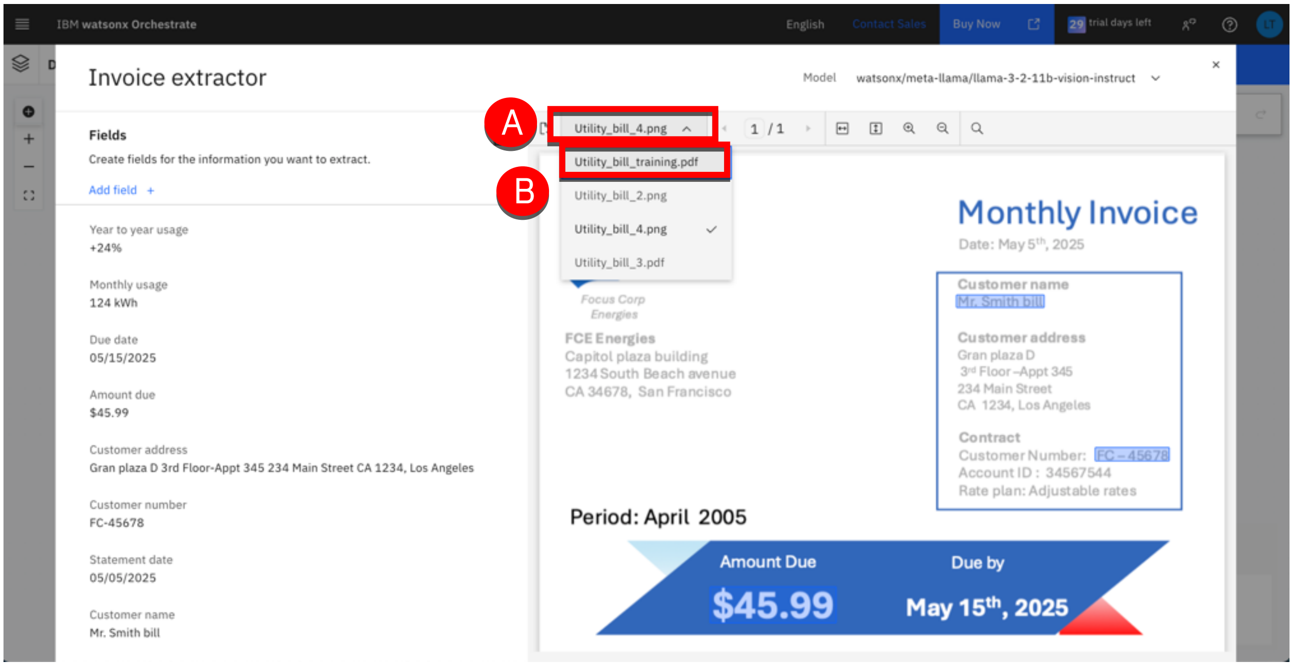


Note: At the time of writing, there is a minor issue with the position of the bounding box showing the extracted values position on the document (as you can see on the previous screen shot). This may be corrected by the time the lab is published so your output may be slightly different.

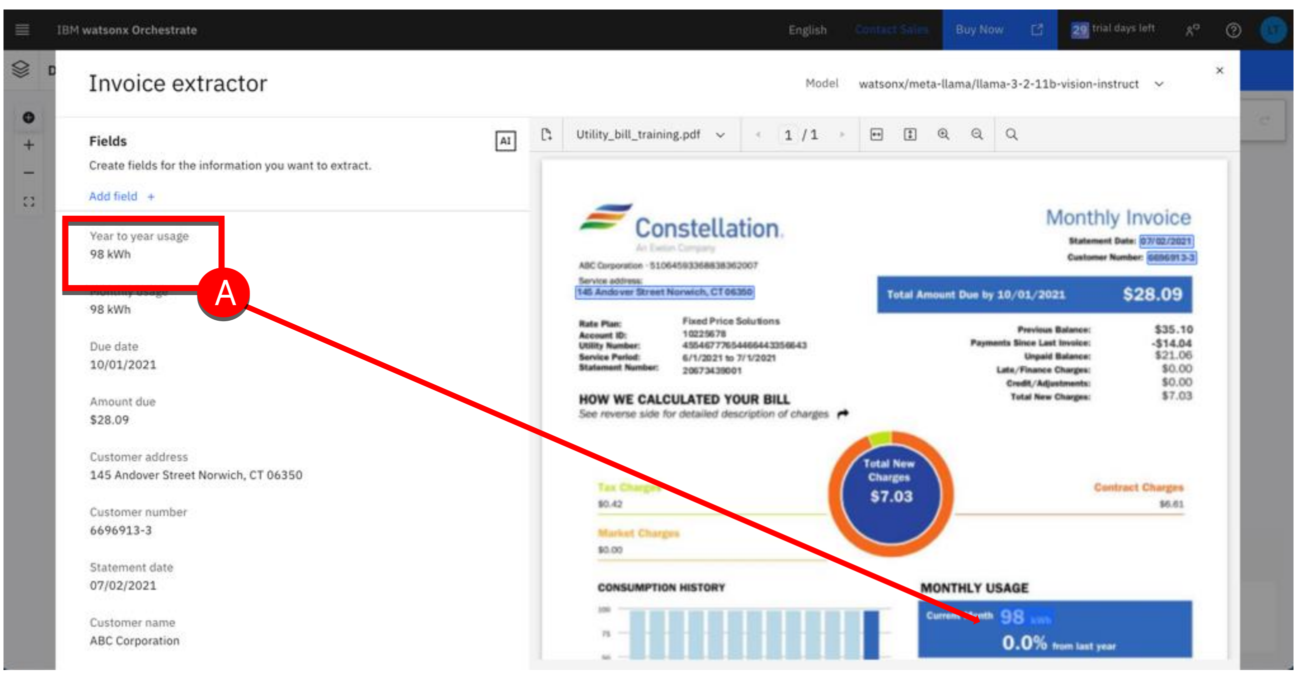
Adding examples

In the previous section, we have seen the benefit of using field descriptions to guide the model. When descriptions do not provide the expected result, you can add examples to improve your extraction results.

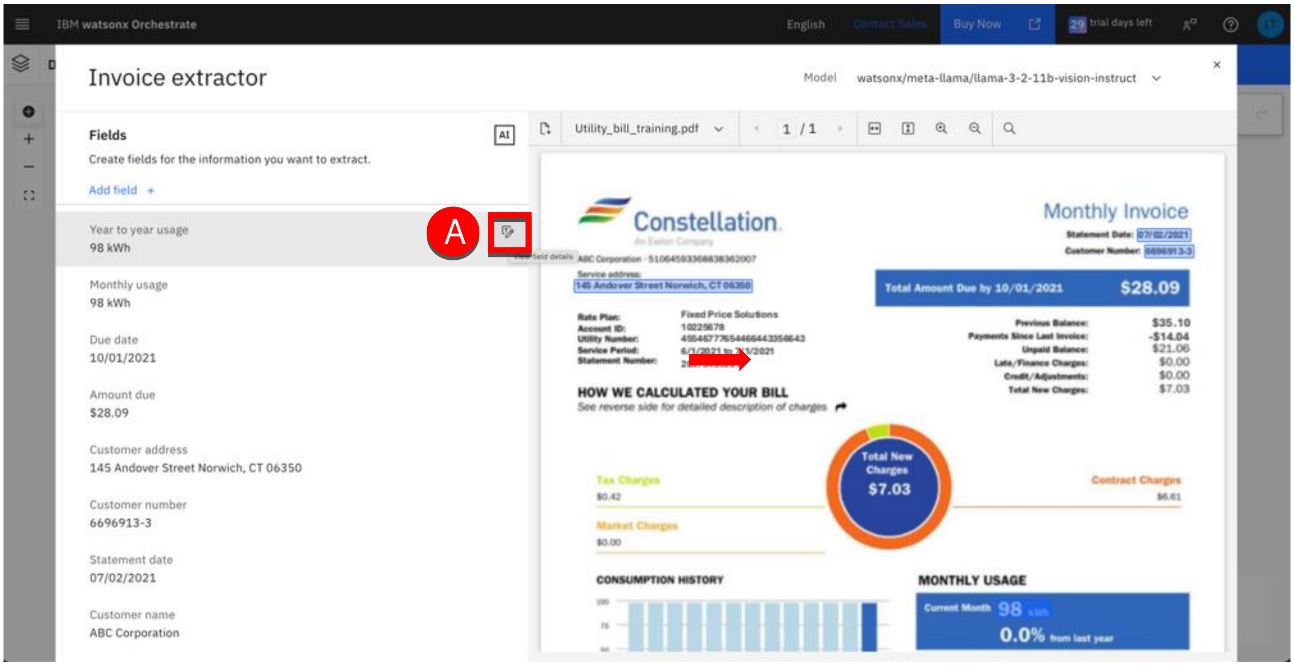
1. Expand the file selector (A). Select **Utility_bill_training.pdf** (B).



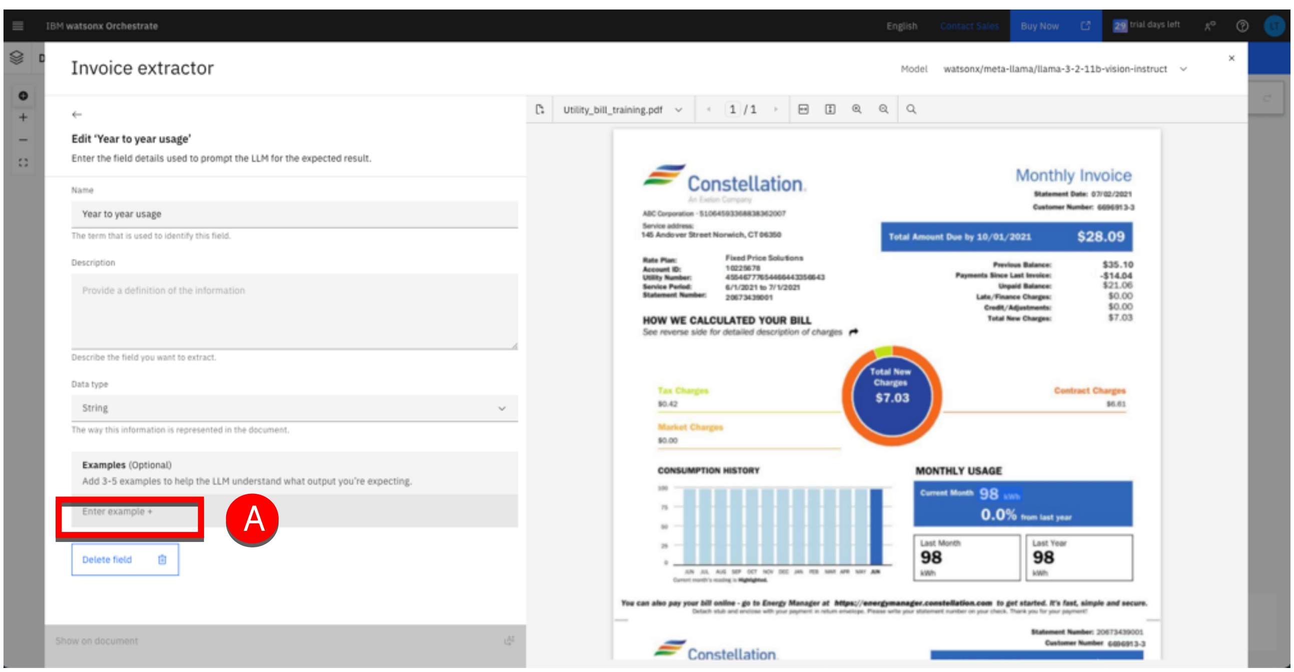
2. The **Year to year usage** is being extracted from the wrong element in the document (A), you will now add examples to help the model to extract the Amount due accurately.



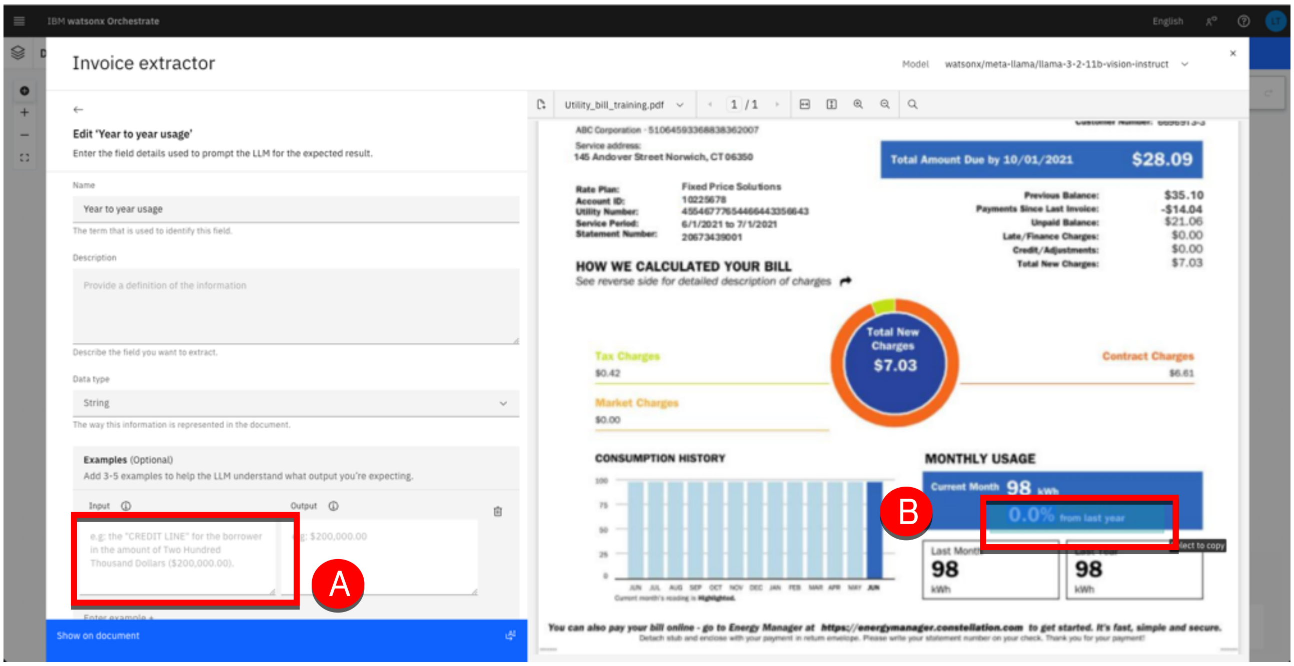
3. Click the **View field details** icon of the Year to year usage field (A).



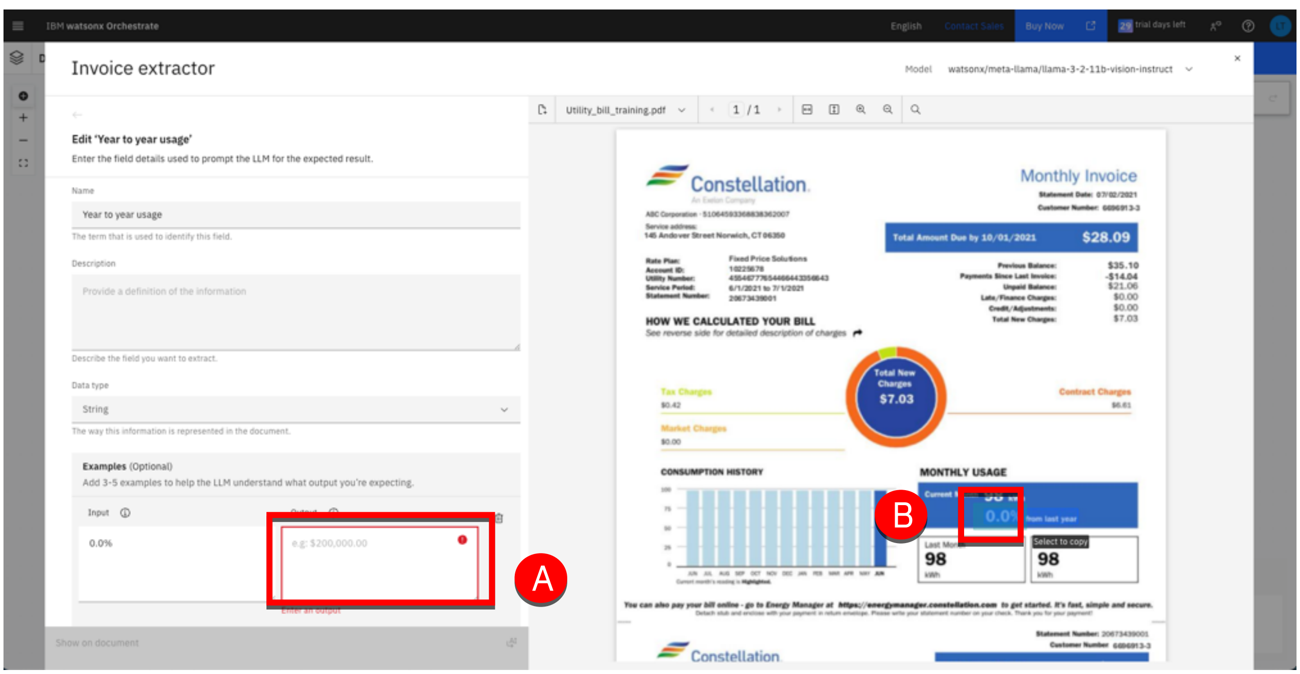
4. Click **Enter example** (A).



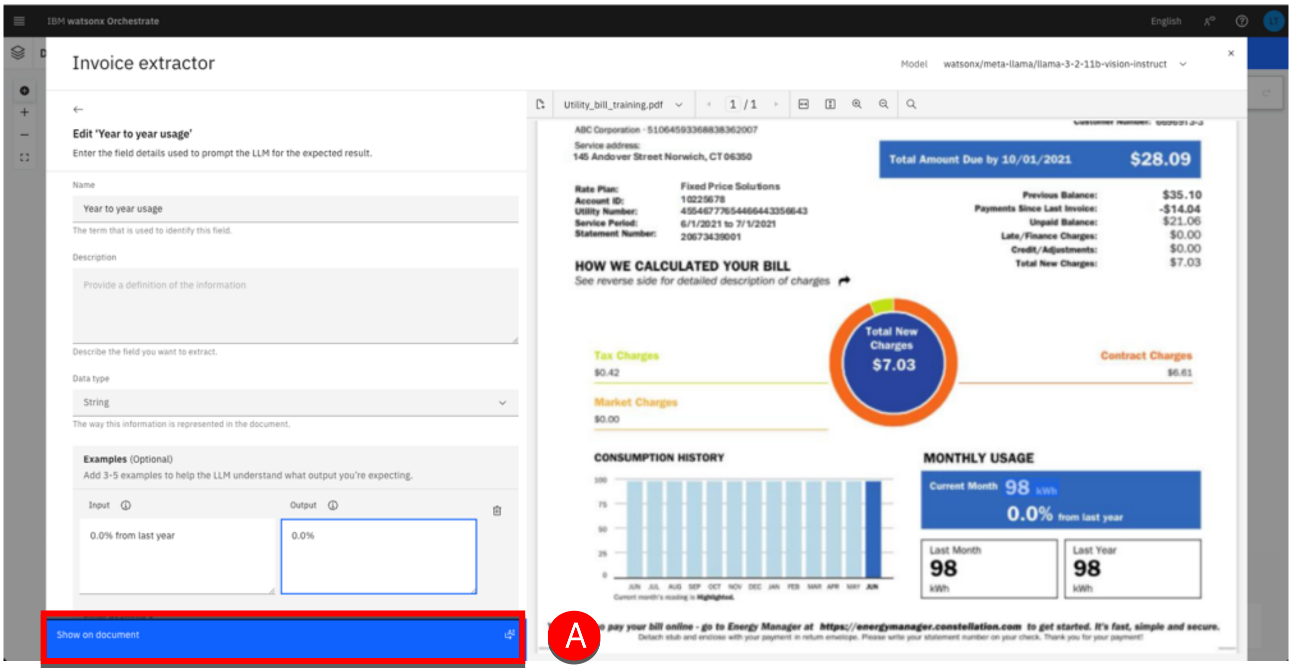
5. Click inside the example **Input** cell (A), then draw a rectangle around the area containing the **Amount due** value in the document (B). This will automatically capture the text contained in your selected area in the Input field.



6. Let's now specify how to extract the output value from that selection. Click the Output cell (A) then just select the percentage value to extract (B).

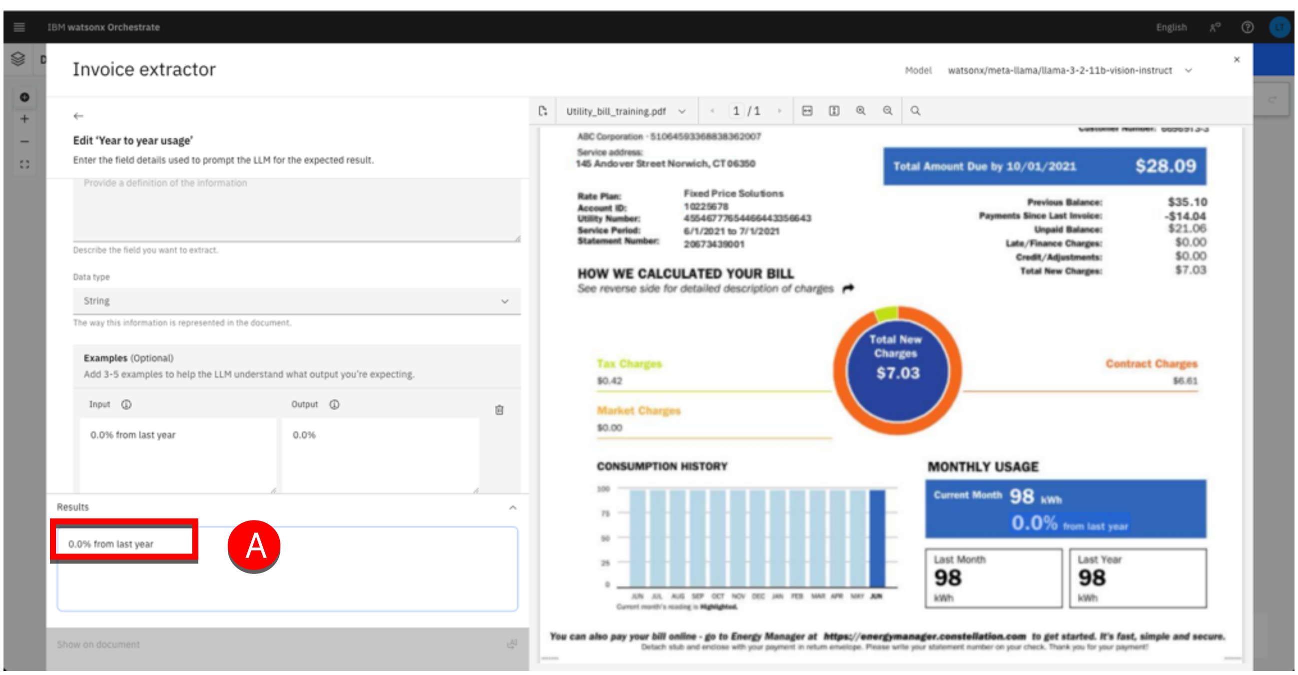


7. Click **Show on document** (A).

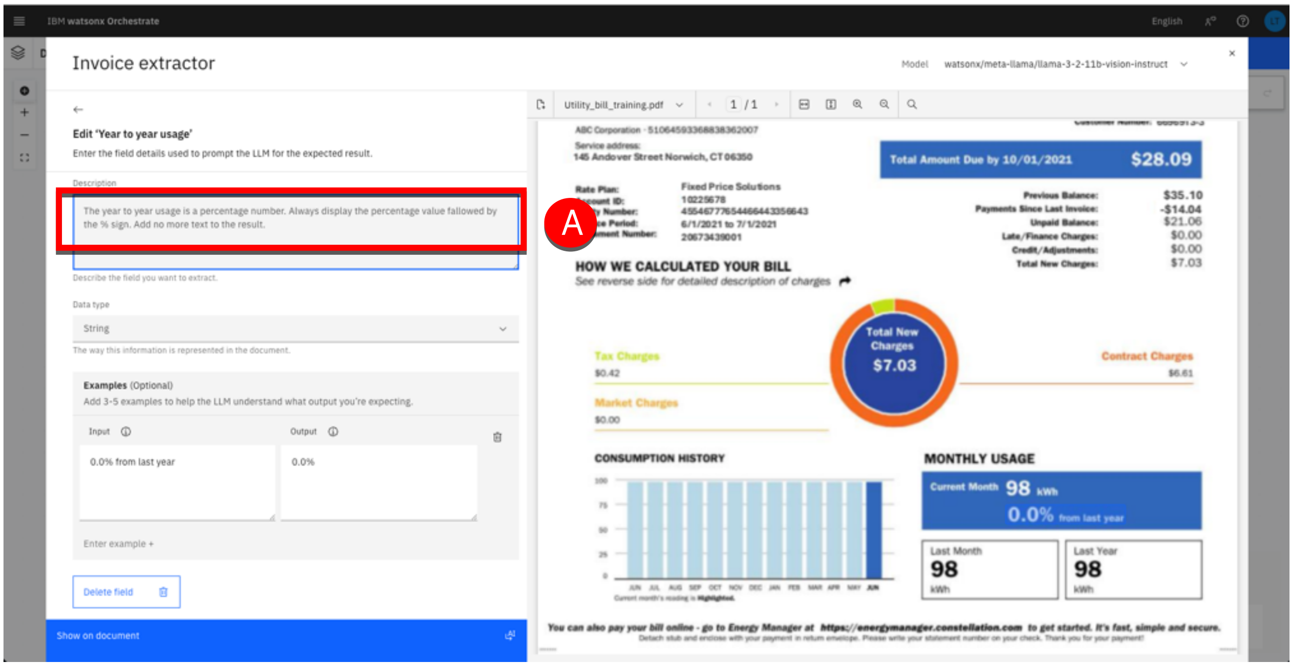


> **Note:** As soon as text under the selected area is recognized, it is added to the input field example. If no text is dynamically selected when surrounding the area, this means the document extractor will not recognize this area as containing text.

8. If the value (A) is not correctly identified (A), a description may be added as shown in the next step.

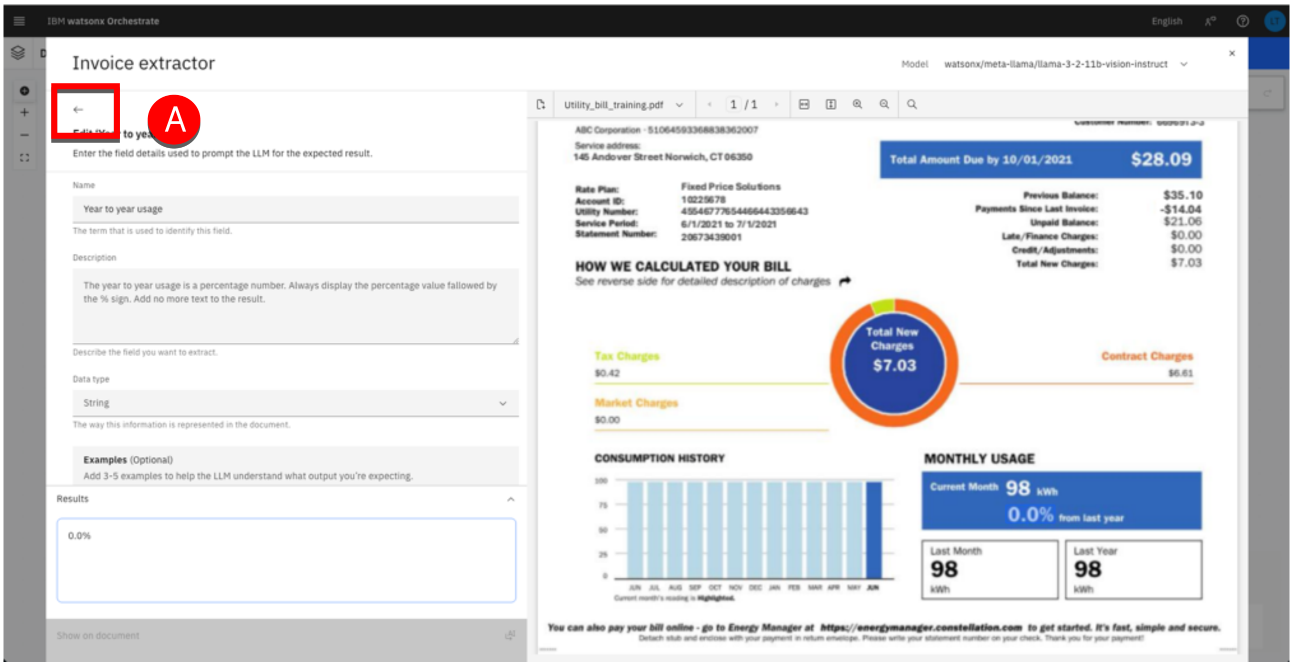


9. If you still have an incorrect output (ie, more text is displayed than expected) provide a description. Type 'The year to year usage is a percentage number. Always display the percentage value fallowed by the % sign. Add no more text to the result.' in the **Description** (A). Click Show on document (B) then wait a few seconds for the result to be updated.

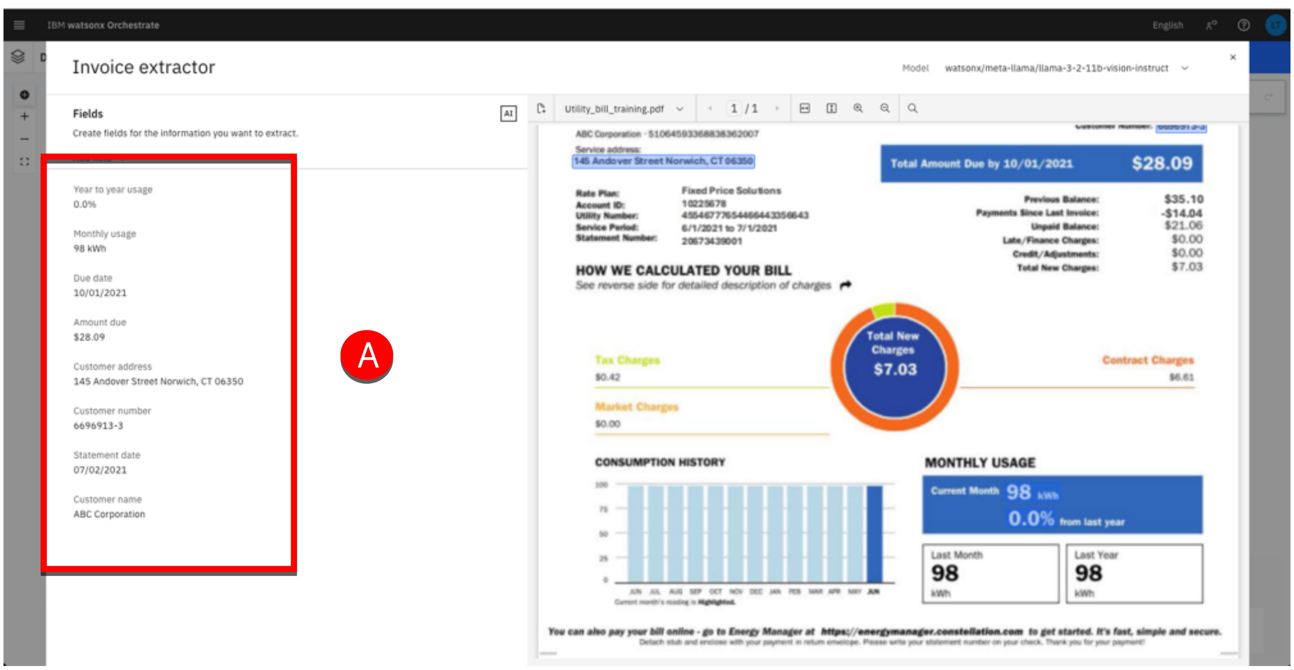


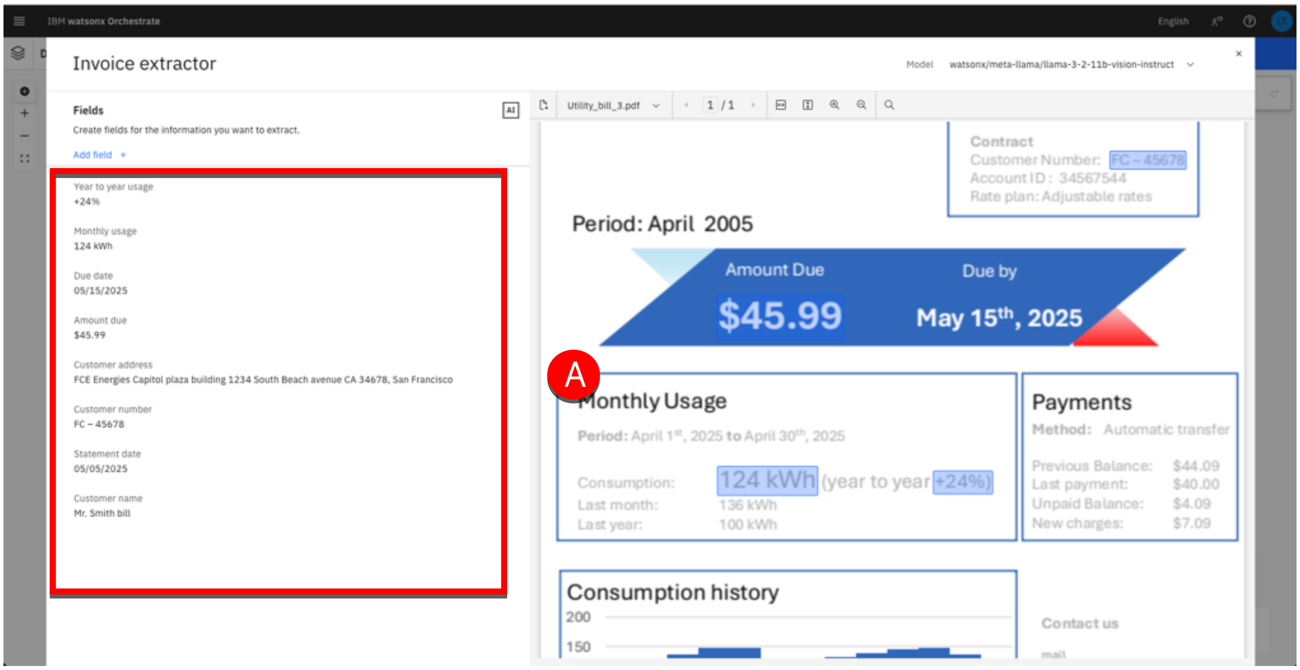
> **Note:** Depending on your environment, it may take some time before the result is correctly displayed. Don't hesitate to add more examples until the correct result is displayed should you model fail to display the expected value. Click anywhere in the interface to refresh the screen and see the updated values if required.

10. Click the **<-** icon (A).

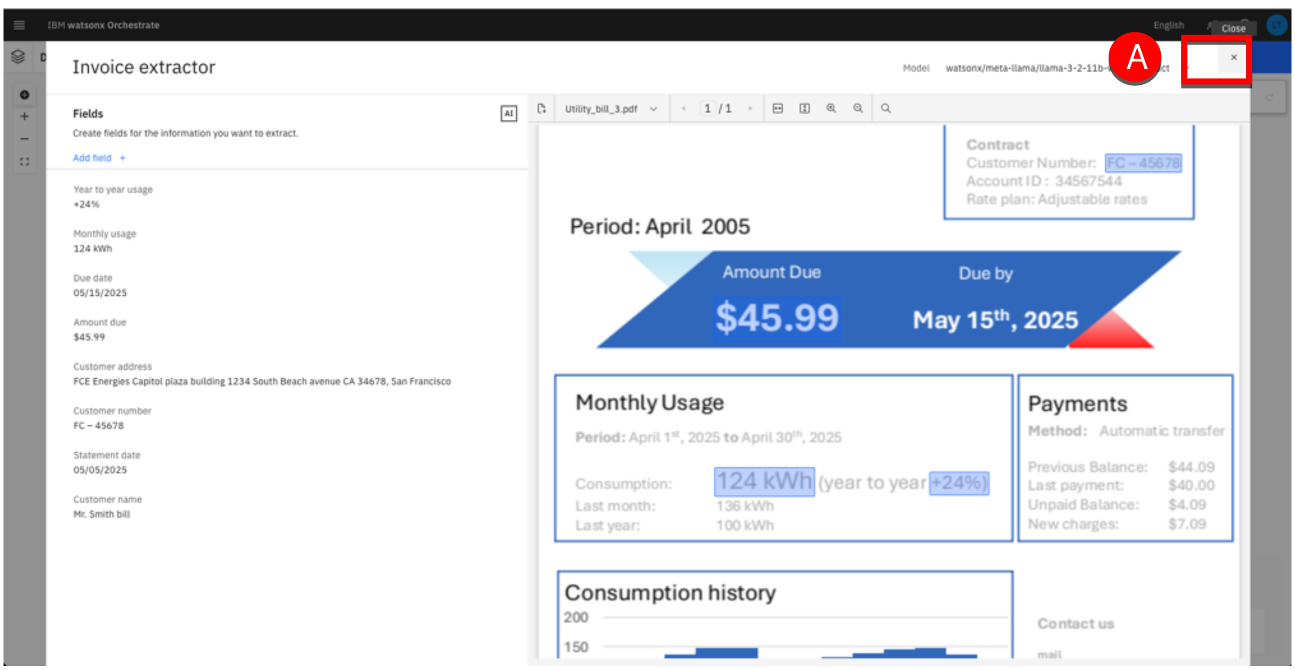


11. Repeat the two improvement methods (Description / Add example) for each field in each of the different document samples you have provided until all the fields are correctly extracted for each file (A).

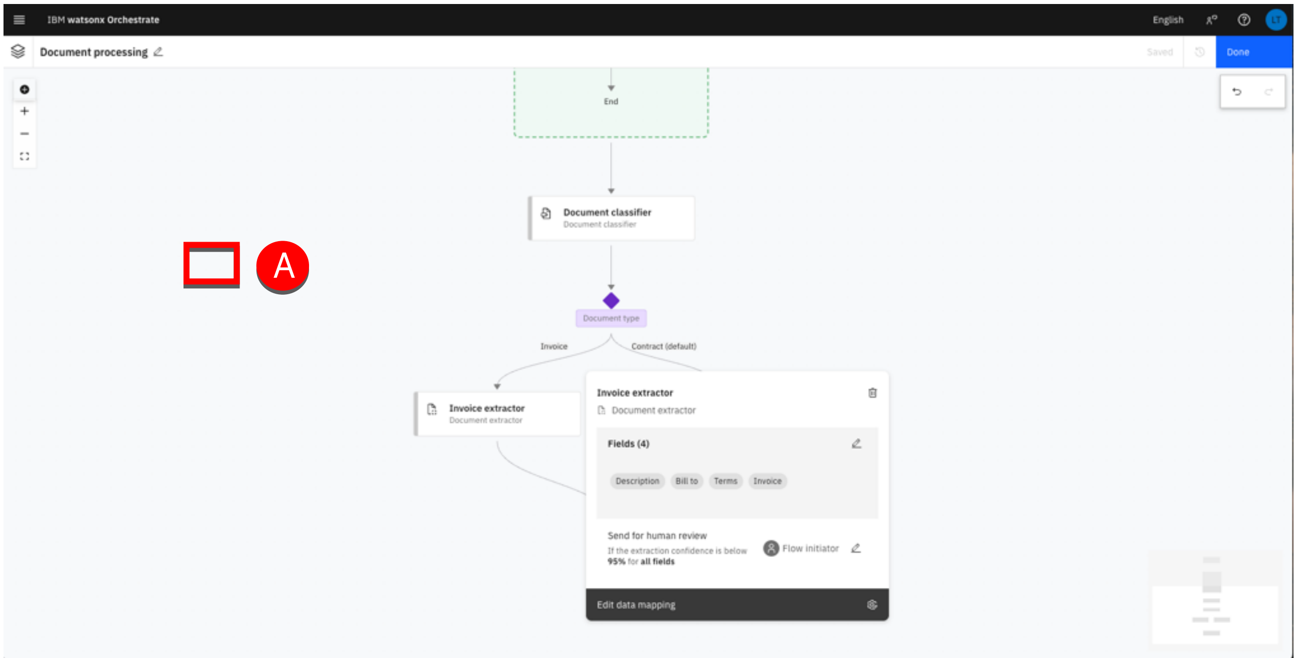




12. Click **X** to close the extractor editor (A).



13. Click on the diagram background to close the **Invoice extractor** property view (A).

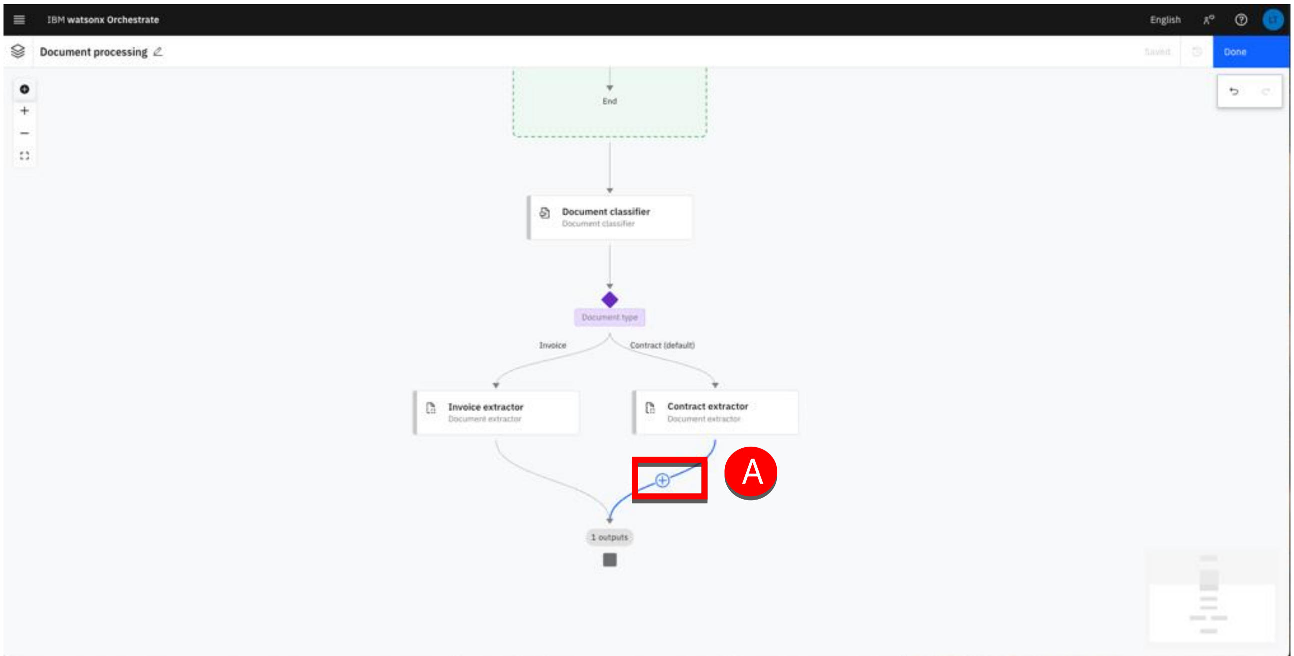


You will now add the user activities for each branch to display the extracted values.

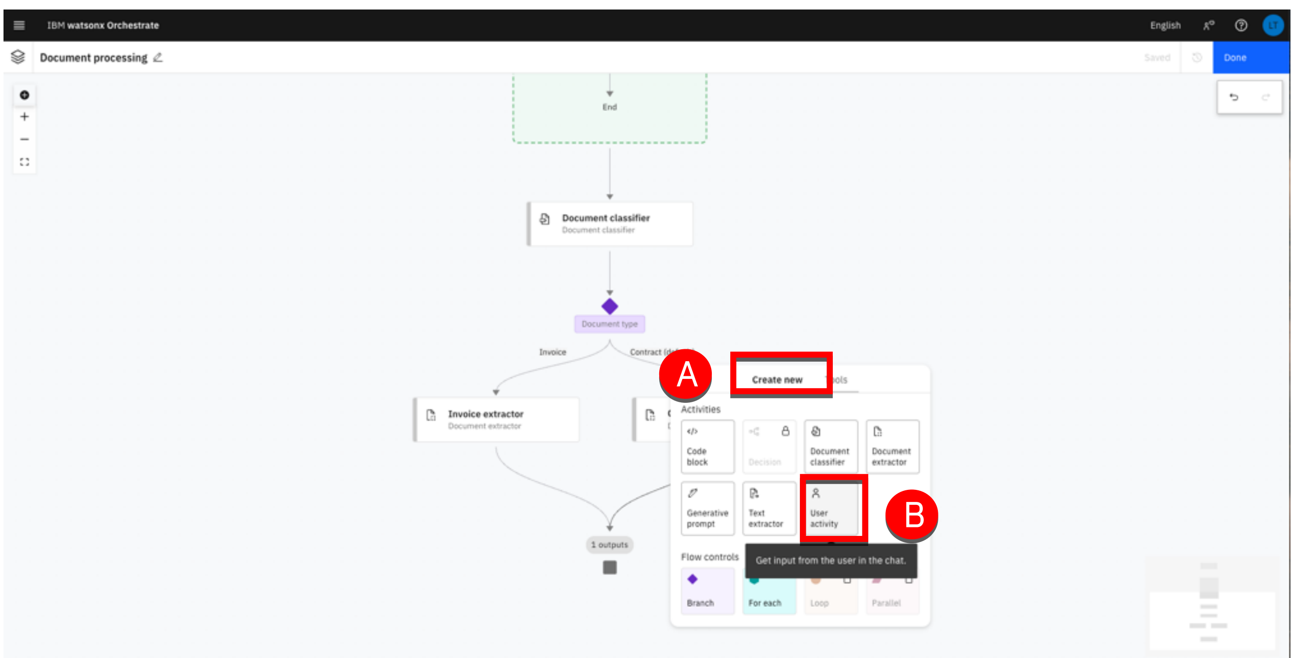
Add a Display to user activity

To complete the flow, we need to add a final activity that presents the extracted values to the user. This step allows us to format and display the results clearly, based on the core fields that we defined when we trained each extractor.

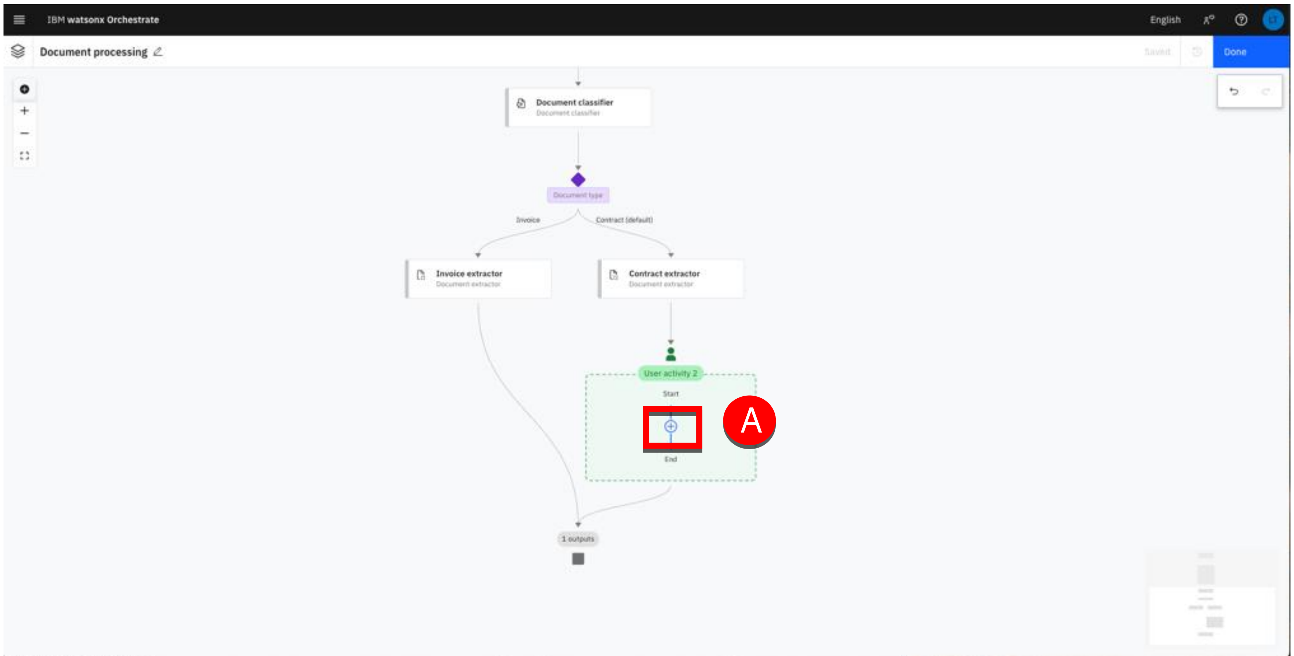
1. Hover the last link in the Contract extractor branch and click + (A).



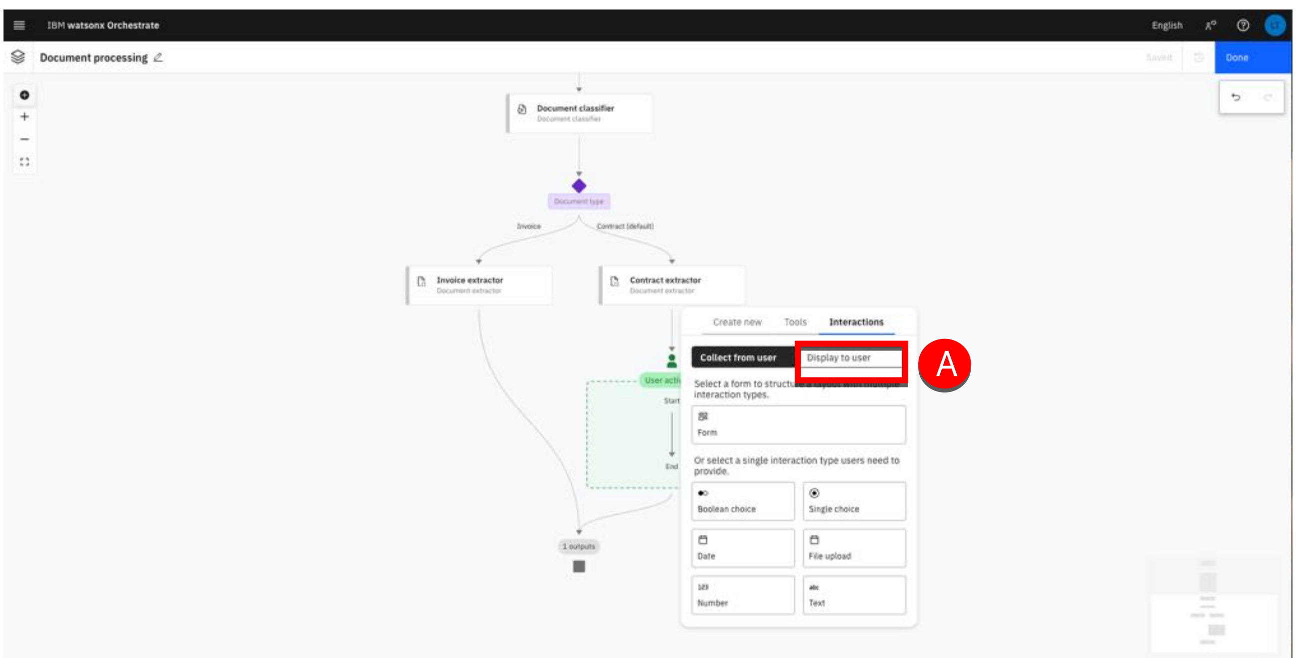
2. Select **Create new** (A) then click **User activity** (B).



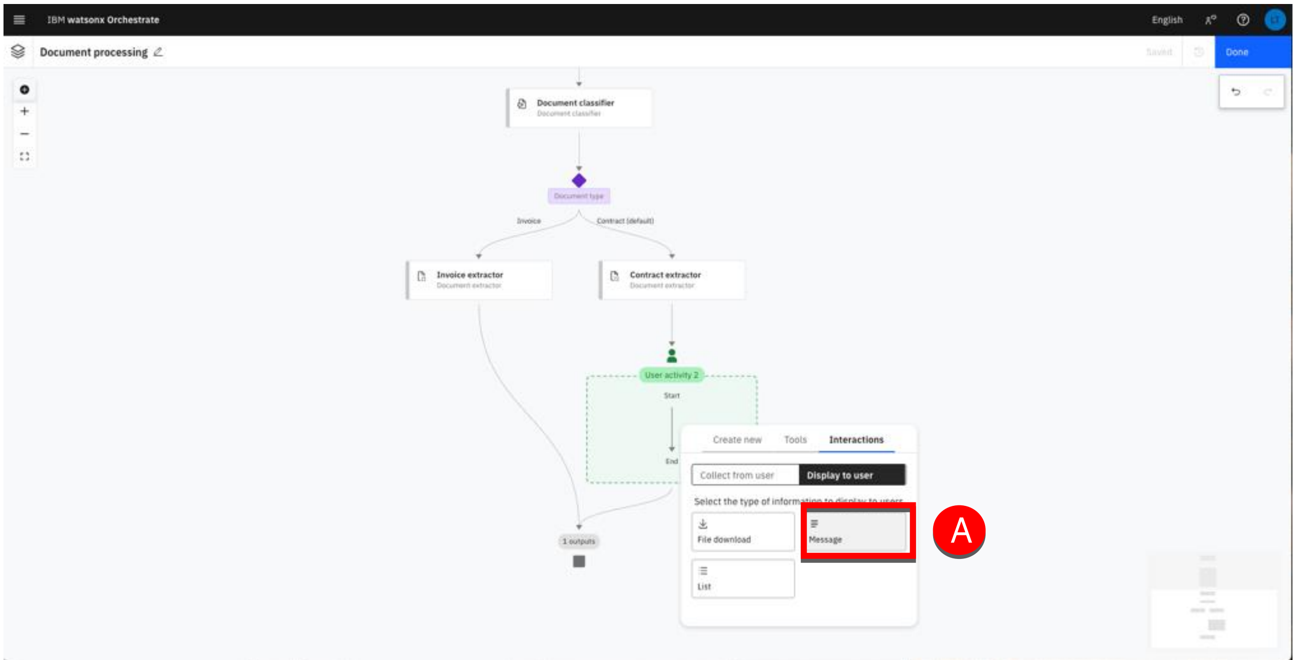
3. Click + (A) to add a new action in the user activity node.



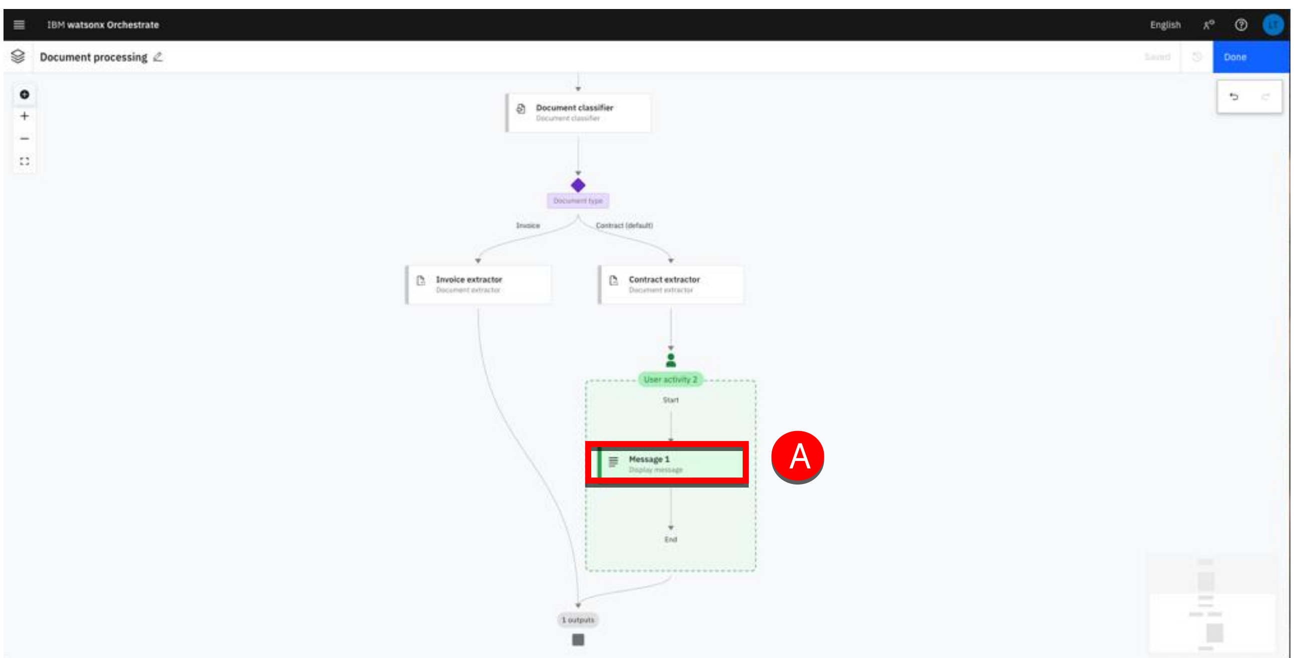
4. Click **Display to user** (A).



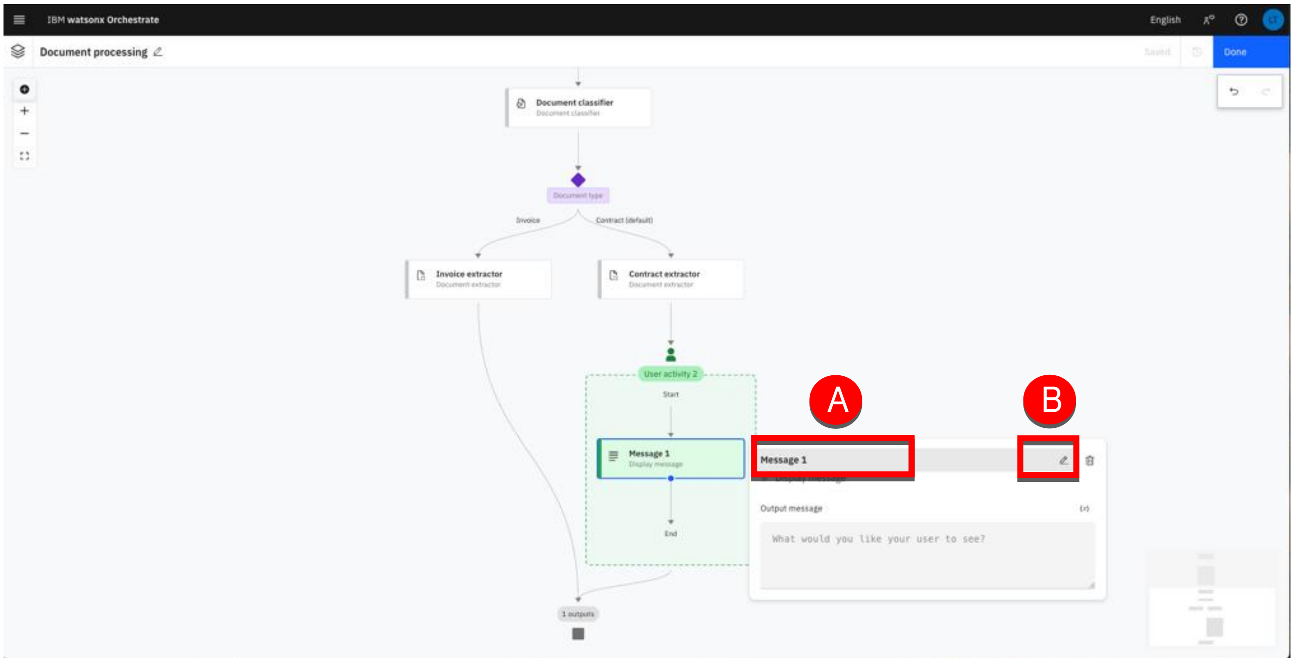
5. Select **Message** (A).



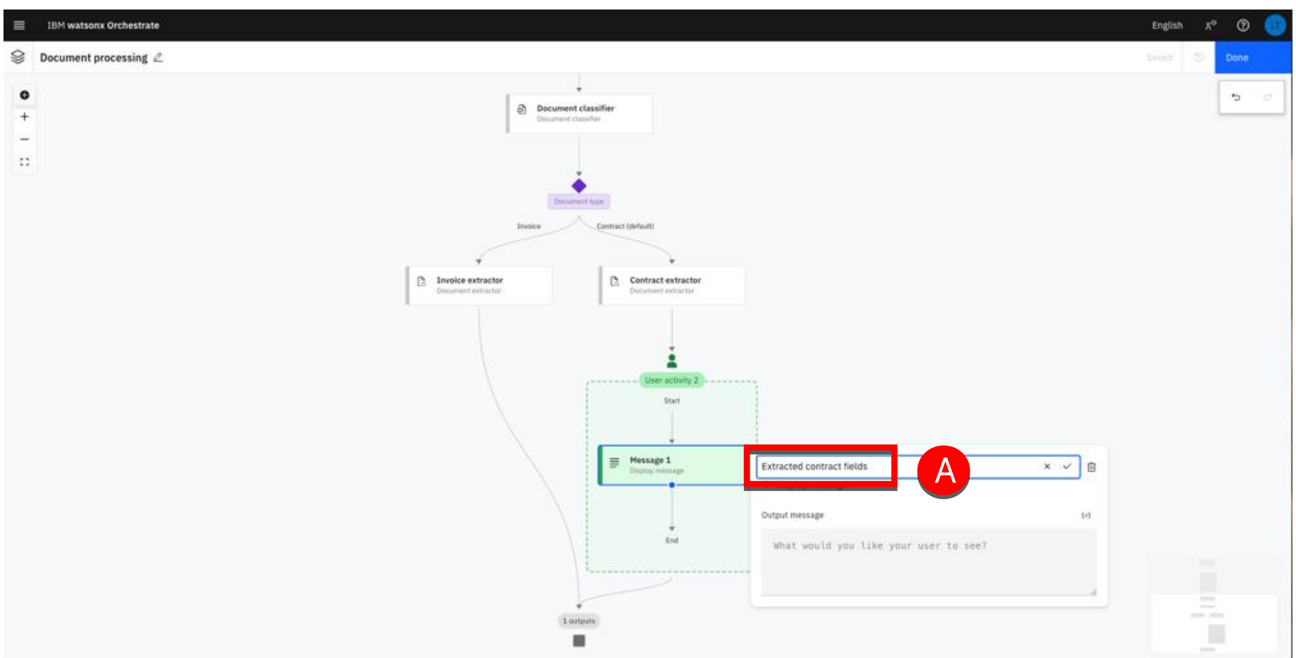
6. Click the **Message** box to edit its content (A).



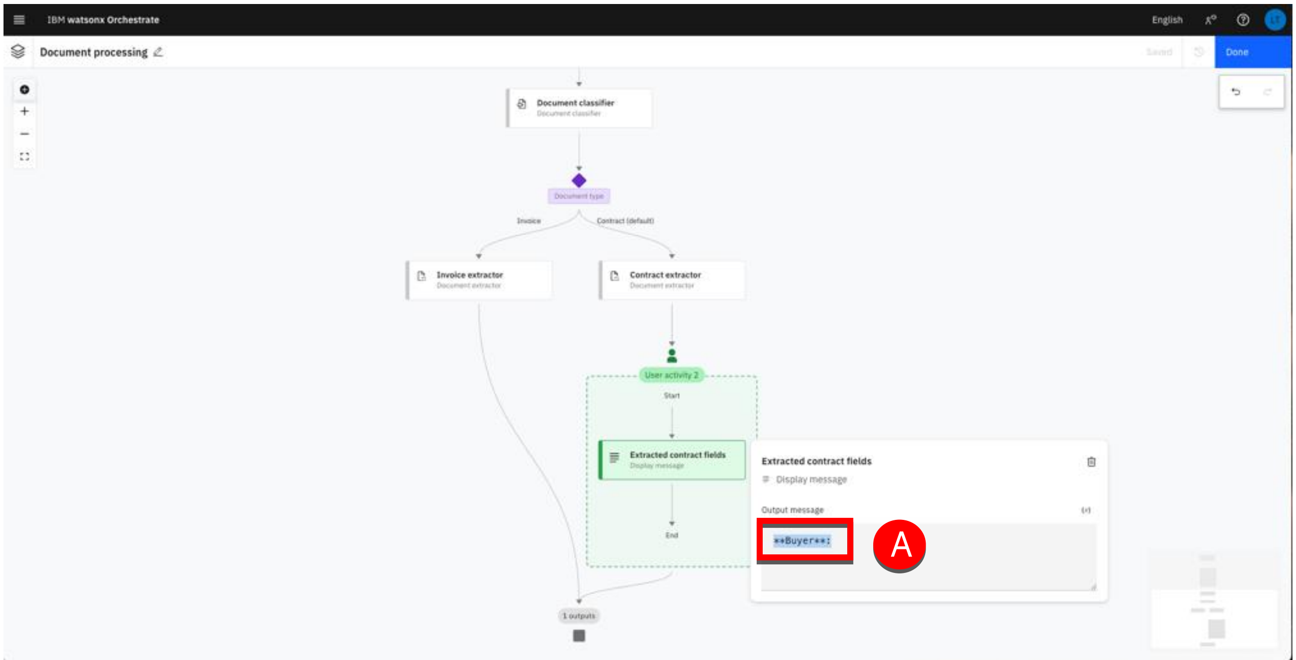
7. Hover the **Message 1** name (A) then click the edit button (B) to rename it.



8. Type ‘`Extracted contract fields`’ (A) and hit Enter.

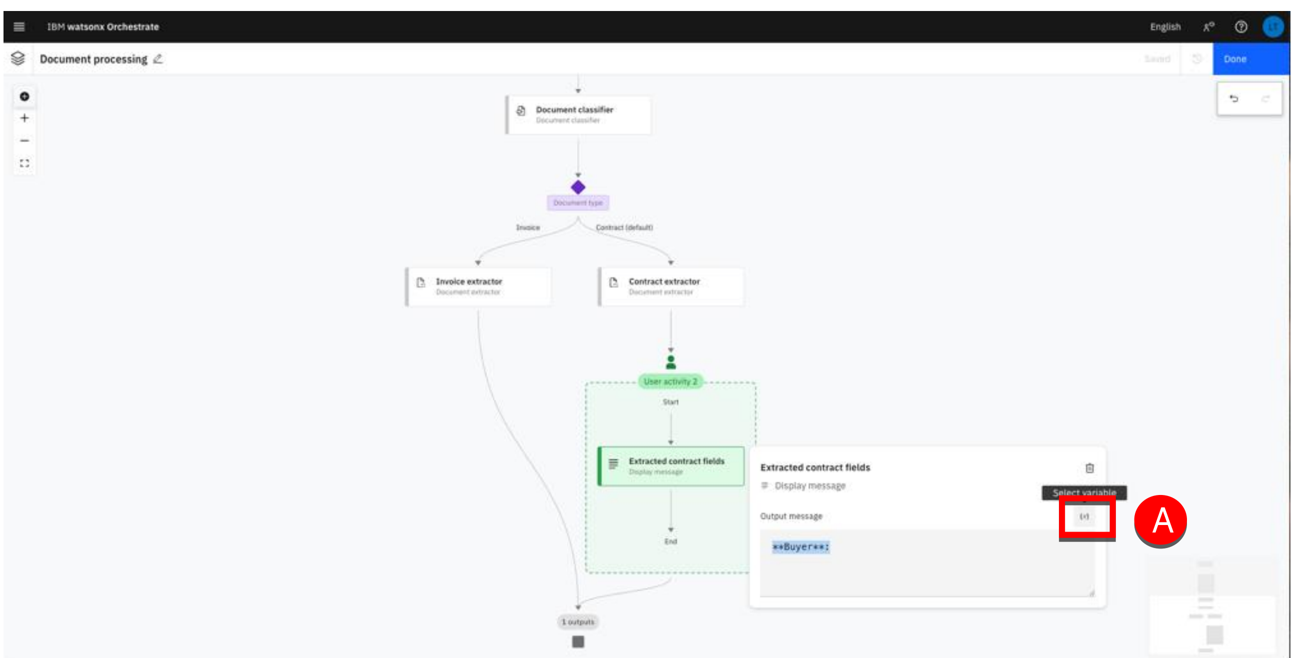


9. Type “**Buyer**:” in the Output message field (A)

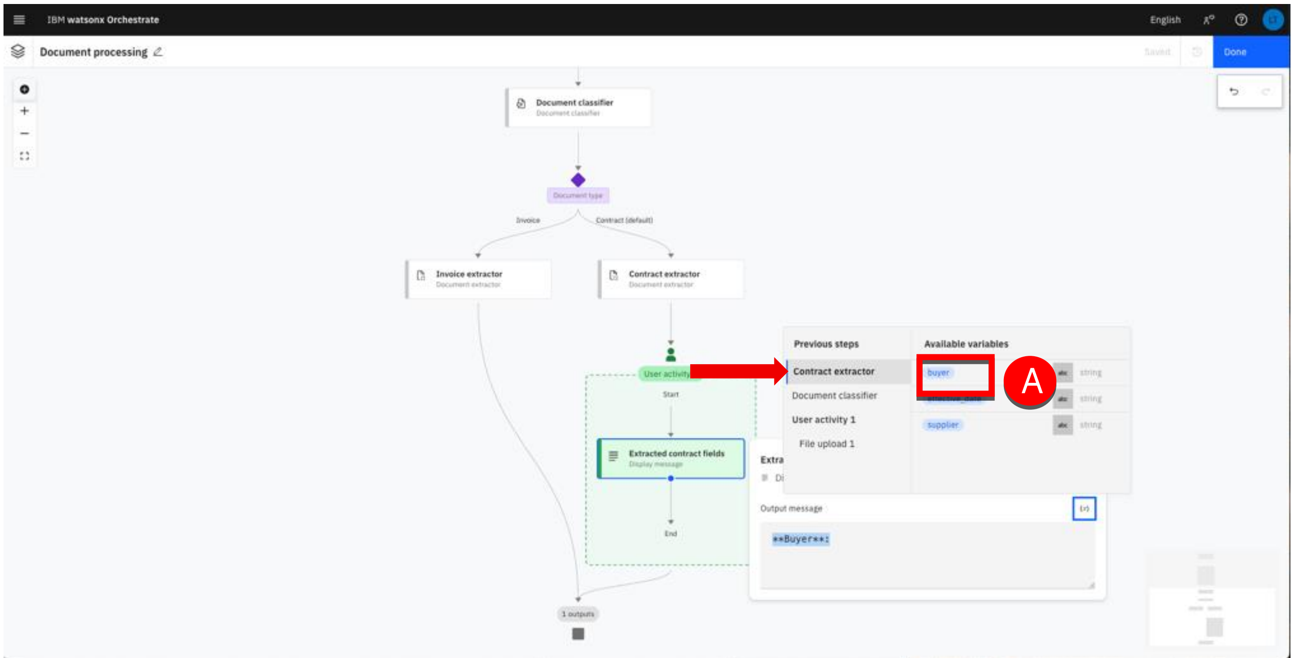


Note: the ‘** ‘are just here to specify that the label must be displayed in Bold.

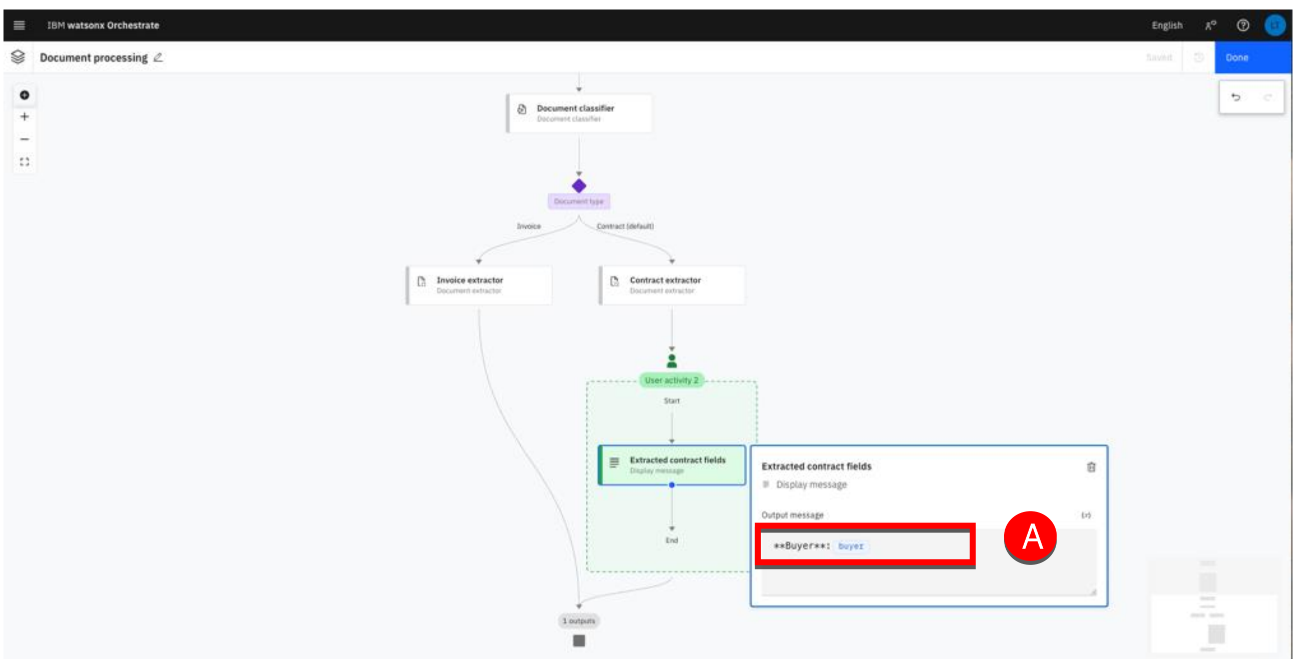
10. To display the results for the field, ‘Buyer’ you must assign a variable to it. Click on the Select variable ‘[x]’ button.



11. Under the ‘**Contract extractor**’ menu click on the **buyer** variable (A).



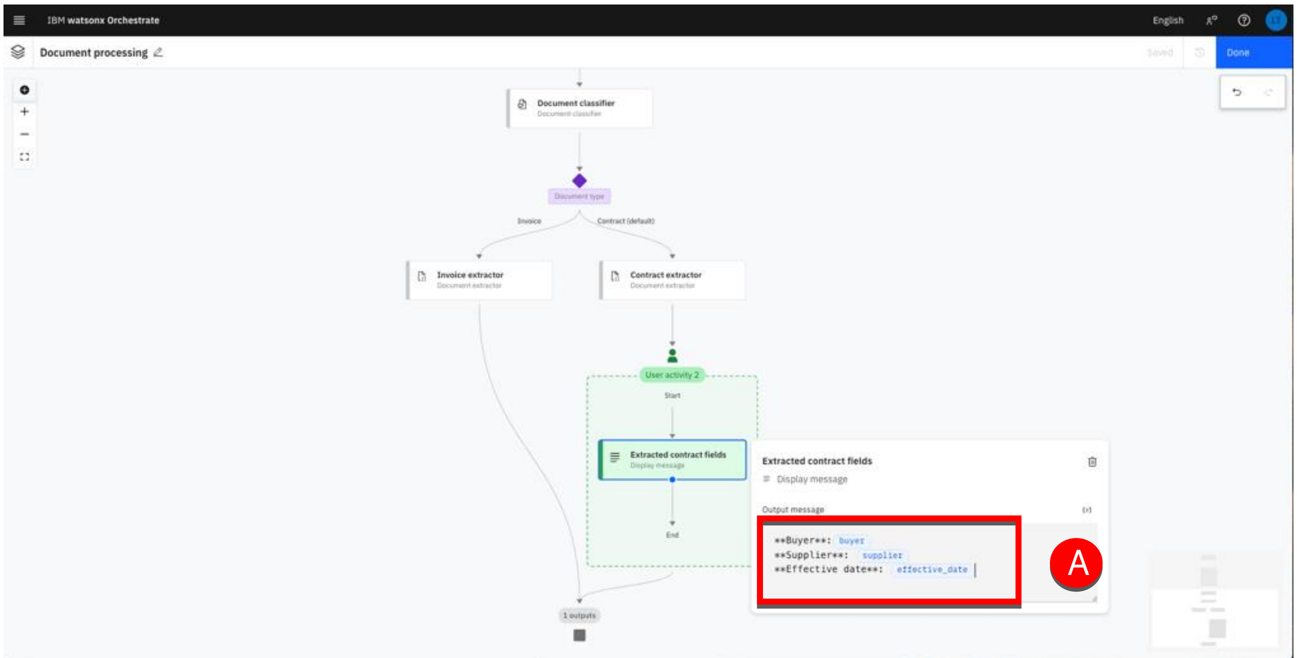
12. Your screen should look like (A):



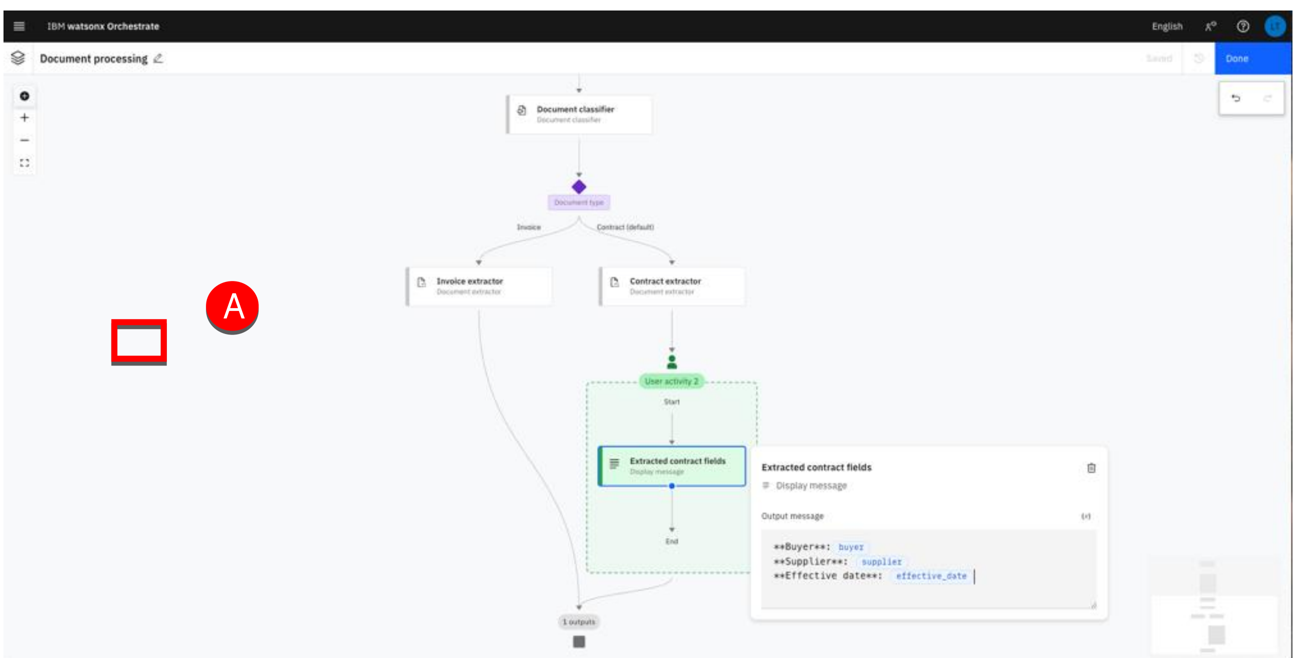
13. Repeat from step 9 to add the remaining fields defined in the Contract extractor:

    - Supplier
    - Effective date

    Your screen should look like (A):

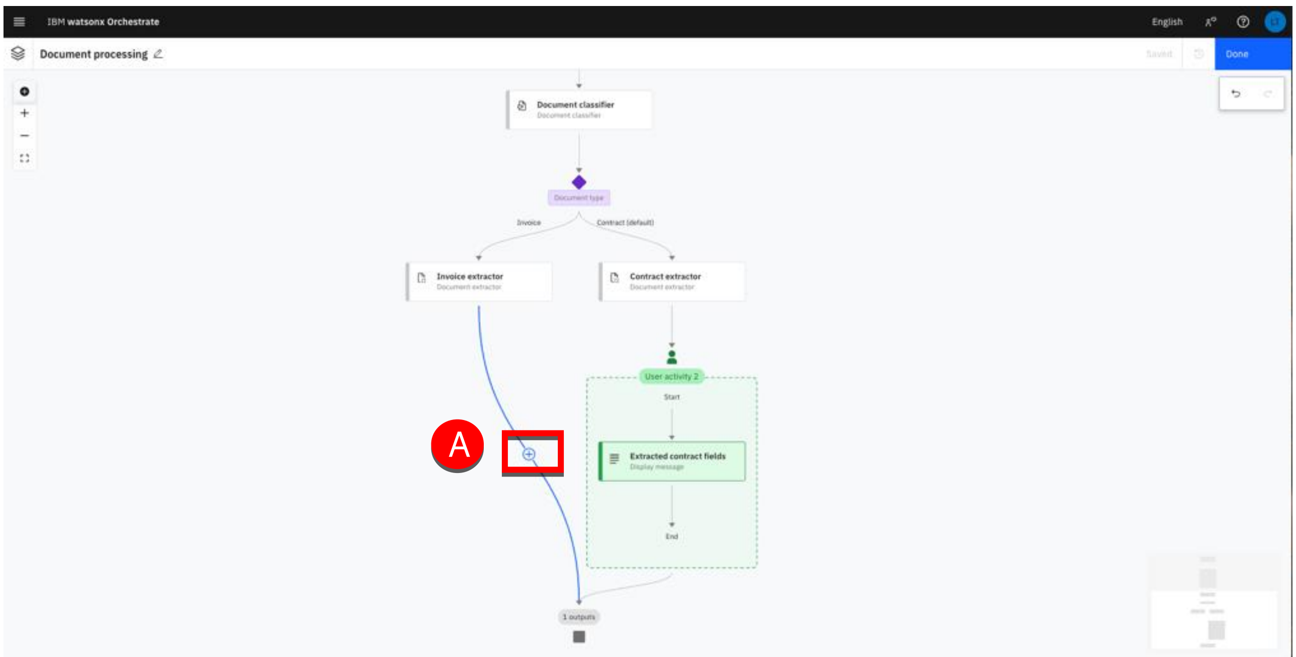


14. Click the background to close the **Extracted contract fields** panel (A).



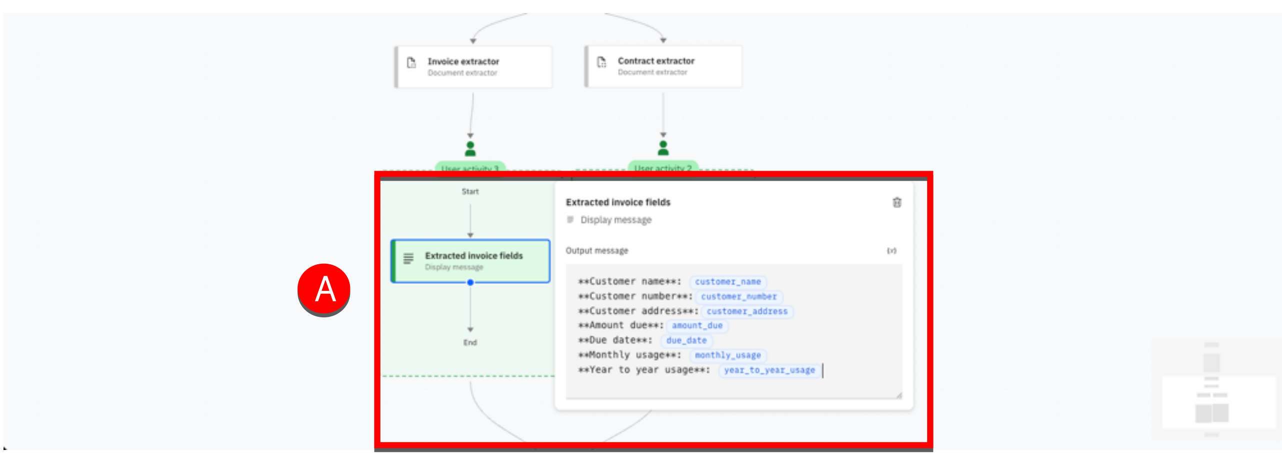
15. Repeating from Step 1, create a user activity in the **Invoice** branch (A) to display the following fields:

    - Customer name
    - Customer number
    - Customer address
    - Amount due
    - Due date
    - Monthly usage
    - Year to year usage



Note: The invoice variables will be under the Invoice extractor folder.

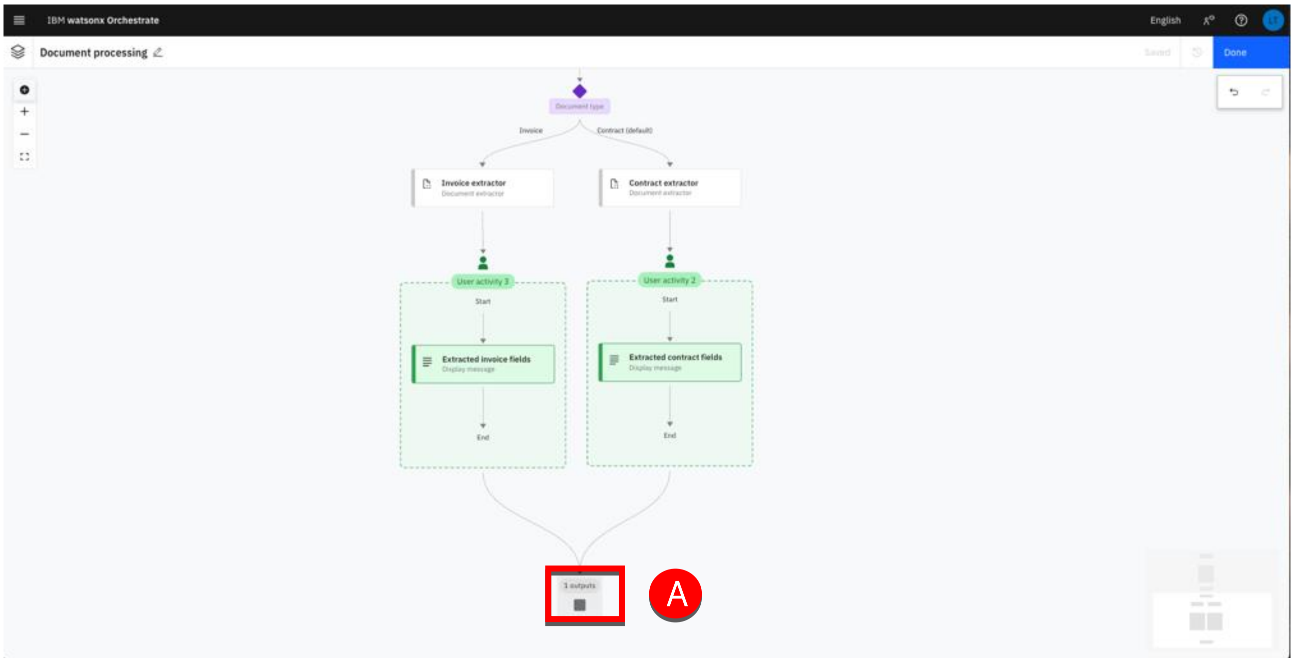
You should get the following result:



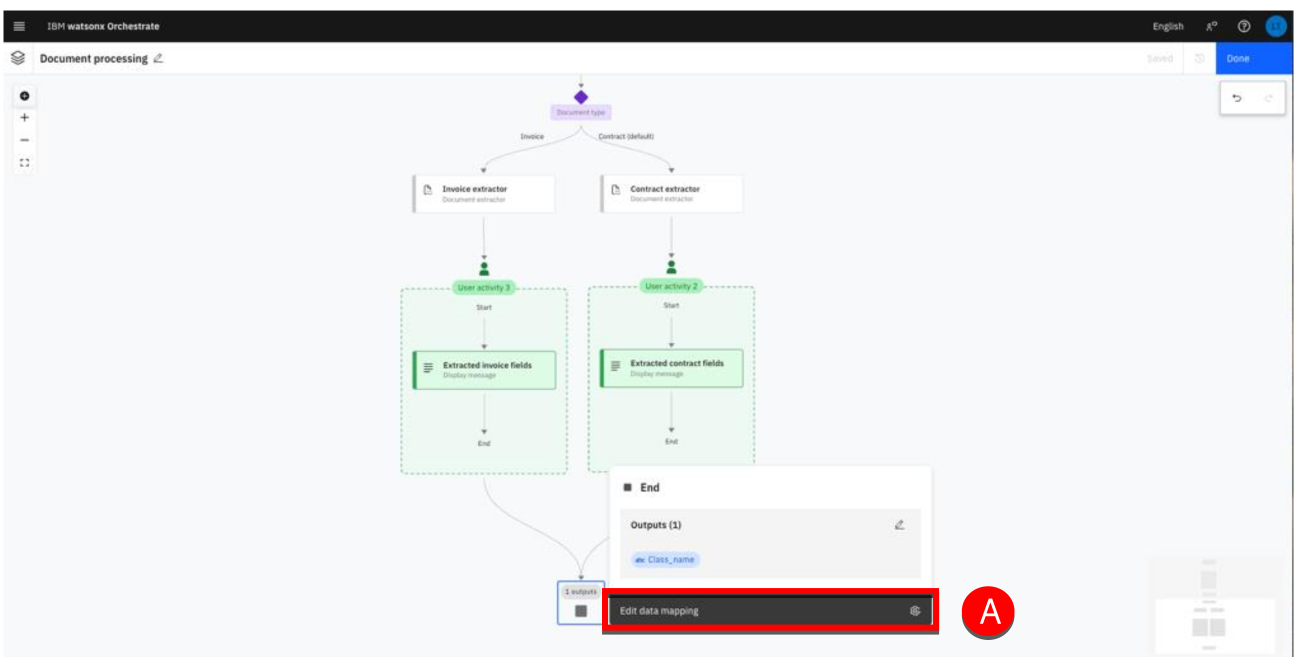
Define the end flow

Now that the output message has been defined for both file types, we can now conclude the flow by updating the end node. The end node will just return the Document type (i.e. Contract/Invoice).

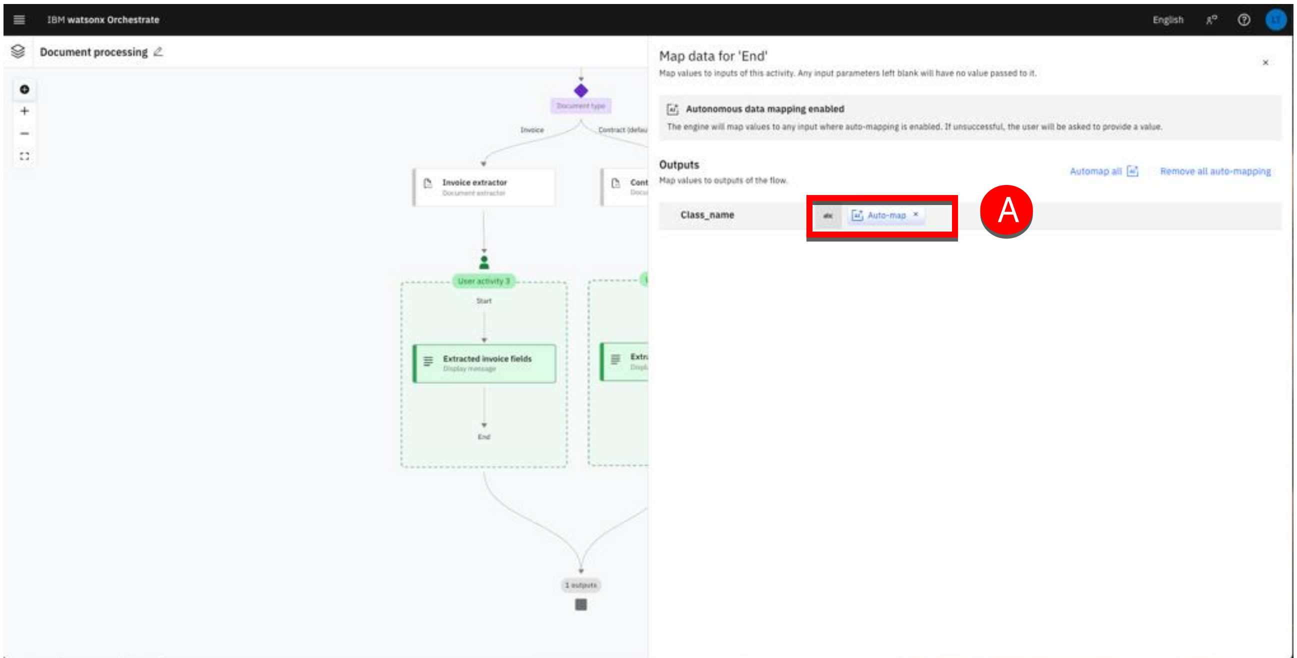
1. Click the **End** node (A).



2. Click **Edit data mapping** (A).

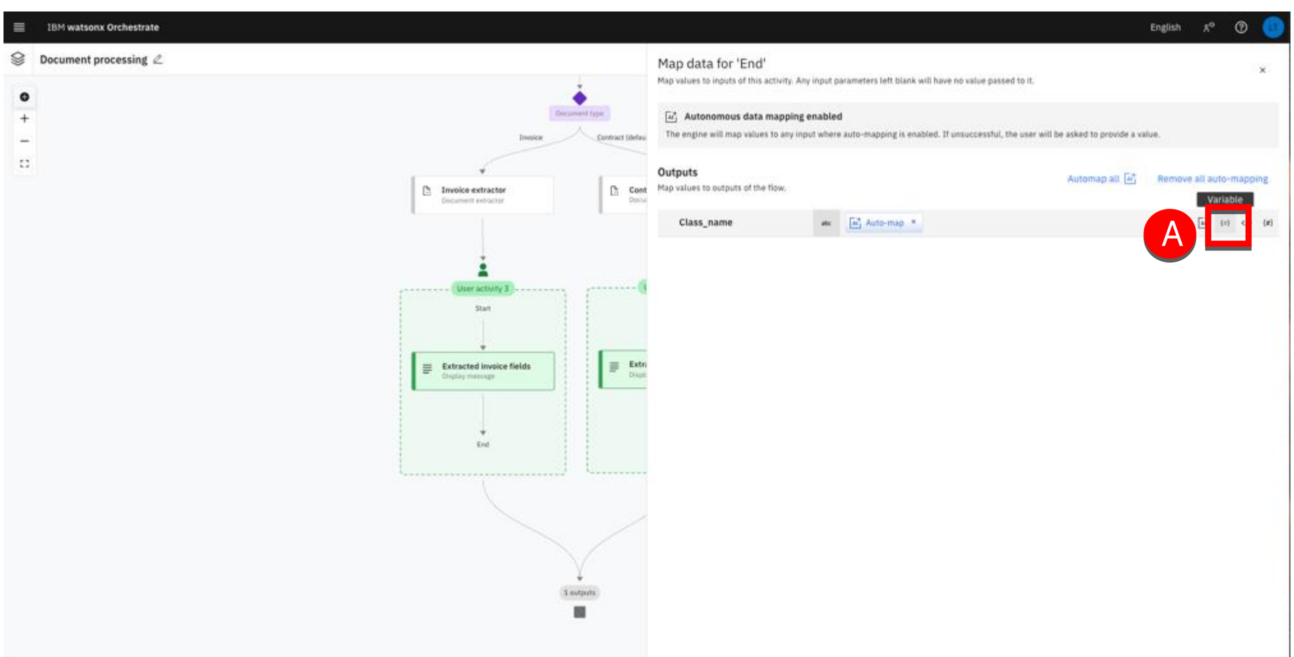


Note that the output, 'class_name' is auto-mapped by the LLM (A). This means that the LLM will attempt to automatically link the input/outputs.

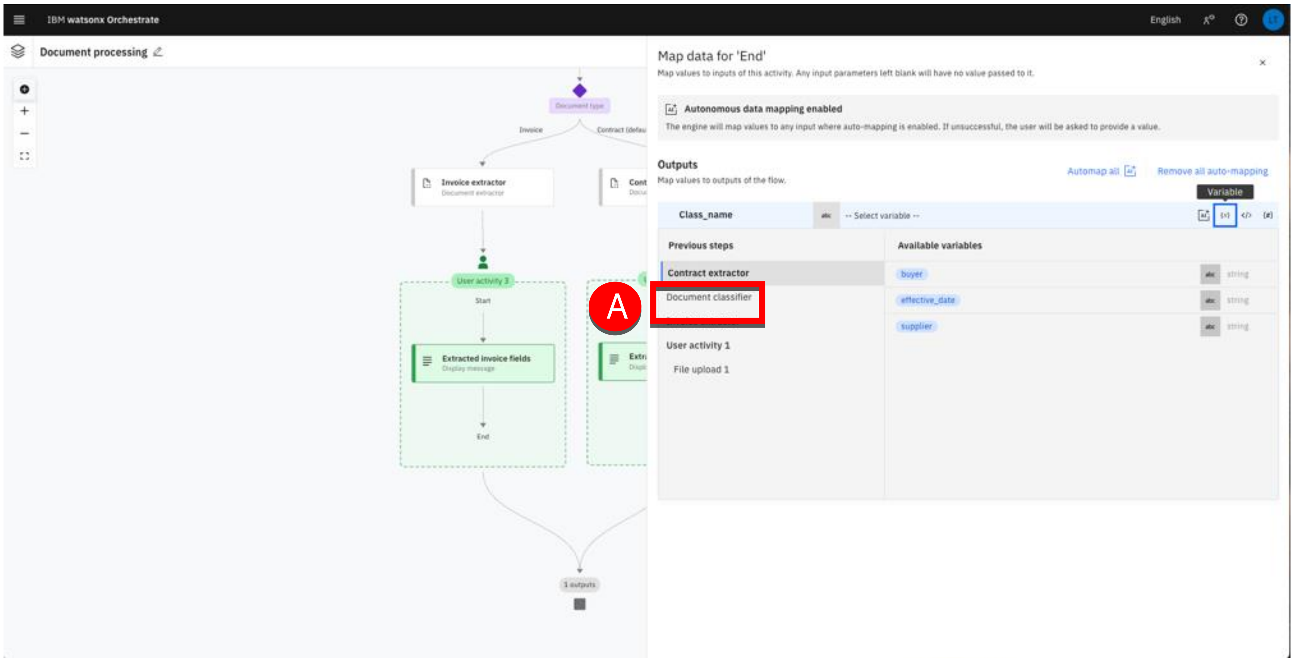


Note: The auto-map feature works effectively for those flows that are less complex. But for the more complex flows, (for instance if you had two 'class_name' outputs) then explicit mapping performs best.

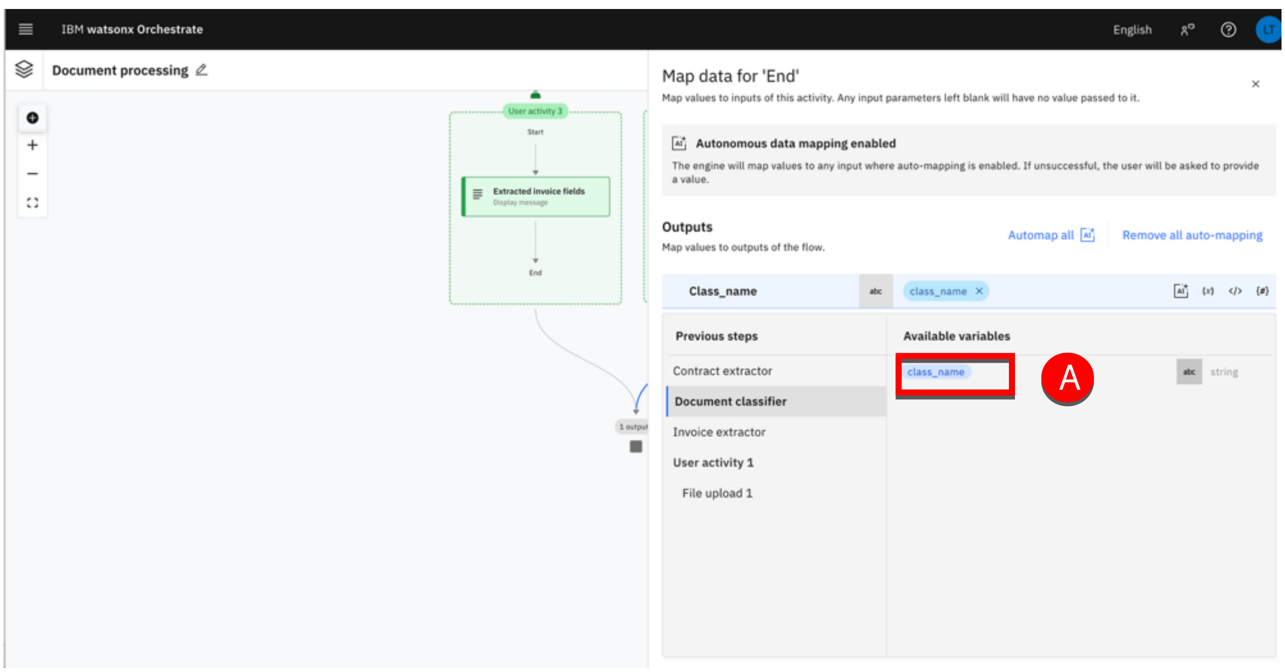
3. Hover the variable row and click on the **Variable** icon (A).



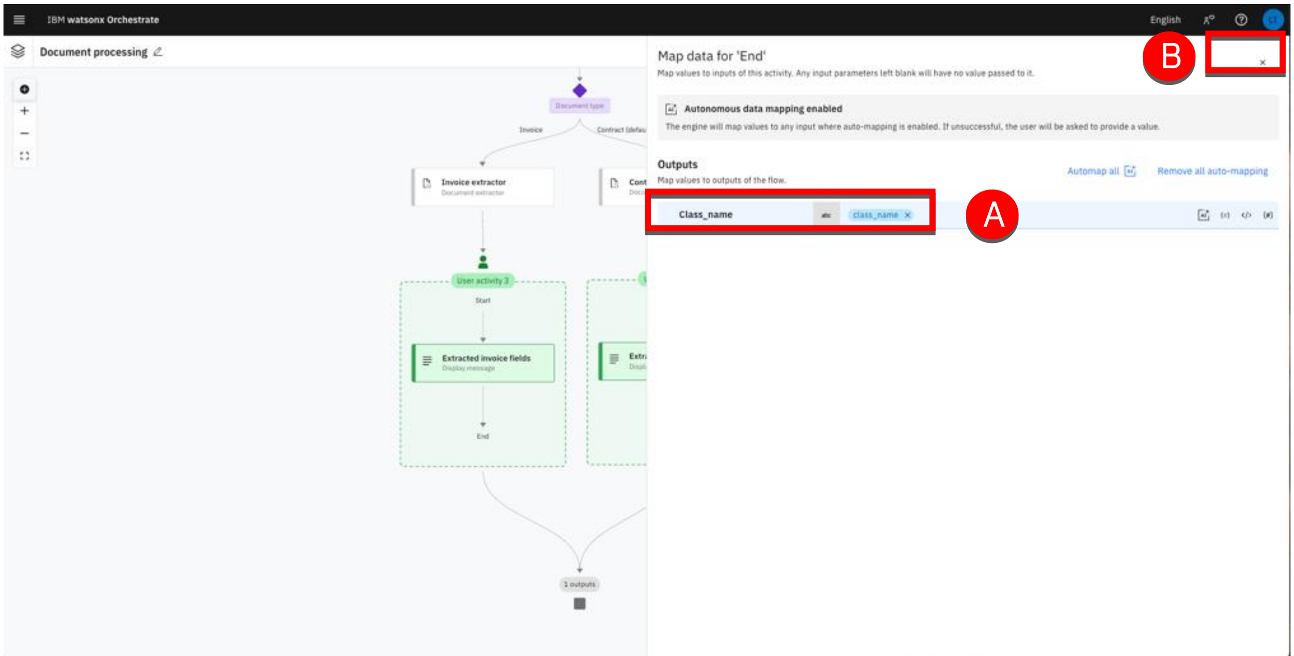
4. Click the **Document classifier** (A).



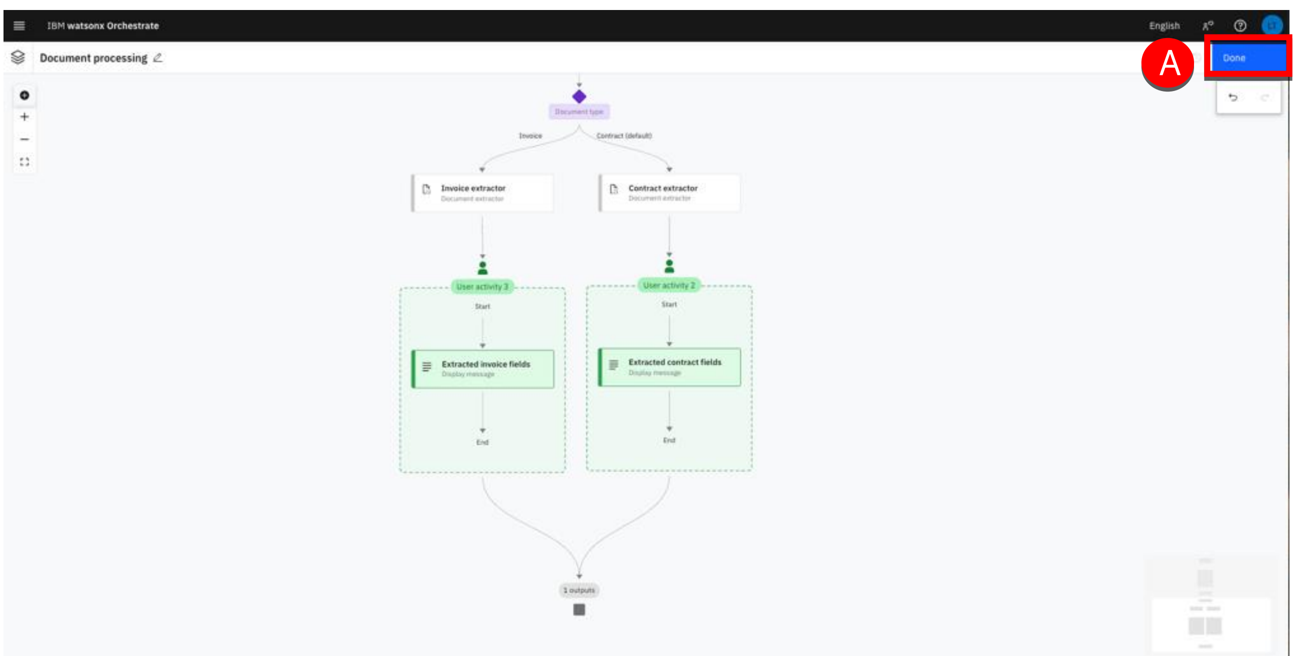
5. Select **class_name** (A).



6. The output ‘class_name’ is now assigned to the ‘class_name’ variable (A). Click x (B) to close the window.



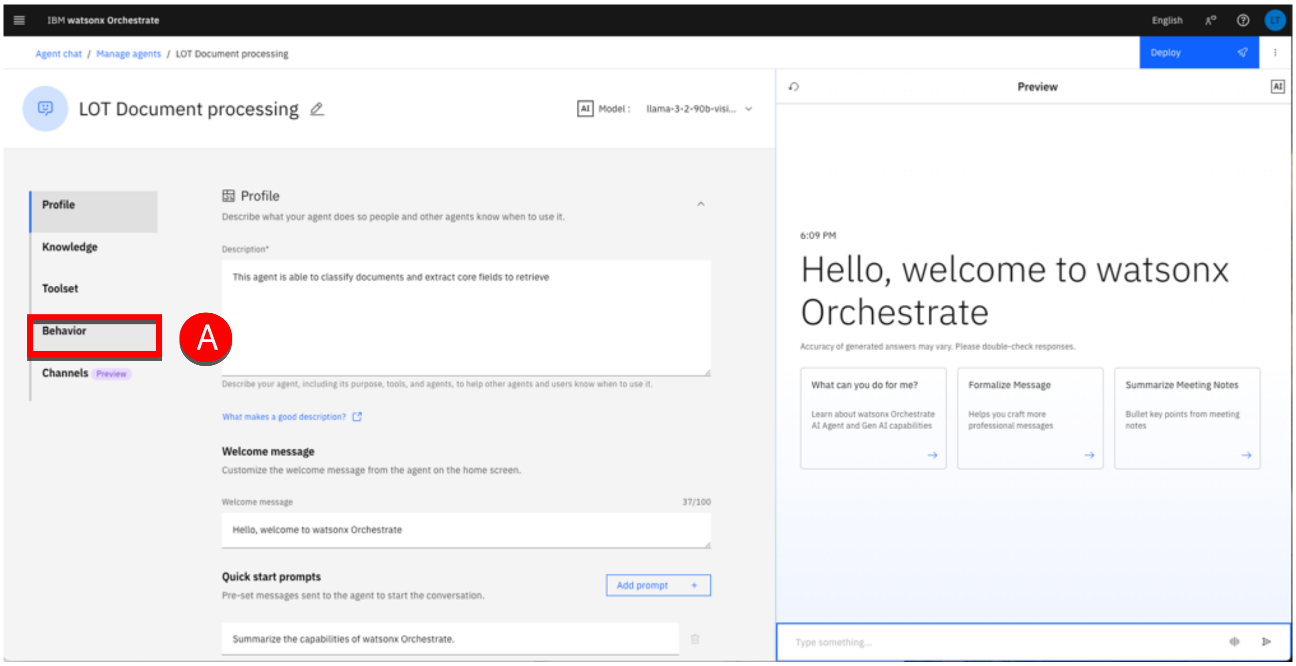
7. Click Done (A).



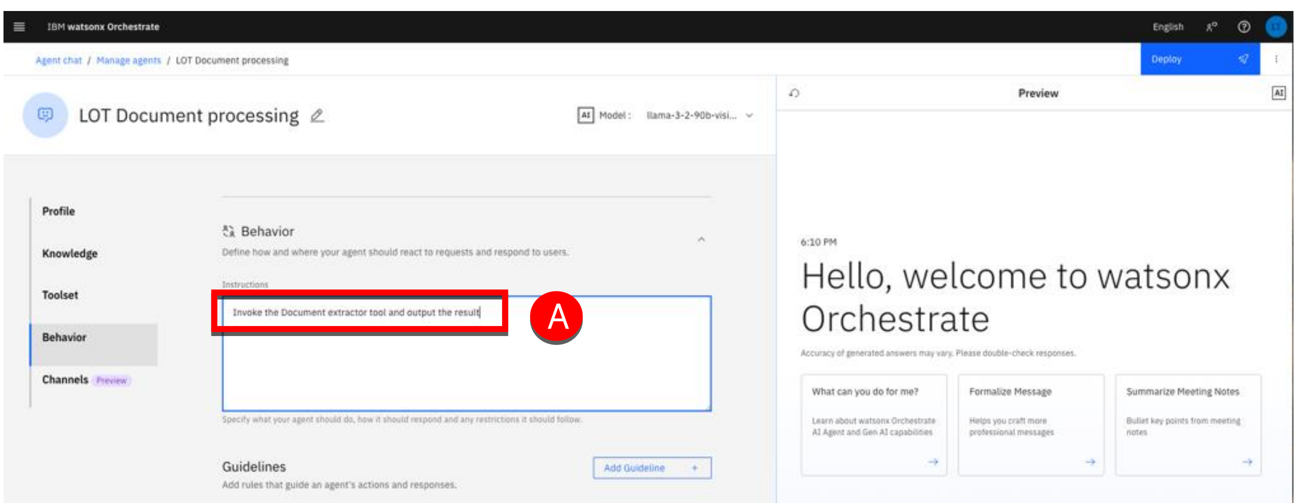
Add a behavior

To trigger the tool you just created, you must update the agent behavior.

1. Click **Behavior** (A).



2. Type 'Invoke the Document extractor tool and output the result' (A) in the Instructions.



Your agent is ready to be tested.

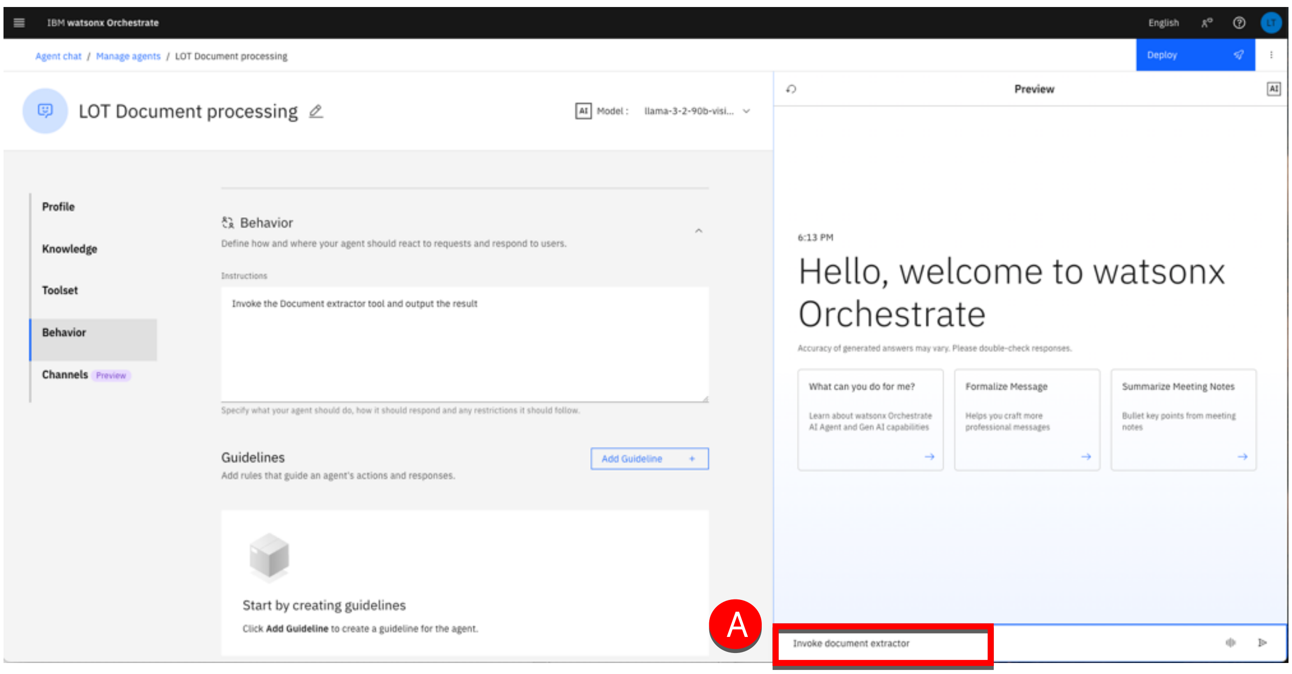
Test your agent

In this section, you will use the Agent preview to test your agent behavior. You will first use the contract.pdf to first test the workflow behavior. As this document has been used to train the contract extractor, there should be no extraction problem.
In a second test, you will use the Utility_Bill_5.pdf to test the agent. As this document has never been used by the invoice extractor, you may have to confirm some extracted value when the extraction confidence score is too low.

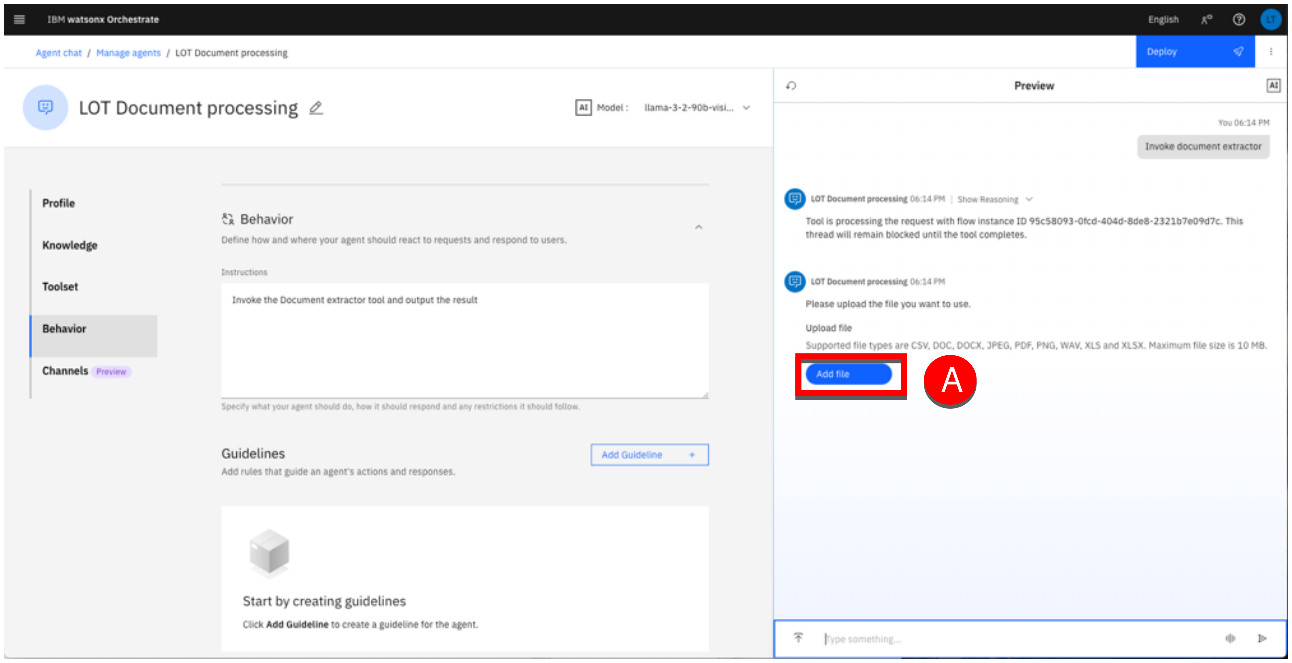
Simple workflow test

Run a simple test for contract extraction.

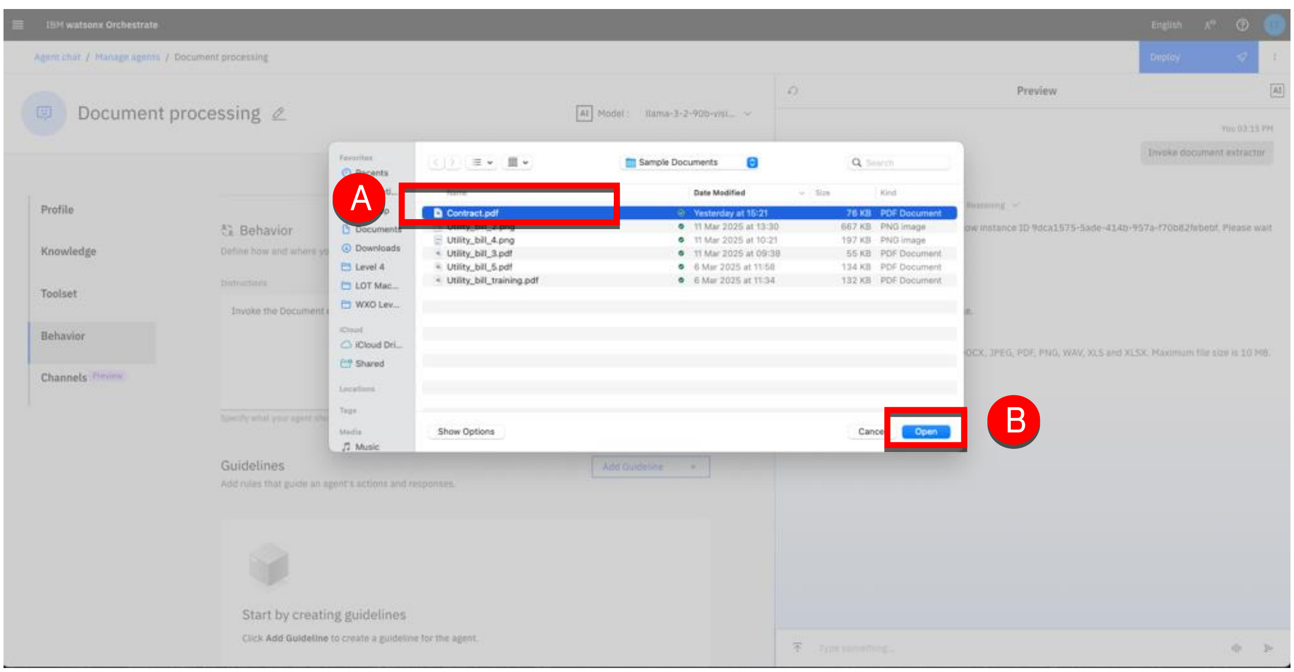
1. Type '`Invoke document extractor`' (A) in the Preview instructions area.



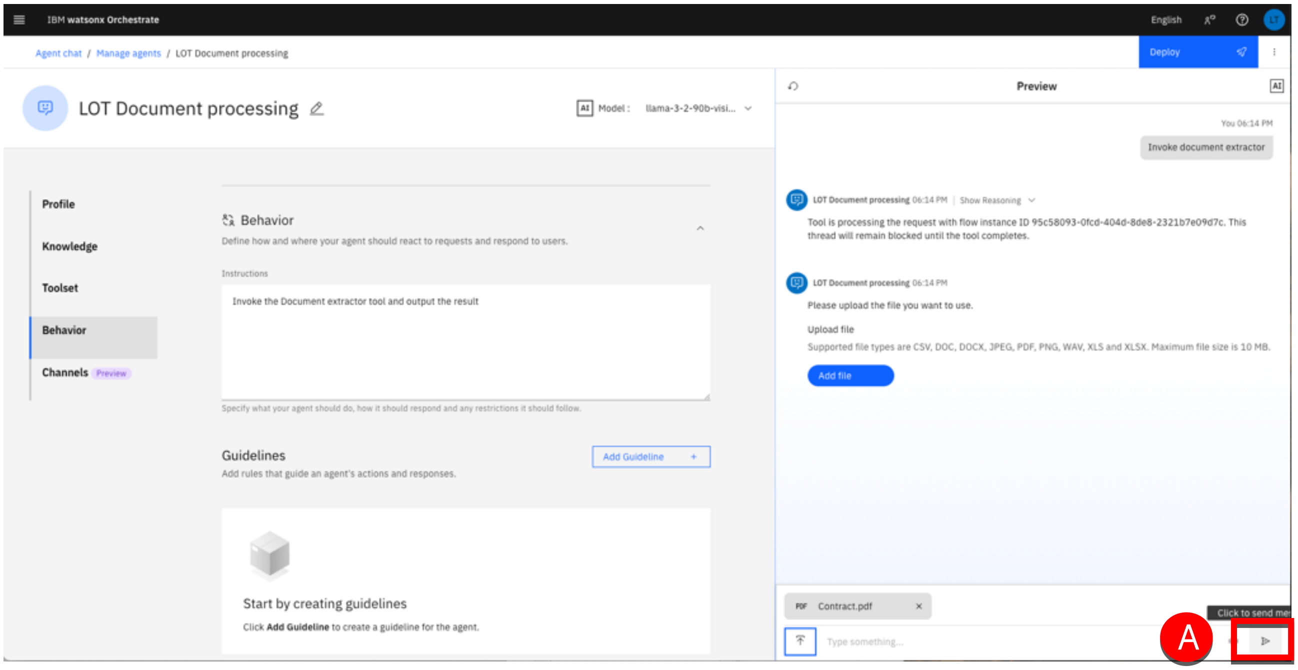
2. Click **Add file** (A).



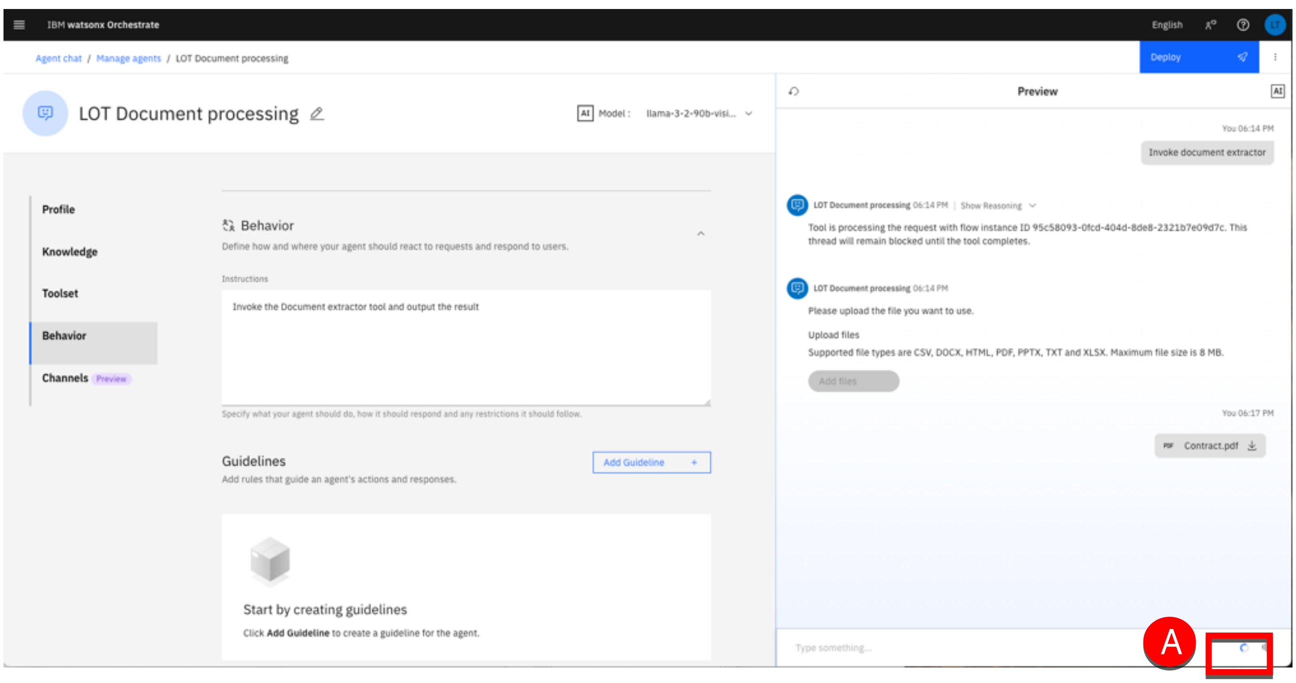
3. Navigate to your **contract.pdf** location and select it (A). Click **Open** (B).



4. Click the send button (A).

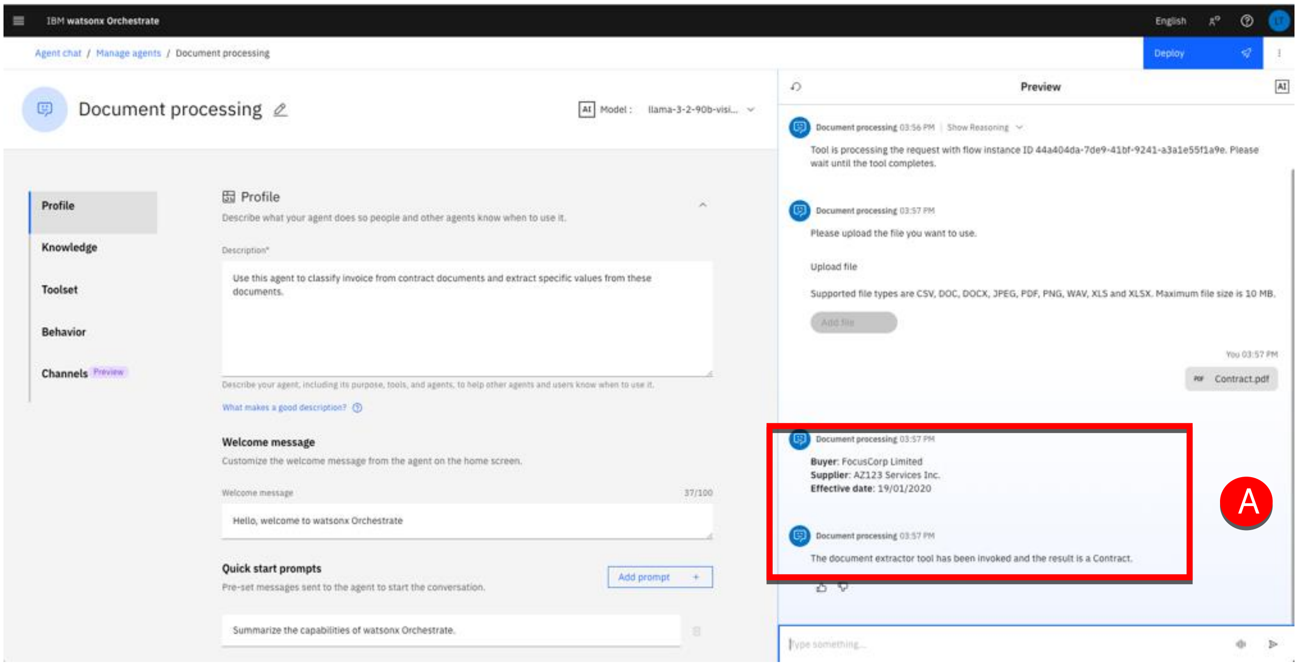


5. Wait for the agent to respond (A). This will take a few seconds.



After uploading your document, the agent will automatically execute the predefined workflow steps in the background. It will extract relevant data, focusing on the fields it has been trained on, and generate a response with the extracted information. Additionally, it will provide insights into the document classification.

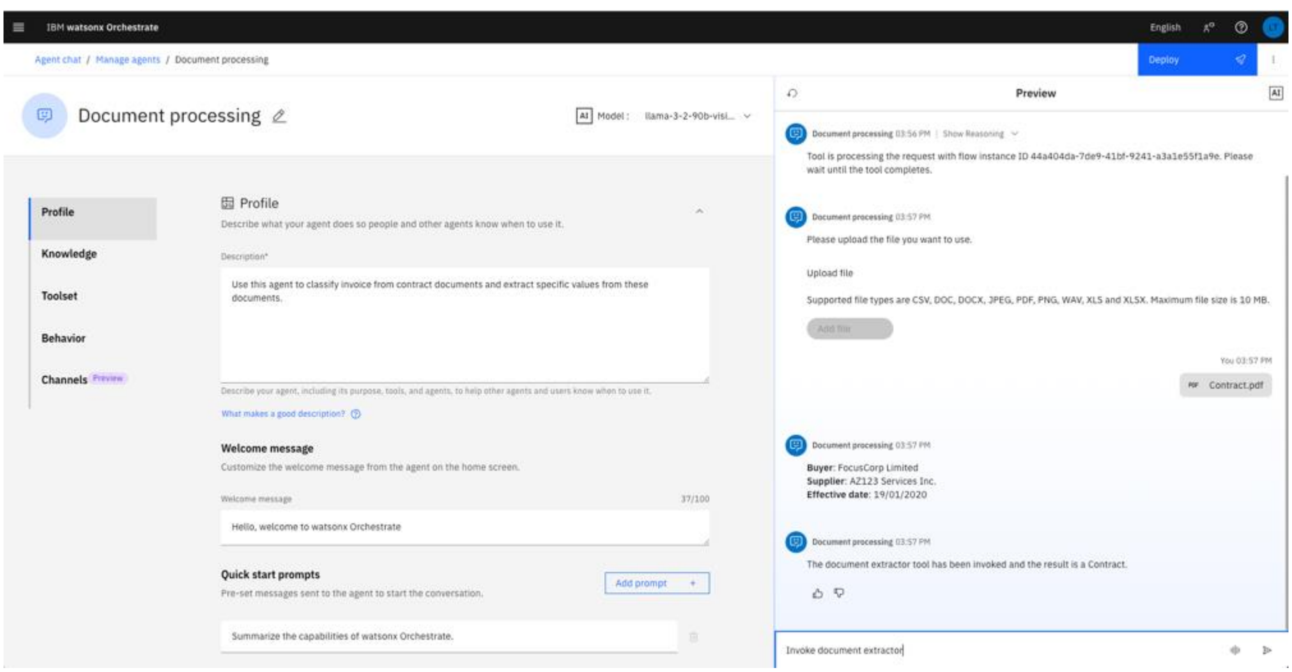
6. After a few seconds the agent will reply with the extracted data and the document type (A).



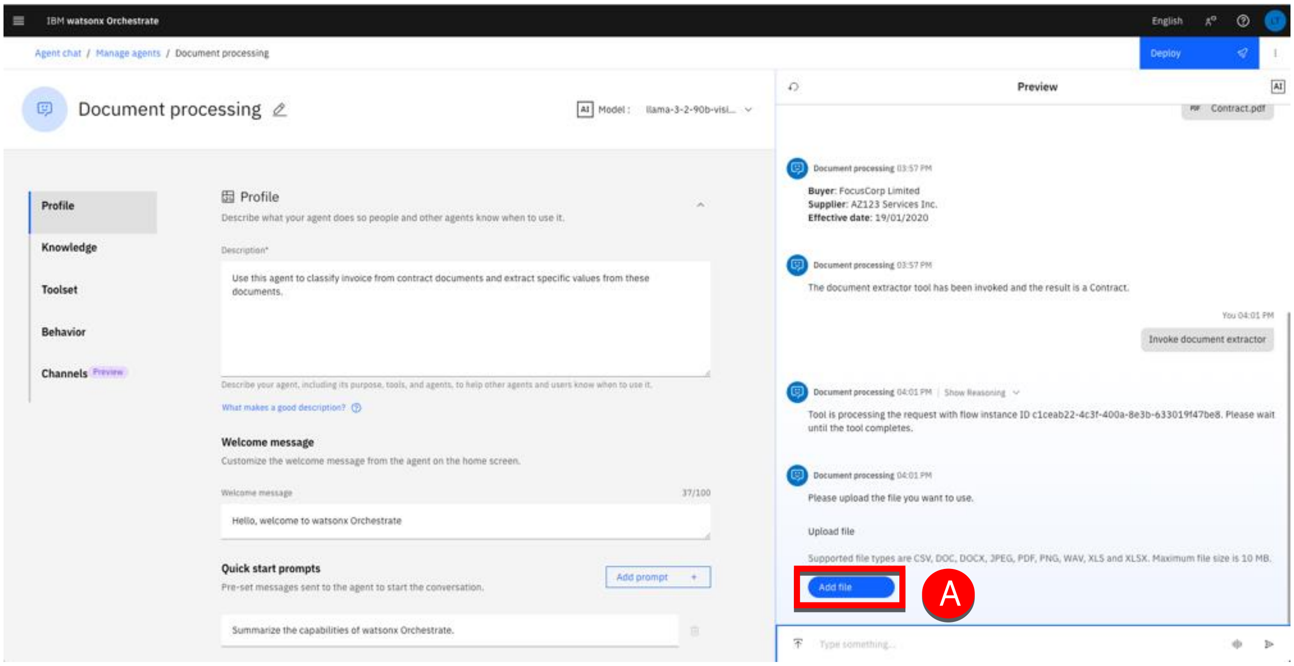
Test with possible extraction validation

Run a more complex test that may require user validation.

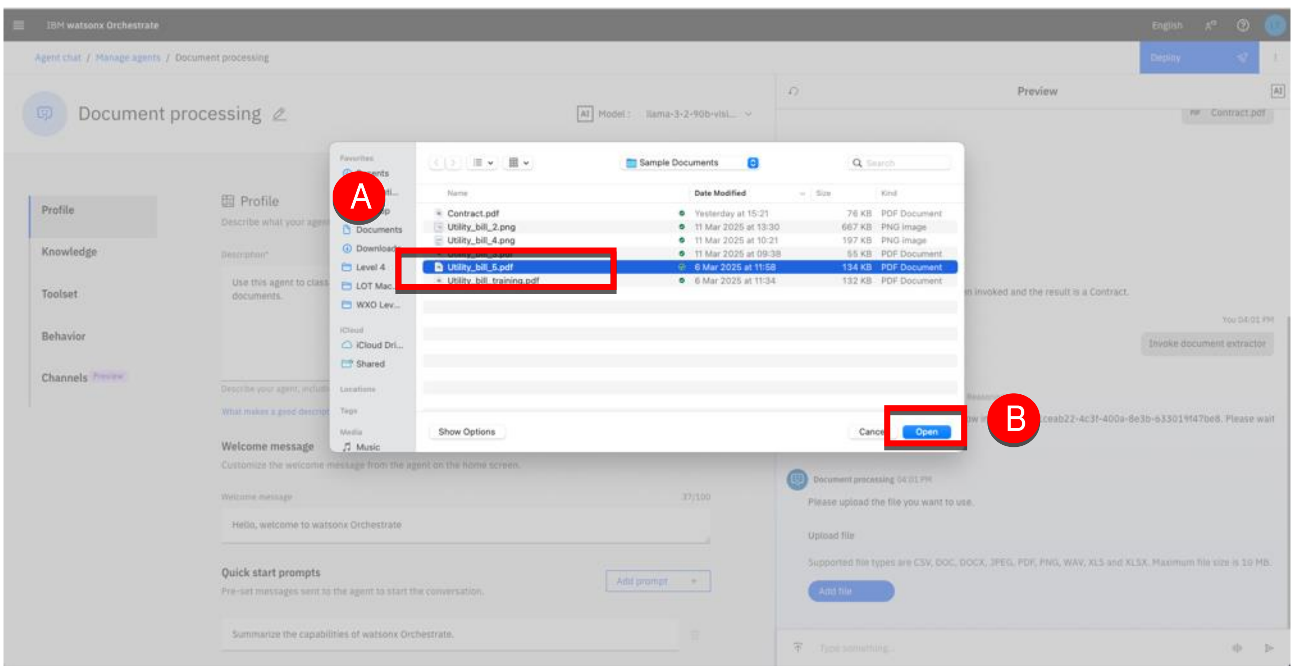
1. Type ‘`Invoke document extractor`’ (A) in the Preview instructions area.



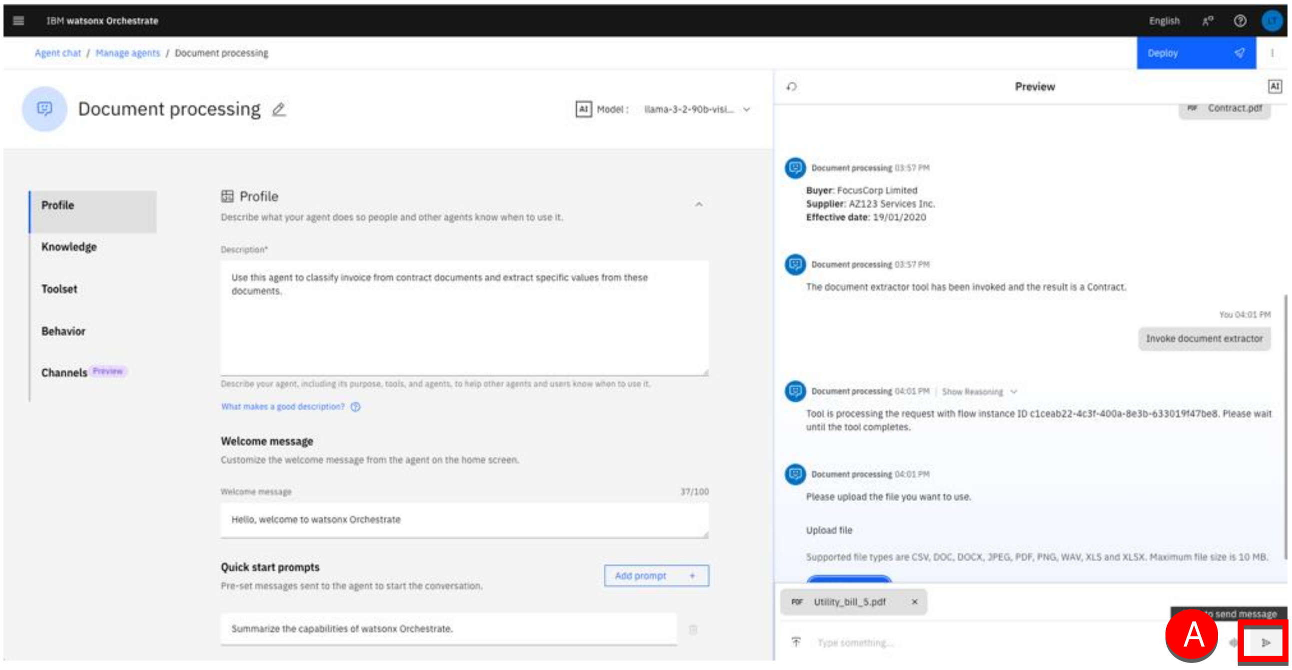
2. Click **Add file** (A).



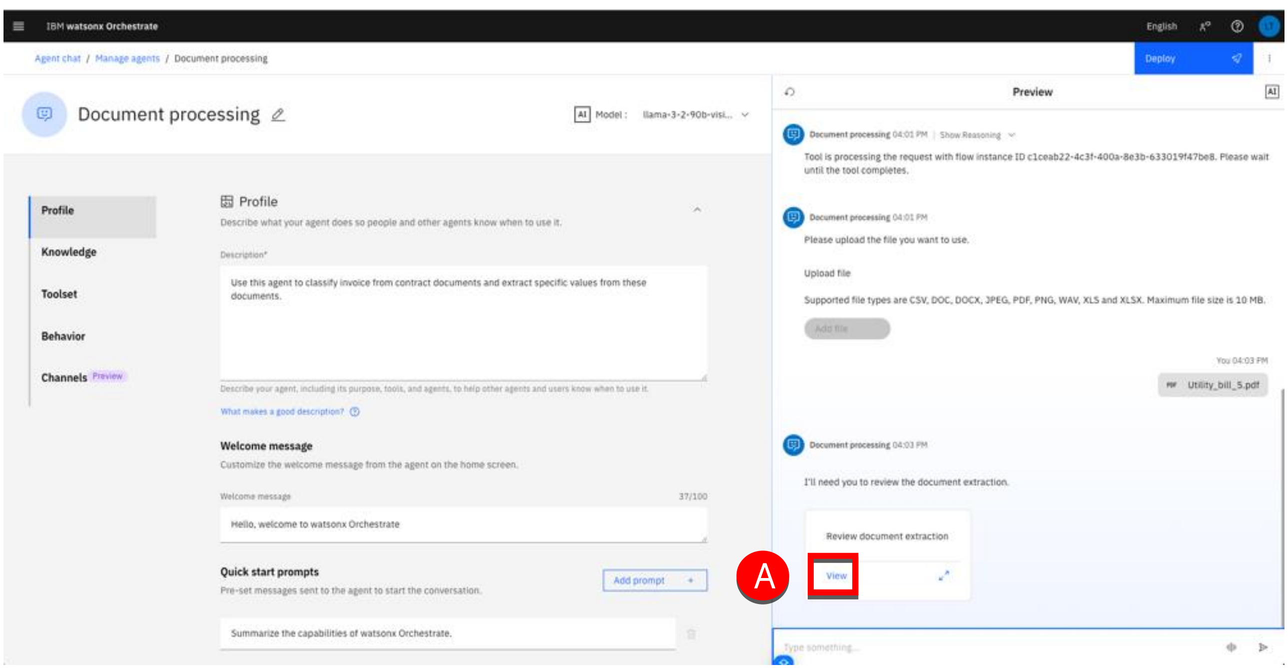
3. Navigate to your **Utility_bill_5.pdf** location and select it (A). Click **Open** (B).



4. Click the send message icon (A).

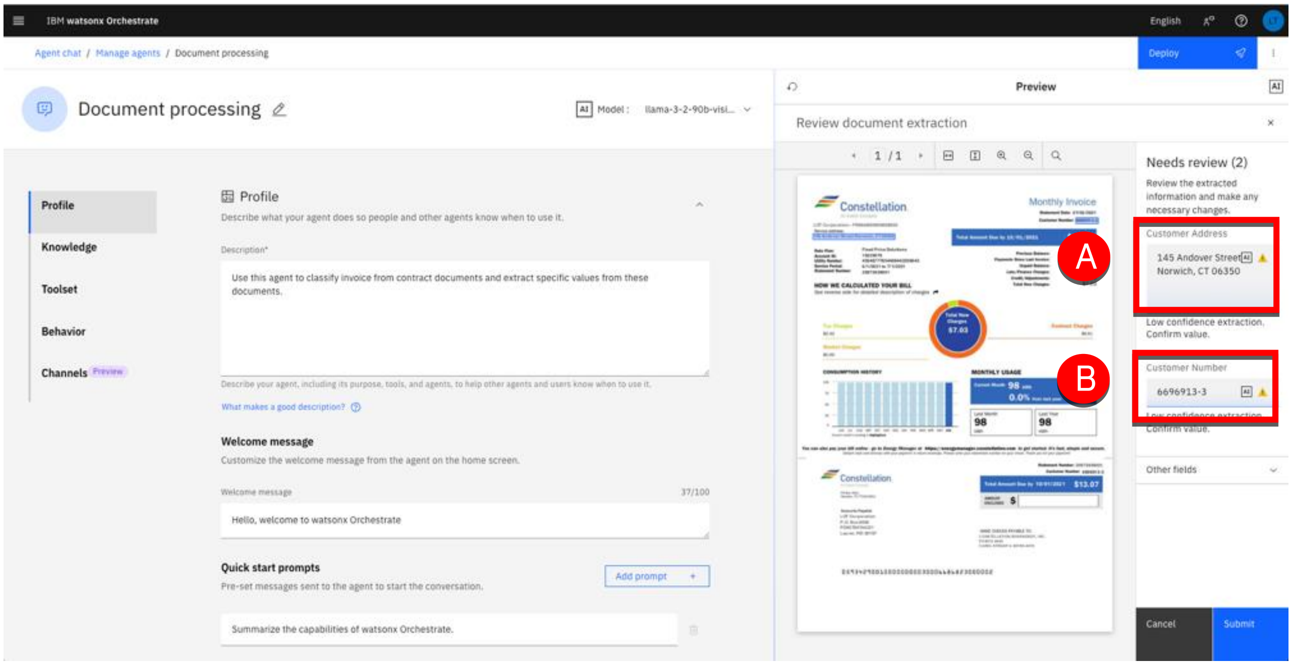


5. For this document the confidence score is too low for automatic extraction. The agent will ask you to validate some of the extracted fields. Click **View** (A).



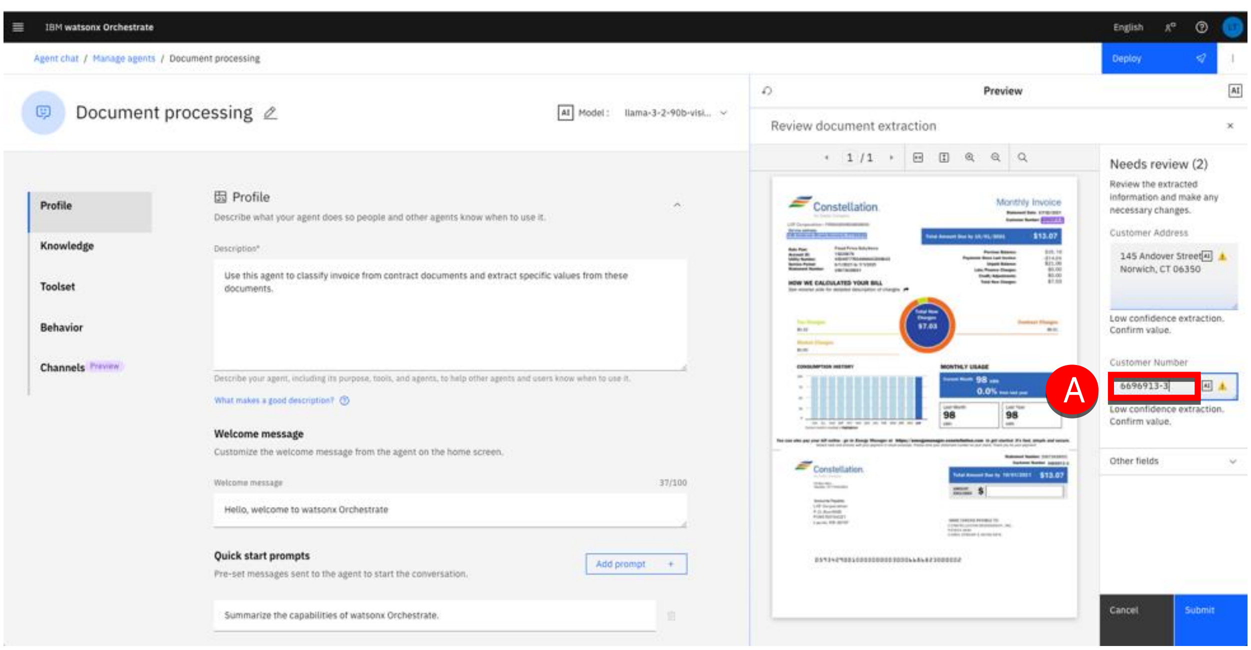
6. In this example, 2 fields require a review:

   - Customer address (A)
   - Customer number (B)

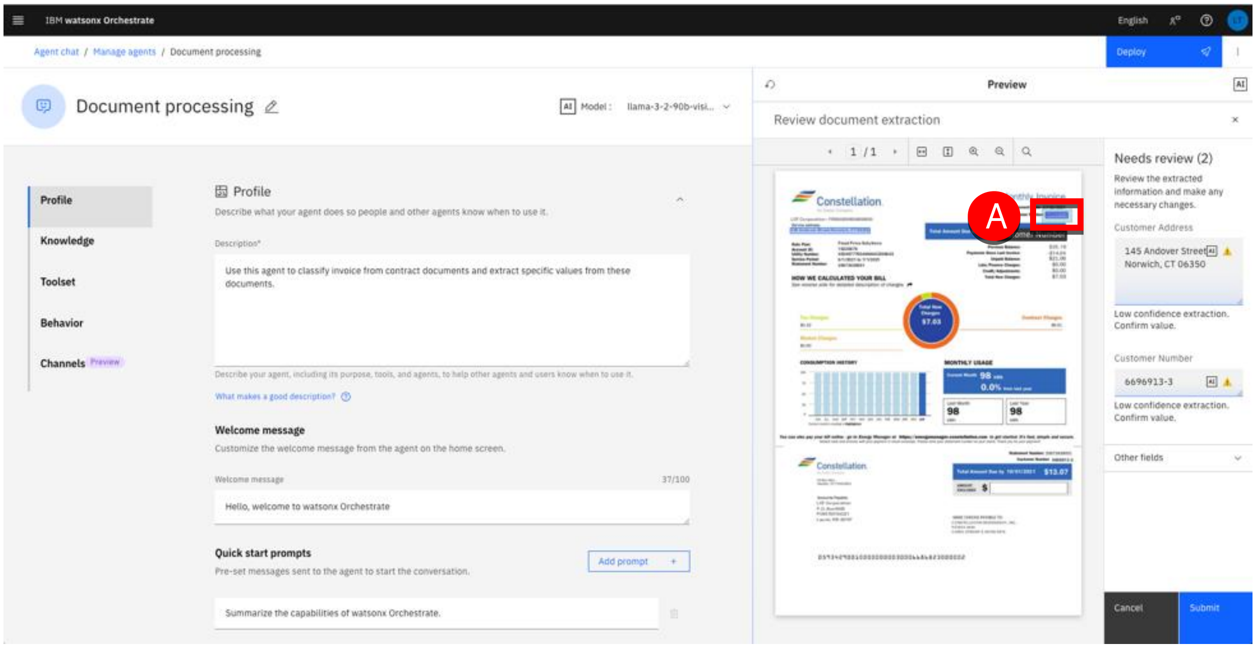


The extraction looks good despite the low confidence score. Should you need to correct an extracted value, follow the following step:

   a. Click the value you need to correct (A).

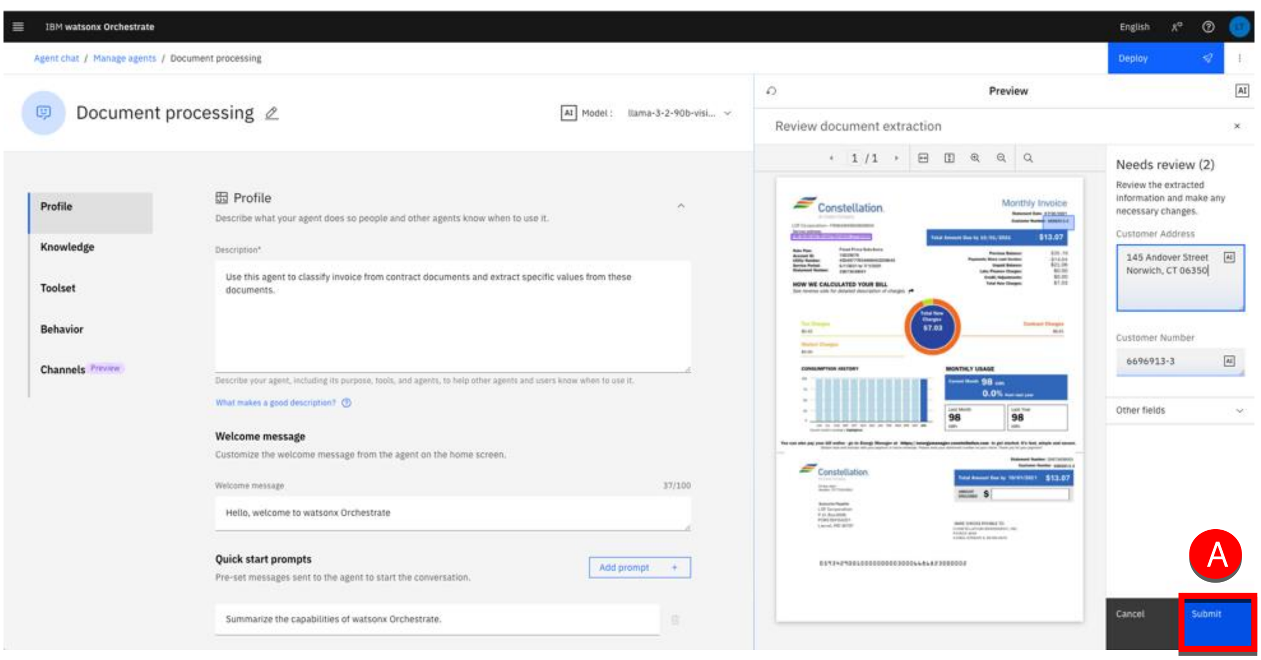


b. Draw a rectangle around the value in the document image (A) to capture the value.

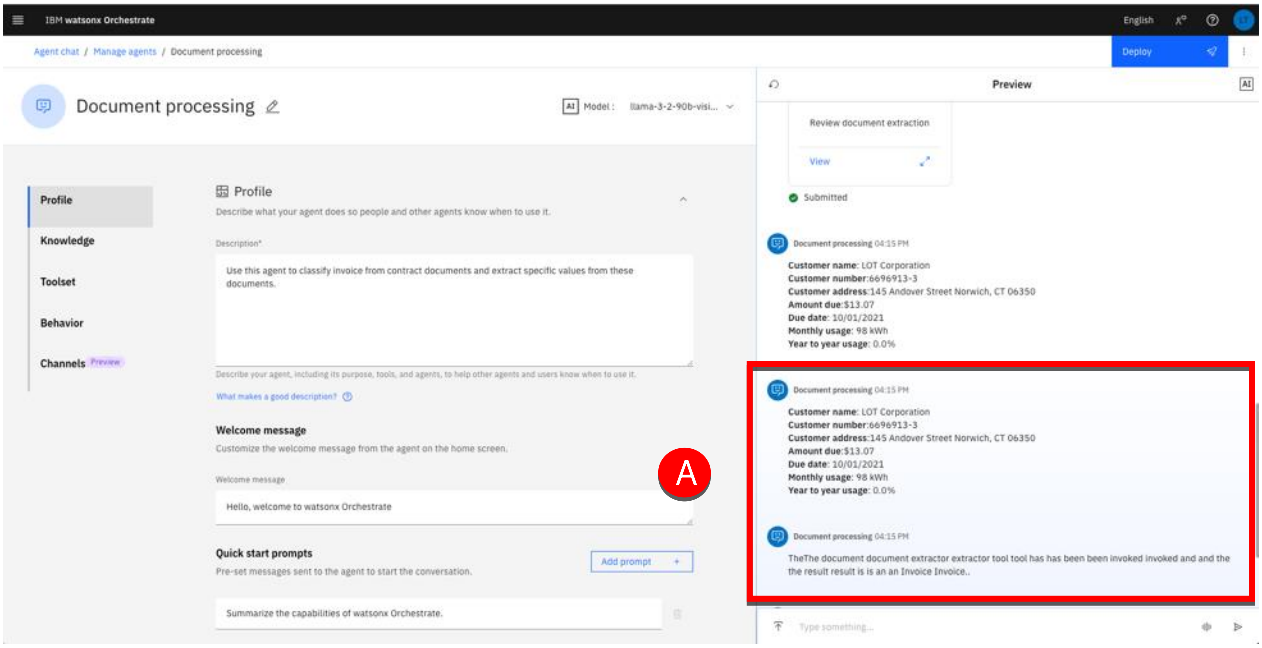


Note: You can also directly type in the value in the field.

c. Click **Submit** (A) when all the fields have been reviewed.



d. Wait for the process to complete and display the final result (A).



Note: At the time we are writing this guide, a reported defect causes the result to be displayed twice when performing manual validation.